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Authors

- Mr. Kamaldeen Adekola

Corresponding Author

Submitting Author

 [ORCID](#)

<https://orcid.org/0009-0008-6180-5546>

[CRediT](#)

Conceptualization, Methodology, Software, Data curation, Investigation, Validation, Formal analysis, Visualization, Project administration, Writing - original draft, Writing - review & editing, Resources

Affiliations

- Department of Engineering Systems and Services, Delft University of Technology, Delft, 2600 GA, Netherlands
- Prof. Laurens de Vries

[CRediT](#)

Validation, Supervision, Funding acquisition, Resources, Writing - review & editing, Conceptualization, Formal analysis, Methodology, Software

Affiliations

- Department of Engineering Systems and Services, Delft University of Technology, Delft, 2600 GA, Netherlands
- Dr. Kenneth Bruninx

[CRediT](#)

Funding acquisition, Validation, Writing - review & editing, Supervision, Resources, Investigation

Affiliations

- Department of Engineering Systems and Services, Delft University of Technology, Delft, 2600 GA, Netherlands

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Coupling_Europe_s_Capacity_Markets_paper_V1.pdf	Main Document - PDF	2.6 MB	Page 4
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Coupling Europe's Capacity Markets

Kamaldeen Adekola, Laurens de Vries, Kenneth Bruninx, *Member, IEEE*

Abstract—European Member States are increasingly introducing national capacity mechanisms (CMs) to manage growing adequacy risks. However, isolated national CMs are inefficient in highly interconnected electricity systems, such as the European system. While progress has been made in facilitating cross-border participation by generation capacity in CMs, existing arrangements are prone to under- or over-investment and do not properly value the contribution of interconnection capacity to Member States' adequacy targets. We argue that national capacity markets, offering long-term contracts, need a second tier: an annual capacity auction coupled across zones, which sets zonal prices for firm, deliverable capacity. We design such an auction by extending the logic of flow-based market coupling to capacity markets. In a comparative analysis of different market design scenarios in an illustrative multi-zone case study, using a bespoke long-run equilibrium problem, we show that the proposed auction removes the circular dependency between imports and available capacity in current cross-border participation schemes, locates investment where it is cheapest while keeping it deliverable under network constraints in defined scarcity situations, and makes the value of additional interconnection capacity explicit.

Index Terms—Flow-based Market Coupling, Capacity Market, Market Coupling, Cross-border Participation

I. INTRODUCTION

THE reliable and cost-effective supply of electricity is fundamental to any modern economy. After the liberalisation of the electricity sector in Europe, energy-only markets were expected to drive private investments [1]. However, the resulting electricity price signals fail to provide sufficient incentives for adequate investment in generation capacity due to market imperfections such as price caps (the "missing money" problem) and market incompleteness (the "missing (risk) markets" problem) [2], [3]. As a result, many governments have introduced capacity mechanisms to offer additional financial incentives for investments in generation capacity [4].

Previously viewed as tools of last resort, capacity mechanisms have increasingly become a common element in European electricity markets [5]. Member states have opted to implement capacity mechanisms as national measures, exercising their sovereign right to steer their energy mix [5]. Recent developments indicate a convergence toward single-buyer, market-wide capacity markets [6], [7]. In these markets, a central authority defines the capacity requirement based on the target reliability standard. Providers submit bids to supply capacity, and the market clears at the intersection of supply offers and the administratively set demand curve [6].

However, capacity mechanisms across Europe still (i) feature non-harmonised design parameters and (ii) allow for limited cross-border participation, despite this being mandated by law [5]. Contract durations vary significantly, typically between 1 and 15 years. Auctions occur at different lead times, ranging from 1 to 5 years [6]. Despite reliability options

becoming the preferred contract type, the specific contract forms, remuneration and penalties for non-delivery vary [7]. Member states conduct their auctions independently, and cross-border participation is limited to implicit participation—reducing capacity procurement based on expected but unremunerated imports, and explicit participation, which limits foreign resources' contribution to a static "maximum entry capacity" (MEC) at each border [7]. Both methods necessitate overly conservative cross-border assumptions that stifle regional investment efficiency [8].

Because the EU electricity market is highly interconnected, distorted incentives for generation and interconnection investment may arise from such national capacity mechanisms [9]. Furthermore, the current mechanisms for cross-border market participation fail to properly value interconnection and its availability during periods of scarcity [10]. Consequently, this results in a high risk of either over- or under-procuring capacity, depending on the Member States' ability to assess the availability of and willingness to rely on imports during periods of scarcity. Moreover, a circular dependency between available capacity and expected imports makes this assessment inherently challenging [8]. The limitations of current patchwork approaches to mitigate the distorted investment incentives arising from national capacity mechanisms have been acknowledged in the literature [10], [11].

Researchers increasingly call for a more harmonised and regional capacity market. Bucksteeg et al. highlight the potential cost savings of regional-level coordination of capacity mechanisms [12]. Menegatti and Meeus propose regionalisation of capacity markets in three sequential steps to minimise overprocurement of resources [8]. Höschle et al. [13] call for a fully integrated, market-based capacity mechanism that clears de-rated capacity, including generation, demand, and storage resources across borders. Capacity-price differentials would drive trade, while congestion rents would signal the need for new interconnectors. However, the existing literature falls short of providing a design for implementing regionalisation of resource adequacy, recognising the political reality that Member States prioritise sovereignty over their national adequacy and capacity mix.

This leads to a clear research question: How should capacity mechanisms in Europe evolve to facilitate the use of cross-border resources? We propose an additional tier: an annual capacity auction, coupled across zones. It complements the multi-year national capacity auctions in which Member States grant long-term contracts, such as reliability options, to de-risk investment. Member States, as holders of these long-term contracts, re-market the contracted volumes as annual products in the coupled auctions (Section II).

The Belgian and Irish capacity mechanisms offer a precedent for this "split" auction [14], [15]. In the Belgian capacity

mechanism, a "Y-4" auction secures the bulk of capacity four years in advance of delivery, allowing derisking capital-intensive investments with longer construction times. This is complemented by a "Y-1" auction held one year prior to delivery, which adjusts volumes based on updated demand forecasts and facilitates participation by existing capacity and fast-to-construct assets (such as battery storage).

Similarly, the German BMWK proposes a Combined Capacity Market [16]. It features an "investment layer" (a centralised capacity market) that provides long-term security for capital-intensive, controllable capacities, and an "adequacy layer" (a decentralised capacity market) where market participants manage residual peak loads through demand-side flexibility or by acquiring capacity certificates. We extend these two-tiered approaches into a European capacity mechanism by coupling their annual layer across borders.

The long-term contracts allow Member States to steer the development of their own energy mix, whereas the annual coupled auctions capitalise on the benefits of a highly interconnected system by facilitating cross-border participation of resources. A key challenge in these annual auctions is the inclusion of intra-zonal and inter-zonal network limitations that may impede delivery of contracted capacity.

We propose leveraging flow-based market coupling (FBMC) logic, currently used to couple Europe's day-ahead and intra-day markets. This allows us to account for network feasibility of flows during scarcity events in annual capacity market auctions (i.e. implicit transmission capacity allocation). Furthermore, we avoid the issues associated with defining Maximum Entry Capacities (MECs), such as simplifying assumptions on the capacity and availability of intra-zonal lines and the availability of generation capacity in neighbouring systems during scarcity situations, which can lead to conservative estimates. As more zones adopt capacity mechanisms, the use of MECs becomes more cumbersome and less accurate [8].

To illustrate the functionality and benefits of this new mechanism, which leverages FBMC logic to couple Europe's capacity markets, we develop a long-run equilibrium model and apply it in a stylised case study. In summary, our contribution is threefold:

Firstly, we propose an annual coupled capacity auction that is compatible with long-term contracts issued by member states to domestic resources to incentivise or de-risk investments. Cross-border participation in the annual capacity auction is enabled via FBMC logic. Combined, this preserves national sovereignty over security of supply and investment decisions while reducing inefficiencies associated with national capacity mechanisms.

Secondly, we develop a long-run equilibrium model of the energy and capacity markets, cast as a non-cooperative, perfectly competitive game, combining the work of Höschle et al. [13] (cross-border capacity markets) and Aravena et al. [17] (FBMC modelling). We replace the static "maximum entry capacity" approximation used in previous studies [11], [13] with an endogenously defined flow-based domain. By embedding flow-based constraints in the capacity market clearing, we ensure the deliverability of the cleared capacity bids w.r.t. network constraints during scarcity events.

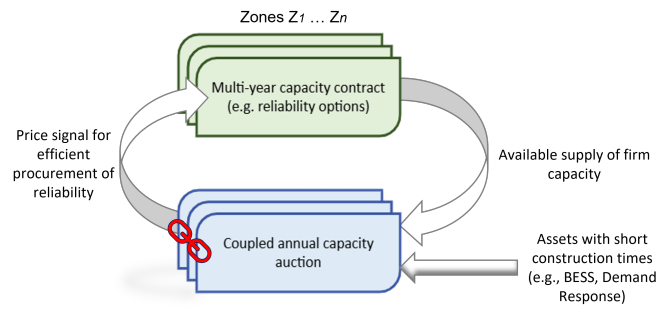


Fig. 1: The proposed annual coupled capacity auction and its interface with national capacity contracts. Member States grant multi-year contracts (e.g., reliability options) to de-risk investment and re-market the contracted volumes as annual products in the coupled auctions.

Finally, we present findings from a stylised case study that allows us to disentangle the impact of various approaches to cross-border participation in capacity markets in a long-run context. We demonstrate how the proposed mechanism facilitates larger cross-border capacity trades, correctly values interconnection capacity and reduces investment costs in generation capacity while maintaining resource adequacy by reallocating investments to more favourable locations within the network.

The remainder of this paper is organised as follows. Section II introduces a conceptual framework for coupling European capacity mechanisms. We present the mathematical formalisation in Section III. Section IV illustrates the proposed design with a numerical example. Section V reflects on the broader implications. We offer concluding remarks in Section VI.

II. COUPLING CAPACITY MARKETS: PROPOSED DESIGN

Based on de Vries et al. [18] and ENTSO-E [7], we highlight the following set of non-exhaustive design criteria for coupling capacity mechanisms:

- Satisfying both short-term and long-term adequacy needs
- Ensuring the deliverability w.r.t. network constraints of capacity offers during scarcity
- Enabling locational signals in capacity remuneration
- Recognising the heterogeneity of national resource adequacy requirements
- Reducing the need for the harmonisation of design parameters across member states

With the above in mind, we propose an annual, coupled European capacity auction as a coordinating layer for national capacity mechanisms awarding long-term contracts (Figure 1). Our primary contribution is the design of the second layer, but we reflect on the requirements it places on the first layer. On the first tier is the "Investment Layer", a long-term national capacity mechanism (e.g., in the form of a centralised, market-wide auction that uses reliability options as contracts). These reliability options should span a duration long enough — typically 10-15 years — to de-risk capital-intensive dispatchable resources, and auctions should be conducted well in advance of delivery (Y-4 or Y-5) to accommodate construction lead times.

Reliability options are preferred as they effectively reduce risk for both generators and consumers. They include a “pay-back” obligation when market prices exceed a predetermined strike price. This creates incentives for resources to remain available during scarcity periods [19]. The Investment Layer remains national in scope as Member States (MS) often have distinct energy policy objectives, such as technology restrictions or carbon targets, and also apply different adequacy standards. Thus, a national mechanism avoids the bottleneck of lengthy negotiations over a harmonised set of design criteria that would be necessary for a cross-border multi-year auction.

However, coordination of the investment tier with the second tier is essential. Member States, acting as the primary contract holders, are obliged to bid the total volume of their multi-year contracted capacity into the second-tier Coupled Annual Capacity Market as annual products. This ensures that long-term investments are fully reflected in the coupled annual capacity market, preventing artificial scarcity in the second tier.

This second tier consists of an annual, coupled capacity market. Firm capacity across all zones participates in a single annual uniform-price auction that implicitly accounts for network flows during anticipated scarcity events. By implicitly allocating transmission capacity in the clearing algorithm (see Section III-D1), we ensure that the cleared capacity can dispatch to meet demand during scarcity events. The uniform-price auction establishes a marginal clearing price for each zone in the coupled region, which reflects the value of one additional megawatt of firm capacity in that zone.

The annual capacity mechanism allows Member States to adjust for over- or under-procurement of long-term reliability options through cross-border trade. Countries that have procured excess options can offer them in the annual capacity auction, which is cleared subject to network constraints. The short-term auction then sends price signals for efficient procurement of reliability options in the long-term auction.

A critical input to the annual, coupled capacity markets is the set of scarcity events, which determine the demand, available generation, and the state of the network throughout the system during critical periods. These scarcity events can either be directly reflected in a stochastic market clearing – similar to how network contingencies are reflected in FBMC – or used to inform a coordinated scarcity assessment. In the latter approach, Regional Coordination Centres (RCCs) perform a Monte Carlo simulation to determine the minimum firm capacity in each zone that satisfies both simultaneous and isolated scarcity events within the coupled region. This is a natural extension of ENTSO-E’s adequacy assessments [20], and in line with ACER’s guidelines that advocate accounting for critical network elements in adequacy assessments [21].

To reflect the network feasibility of flows from contracted capacity to serve demand during scarcity events, we extend the logic behind flow-based market coupling (FBMC) to an annual, coupled, European cross-border capacity market for annual capacity contracts. Rather than computing border-to-border maximum entry capacities (MECs), which suffer from circular dependency with neighbours’ available capacity, FBMC explicitly models flows on critical network elements resulting from cross-border trade [22]. It enables the coupling

of European capacity markets within a common grid model, capturing the physical flows induced by zonal net positions, such as loop flows and internal bottlenecks. In doing so, it allows the deliverability of both foreign and domestic capacity to be assessed under defined scarcity conditions (Section II).

The short lead time and duration of the awarded capacity contracts efficiently signal short-term adequacy needs and promote the participation of flexible assets with short construction times, such as storage and demand-side response. Additionally, it mitigates the private financial risk associated with cross-border trade of long-term capacity obligations. Generation companies show limited appetite for long-term cross-border trade, as securing transmission rights years in advance is often impossible due to the limited liquidity. By implicitly allocating transmission capacity in an annual capacity market clearing, we remove the burden on generators to secure transmission rights. Instead, the risk of regional price differences is shifted to the state, which acts through the TSOs, who then capture a “capacity congestion rent” – the capacity price spread multiplied by the volume of reliability obligations exchanged across the zones. The collected rent can be reinvested in maintaining and further developing interconnection infrastructure or reducing network tariffs for consumers.

The proposed design respects the heterogeneity of adequacy requirements within the coupled region. Member States can impose their own reliability standards by defining the expected scarcity events for which annual capacity obligations must be available. They also determine their reliance on neighbouring countries through the interaction between the two layers of the proposed mechanism. For example, a state favouring national autonomy can prioritise multi-year capacity contracts. Thus, we strike a balance between national autonomy and regional integration. Consequently, the degree of regional efficiency will be determined by the level of coordination between the first and second layers in Figure 1.

Furthermore, the proposed design requires limited harmonisation of design choices and capacity market parameters across member states. For example, the responsibility for purchasing the required firm capacity demand in the annual auctions may be centralised, passed to load-serving entities and large electricity consumers in a decentralised capacity market, or a combination of both approaches. The multi-year contracts awarded to domestic resources may be bespoke, as it is the government that re-markets them as annual “European” reliability options. This allows for reflecting context-specific design requirements within a regional capacity mechanism.

Lastly, the proposed design can mitigate some common distortions associated with national capacity mechanisms. In the annual coupled auction, only physically deliverable resources can be selected. Consequently, a Member State cannot re-market the reliability options it holds from first-layer investments if assets are inefficiently located and unable to help its neighbours during scarcity. Furthermore, the combination of coordinated scarcity assessment, implicit transmission allocation and uniform pricing in the annual auction ensures participants are remunerated for their contribution to regional adequacy, potentially mitigating the capacity displacement effect observed in national capacity markets [23].

In Section III and IV, we zoom in on the second layer: a coupled annual capacity market, and abstract from the first layer, assuming a perfect propagation of price and investment signals to the first layer. Thus, our results do not capture the effect of distortions that arise within the first layer, such as overprocurement, cost-allocation and free-riding. We reflect on the impact of these limitations on our analysis in Section V. We capture our proposed design in a mixed complementarity problem and apply it to a case study, demonstrating how this can facilitate the coupling of European capacity markets.

III. MIXED COMPLEMENTARITY PROBLEM

We consider four types of agents: generators, consumers, a transmission system operator that couples the energy and capacity markets across borders, and a market clearing agent that clears the zonal energy and capacity markets. We assume perfectly competitive, zonal electricity and capacity markets. In the electricity market, generators self-dispatch to maximise profit, while consumers maximise their utility. The energy market operator clears the coupled zonal energy markets under flow-based market coupling. We do not include interconnection investments in the network operator's decision problem.

In the capacity market, consumers, through a central entity, specify their demand for capacity, and generators offer firm capacity, which is limited to their investments. We consider several scenarios for the capacity market operator that differ in how cross-border participation is implemented. These scenarios include our proposed flow-based coupling of capacity markets (FBMC).

Under FBMC, the feasible set of zonal net positions depends on installed capacity. Consequently, the Mixed Complementarity Problem (MCP) corresponding to this game describes a Generalised Nash Equilibrium. L  t   et al. [24] show that, under flow-based coupling, the classical equivalence between a central planner's optimisation problem and the game described above breaks down. Embedding the network constraints and generators in a single problem allows optimally siting generators within a market zone to manage congestion. The planner's solution is that of a market completed by locational payments, which are absent in real-world zonal markets and the game described above [24]. We therefore solve the MCP using a price-search algorithm based on the Alternating Direction Method of Multipliers (ADMM) [25].

We now outline the decision problems of each agent:

A. Generation agents

Each generator $i \in \mathcal{I}$, located in zone $z \in \mathcal{Z}$ maximises profit (Eq. (1a)) by choosing $x_i = (g_{i,z,t}, y_{i,z}, y_{i,z}^{\text{cm}})$: (i) its dispatch $g_{i,z,t}$ for all time steps $t \in \mathcal{T}$, incurring variable operational cost $C_{i,z}^g(g_{i,z,t}) = \frac{A_{i,z}}{2} g_{i,z,t}^2 + B_{i,z} g_{i,z,t}$, where $A_{i,z}$ and $B_{i,z}$ are technology-specific parameters; (ii) total capacity $y_{i,z}$ in zone z at cost $C_{i,z}^f$, the annualised fixed cost of new investment per MW; and (iii) for technologies eligible to participate in the capacity market, $i \in \mathcal{I}^{\text{cm}} \subseteq \mathcal{I}$, its capacity offer $y_{i,z}^{\text{cm}}$, up to its available capacity.

$\lambda_{z,t}^e$ and $\lambda_{z,t}^{\text{cm}}$ denote the zonal electricity market clearing price (  /MWh) and the zonal capacity market clearing price

(  /MW). Existing installed capacity is denoted by $Y_{i,z}$. The availability factor $AF_{i,z,t} \in [0, 1]$ specifies the fraction of installed capacity that is available to be dispatched in period t . Time-aggregation weights W_t scale the representative periods.

$$\max_{x_i \in \mathcal{X}_i} \sum_{t \in \mathcal{T}} W_t (\lambda_{z,t}^e g_{i,z,t} - C_{i,z}^g(g_{i,z,t})) - C_{i,z}^f (y_{i,z} - Y_{i,z}) + \lambda_{z,t}^{\text{cm}} y_{i,z}^{\text{cm}} \quad (1a)$$

$$\text{s.t. } 0 \leq g_{i,z,t} \leq AF_{i,z,t} \cdot y_{i,z}, \quad \forall t \in \mathcal{T}, \quad (1b) \quad 58$$

$$0 \leq y_{i,z}^{\text{cm}} \leq y_{i,z} \quad (1c) \quad 59$$

$$Y_{i,z} \leq y_{i,z} \quad (1d) \quad 59$$

B. Consumer agents

Each representative consumer in zone z maximises its surplus (Eq. (2a)) by choosing total electricity demand $d_{z,t}^e$, which consists of an inelastic component $d_{z,t}^{\text{inel}}$ and an elastic component $d_{z,t}^{\text{ela}}$, and unserved energy $\text{ens}_{z,t}$. We denote by $D_{z,t}^e$ the reference electricity demand in zone z and period t . A fraction $\epsilon_z \in (0, 1)$ of this demand is elastic. Consumers have a maximum willingness to pay for electricity, WTP_z . $\chi_c = (d_{z,t}^{\text{ela}}, d_{z,t}^{\text{inel}}, \text{ens}_{z,t}) \in \mathcal{X}_c$ is the decision vector of the consumer agents:

$$\max_{\chi_c \in \mathcal{X}_c} \sum_{t \in \mathcal{T}} W_t \left[(\text{WTP}_z - \lambda_{z,t}^e) d_{z,t}^e - \frac{\text{WTP}_z}{2 \epsilon_z D_{z,t}^e} (d_{z,t}^{\text{ela}})^2 \right] \quad (2a) \quad 70$$

$$\text{s.t. } d_{z,t}^{\text{inel}} + d_{z,t}^{\text{ela}} = d_{z,t}^e, \quad \forall t \in \mathcal{T}, \quad (2b) \quad 71$$

$$d_{z,t}^{\text{inel}} + \text{ens}_{z,t} = (1 - \epsilon_z) D_{z,t}^e, \quad \forall t \in \mathcal{T}, \quad (2c) \quad 71$$

$$0 \leq d_{z,t}^{\text{ela}} \leq \epsilon_z D_{z,t}^e, \quad d_{z,t}^{\text{inel}}, \text{ens}_{z,t} \geq 0, \quad \forall t \in \mathcal{T} \quad (2d) \quad 71$$

C. Transmission system operator - energy market

The transmission system operator allocates cross-border transmission capacity in the coupled zonal electricity markets. Acting as a price-taking agent, it takes electricity prices $\lambda_{z,t}^e$ and determines the zonal net positions $p_{z,t}^E$ that maximise congestion rent [24]. Network feasibility is imposed through flow-based market coupling, which we capture via the exact projection method of Aravena et al. [17]. The set of decision vectors for the energy market operator is denoted by $x_E = (p_{z,t}^E, f_{t,\ell}, \hat{g}_{i,n,t}, \hat{y}_{i,n}) \in \mathcal{X}_E$.

The transmission system operator's decision problem is presented in Problem (3). This problem is solved independently for every time step $t \in \mathcal{T}$. Eqs. (3c)–(3e) ensure that, for all periods t , there exists an auxiliary nodal dispatch $\hat{g}_{i,n,t} \leq \sum \hat{g}_{i,n,t}$ that meets nodal demand $d_{n,t}^e$. Zonal electricity demand is allocated to a nodal demand variable $d_{n,t}^e$ through coefficients $S_{n,z} \in [0, 1]$, which satisfy $\sum_{n \in \mathcal{N}_z} S_{n,z} = 1$ for each zone z . $\hat{y}_{i,n}$ is the nodal allocation of installed capacity. The nodal allocation of existing capacity is fixed by the same coefficients used to allocate demand, $\hat{y}_{i,n} = S_{n,z} y_{i,z}$, while new capacity is sited freely subject to Eq. (3e).

The transmission network consists of a set of nodes $\mathcal{N}$ and transmission lines $\mathcal{L}$. The subset of nodes belonging to zone

1 $z \in \mathcal{Z}$ is denoted by $\mathcal{N}_z$. Each line $\ell \in \mathcal{L}$, has a thermal
2 capacity $F_\ell^{\max}$. The nodal injection at node n is denoted by
3 $r_{n,t}$, and the corresponding line flows on ℓ are given by a DC
4 load-flow approximation using the power transfer distribution
5 factors $\text{PTDF}_{\ell,n}$.

6 The exact projection method [17] enforces network feasi-
7 bility of flows by requiring each admissible zonal net position
8 $p_{z,t}^E$ to lie within the flow-based domain. The domain is
9 constructed by aggregating the nodal injections $r_{n,t}$ that satisfy
10 the power balance (Eq. (3g)) and thermal limits (Eq. (3i)).
11 By mapping the nodal constraints into the zonal space via
12 (Eq. (3f)), we project exactly the nodal feasible set onto the
13 zonal net positions.

$$\max_{x_E \in \mathcal{X}_E} \sum_{z \in \mathcal{Z}} \lambda_{z,t}^E p_{z,t}^E \quad (3a)$$

$$14 \text{ s.t. } d_{n,t}^E = S_{n,z} d_{z,t}^E, \quad \forall n \in \mathcal{N}_z, z \in \mathcal{Z}, t \in \mathcal{T} \quad (3b)$$

$$15 \sum_{i \in \mathcal{I}} \hat{g}_{i,n,t} - d_{n,t}^E = r_{n,t}, \quad \forall n \in \mathcal{N}, t \in \mathcal{T} \quad (3c)$$

$$\hat{g}_{i,n,t} \leq A F_{i,z,t} \hat{y}_{i,n}, \quad \forall i \in \mathcal{I}, n \in \mathcal{N}_z, z \in \mathcal{Z}, t \in \mathcal{T} \quad (3d)$$

$$16 \sum_{n \in \mathcal{N}_z} \hat{y}_{i,n} = y_{i,z}, \quad \forall i \in \mathcal{I}, z \in \mathcal{Z} \quad (3e)$$

$$p_{z,t}^E = - \sum_{n \in \mathcal{N}_z} r_{n,t}, \quad \forall z \in \mathcal{Z}, t \in \mathcal{T} \quad (3f)$$

$$17 \sum_{n \in \mathcal{N}} r_{n,t} = 0, \quad \forall t \in \mathcal{T} \quad (3g)$$

$$f_{t,\ell} = \sum_{n \in \mathcal{N}} \text{PTDF}_{\ell,n} \cdot r_{n,t}, \quad \forall \ell \in \mathcal{L}, t \in \mathcal{T} \quad (3h)$$

$$18 -F_\ell^{\max} \leq f_{t,\ell} \leq F_\ell^{\max}, \quad \forall \ell \in \mathcal{L}, t \in \mathcal{T} \quad (3i)$$

19 D. Transmission system operator - capacity market

20 The transmission system operator maximizes the capacity
21 congestion rent in coupled capacity auctions, while certifying
22 the deliverability of cross-border capacity obligations w.r.t.
23 network constraints in anticipated scarcity conditions. Scarcity
24 scenarios $s \in \mathcal{S}$ represent stressed system states (e.g., simul-
25 taneous peak load or low renewable output) that define the
26 critical periods against which deliverability of procured local
27 and foreign capacity w.r.t. network constraints must be demon-
28 strated. The administratively set zonal demand for firm capac-
29 ity is denoted by D_z^{cm} . The set of decision variables of the
30 capacity market operator is $x_{\text{CO}} = (p_z^{\text{CO}}, \hat{g}_{s,n}^{\text{sc}}, f_{s,\ell}^{\text{cm}}, \hat{y}_n^{\text{cm}}) \in$
31 $\mathcal{X}_{\text{CO}}$, where p_z^{CO} denotes the net cross-border capacity obli-
32 gation at zone z , $\hat{g}_{s,n}^{\text{sc}}$ the auxiliary dispatch at node n in
33 scarcity scenario s , $f_{s,\ell}^{\text{cm}}$ the anticipated network flows during
34 defined scarcity period, and $\hat{y}_n^{\text{cm}}$ is the nodal allocation of
35 the total capacity obligation Y_z^{cm} by generators in zone z .
36 This nodal allocation cannot exceed the total available capacity
37 in the node $\hat{y}_{i,n}$ (Eqs. (4c) and (4d)). This nodal allocation
38 is determined in Problem (3) and, hence, is a parameter in
39 Problems (4) and (5).

40 We compare two approaches to explicit cross-border ca-
41 pacity trade, differing mainly in how the feasible domain
42 for cross-border trade is defined: (i) a flow-based domain

computed endogenously at clearing, which captures exactly the
set of net positions deliverable across all scarcity scenarios;
and (ii) an ex-ante defined Net Transfer Capacity (NTC)
domain, which imposes bilateral maximum entry capacities
(MECs) on each border.

1) *Flow-based coupling of capacity markets:* Our proposed
approach consists (i) a coupled capacity market clearing and
(ii) a joint scarcity assessment (Figure 2). The flow-based
coupling of the capacity markets is formalised in Problem (4).

The capacity market coupling allows net cross-border obli-
gations p_z^{CO} only if they are certified to be deliverable to
the relevant load centres in each anticipated scarcity scenario
 $s \in \mathcal{S}$. It does so by limiting the transmission capacity
made available for cross-zonal capacity exchange, ensuring
the feasibility of network flows during all scarcity situations.

In each scenario s , the zones in $\mathcal{Z}_s^{\text{sc}}$ are placed at their
scarcity demand, above the capacity demand $D_{s,n}^{\text{cm}} = S_{n,z} D_z^{\text{cm}}$
allocated to them by the joint assessment, while the remaining
zones are placed at their coincident demand (Figure 2, right).
The transmission system operator determines how much cross-
zonal capacity exchange can be supported while remaining
deliverable across the prescribed scarcity scenarios. To this
end, we required that there must exist a nodal dispatch $\hat{g}_{s,n}^{\text{sc}}$
that serves demand while respecting the network constraints,
including intra-zonal lines at state-N thermal limits in each
scenario. The resulting zonal net positions must remain con-
sistent with the cross-zonal capacity obligations. In particular,
a zone that has sold capacity abroad must be able to sustain
the corresponding export position when that capacity is called
during the importing zone's scarcity event. The auction yields,
for each zone z , a single annual capacity price $\lambda_{z,\text{cm}}^{\text{cm}}$, the
domestic capacity obligation Y_z^{cm} , and the net cross-border
capacity obligation p_z^{CO} .

$$\max_{x_{\text{CO}} \in \mathcal{X}_{\text{CO}}} \sum_{z \in \mathcal{Z}^{\text{cm}}} \lambda_{z,\text{cm}}^{\text{cm}} p_z^{\text{CO}} \quad (4a) \quad 76$$

$$\text{s.t. } \hat{g}_{s,n}^{\text{sc}} \leq \hat{y}_n^{\text{cm}}, \quad \forall s \in \mathcal{S}, n \in \mathcal{N} \quad (4b) \quad 77$$

$$\hat{y}_n^{\text{cm}} \leq \sum_{i \in \mathcal{I}^{\text{cm}}} \hat{y}_{i,n}, \quad \forall n \in \mathcal{N}_z, z \in \mathcal{Z} \quad (4c) \quad 78$$

$$\sum_{n \in \mathcal{N}_z} \hat{y}_n^{\text{cm}} = Y_z^{\text{cm}}, \quad \forall z \in \mathcal{Z} \quad (4d) \quad 79$$

$$\hat{g}_{s,n}^{\text{sc}} - D_{s,n}^{\text{cm}} = r_{s,n}, \quad \forall s \in \mathcal{S}, n \in \mathcal{N}_z \quad (4e) \quad 80$$

$$- \sum_{n \in \mathcal{N}_z} r_{s,n} \leq p_z^{\text{CO}}, \quad \forall s \in \mathcal{S}, z \in \mathcal{Z} \setminus \mathcal{Z}_s^{\text{sc}} \quad (4f) \quad 81$$

$$\sum_{n \in \mathcal{N}} r_{s,n} = 0, \quad \forall s \in \mathcal{S}, \quad (4g) \quad 82$$

$$\sum_{z \in \mathcal{Z}} p_z^{\text{CO}} = 0, \quad (4h) \quad 83$$

$$-F_\ell^{\max} \leq f_{s,\ell}^{\text{cm}} \leq F_\ell^{\max}, \quad \forall s \in \mathcal{S}, \ell \in \mathcal{L} \quad (4i) \quad 84$$

$$f_{s,\ell}^{\text{cm}} = \sum_{n \in \mathcal{N}} \text{PTDF}_{\ell,n} \cdot r_{s,n}, \quad \forall s \in \mathcal{S}, \ell \in \mathcal{L} \quad (4j) \quad 85$$

2) *Cross-border capacity market using NTC:* The cross-
border capacity market using NTC serves as a best-case proxy
for what is observed in European capacity markets [5]. In
line with current practice in explicit cross-border capacity

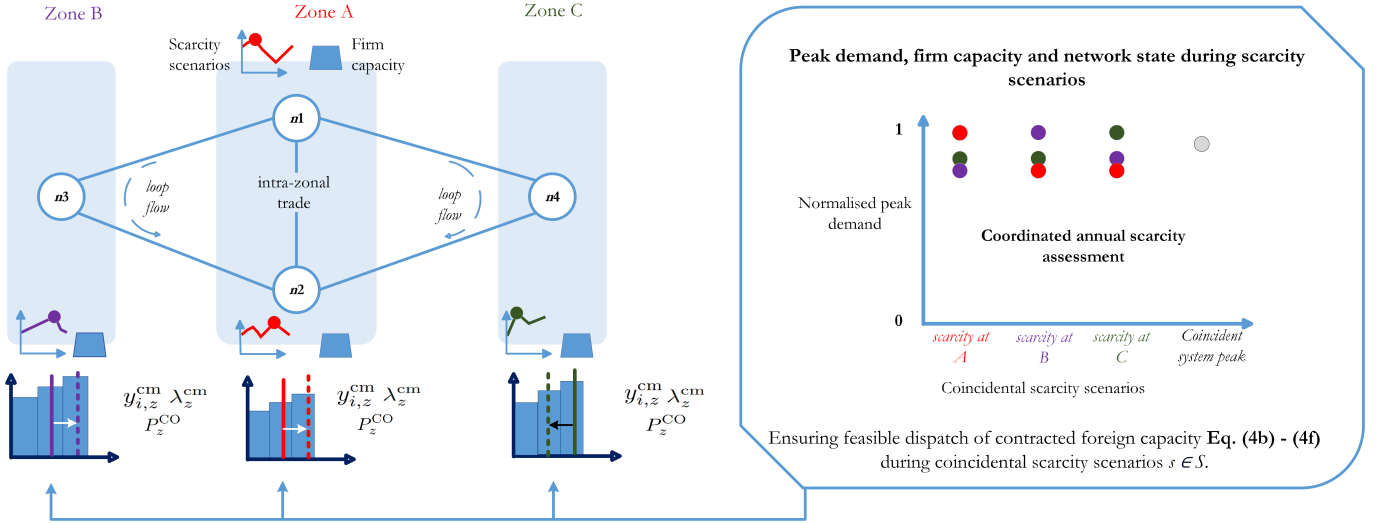


Fig. 2: Left: The annual, coupled capacity auction (zones A, B, and C) clears capacity offers $y_{i,z}^{cm}$, produces capacity market prices λ_z^{cm} , and net cross-border capacity obligations p_z^{CO} subject to the flow-based network constraints. Firm capacity may contribute to adequacy in different zones depending on the scarcity conditions and transmission constraints. Right: A joint assessment of the scarcity events across zones determines the maximum coincidental system capacity requirement, lowering zonal capacity demands compared to isolated national assessments. "Scarcity at A" represents a scarcity situation in zone A and the state of other zones within the coupled region during A's critical scarcity event.

participation, we bound bilateral capacity trade within an ex-ante NTC domain $\mathcal{P}^{ATC}$. This domain specifies the maximum entry capacity at each border before the capacity market auction. Since our model excludes pre-allocated capacity, the NTC is equivalent to the available transfer capacity (ATC).

For comparability with the flow-based approach, we compute the largest NTC domain that fits entirely within the flow-based feasibility polytope. We obtain this domain by solving a separate optimisation problem following Aravena et al. [17]. The resulting set, $\mathcal{E}^{ATC}$, gives the maximum net transfer capacity on each border that still yields feasible network flows during the scarcity scenarios considered in the determination of the flow-based domain (Section III-D1).

Note that in reality, the NTC domain influences agents' investment decisions, and these decisions in turn reshape the domain. To isolate the effect of the chosen cross-border capacity calculation method (NTC versus flow-based), we fix the investment levels used to construct the NTC domain to those obtained under the flow-based approach. Consequently, we isolate the inefficiency of using NTC rather than flow-based for capacity market clearing. Put differently, we do not capture how investing in the wrong location lowers permissible cross-border exchanges. It is thus a theoretical upper bound to the performance of NTC-coupled markets w.r.t flow-based market coupling.

The capacity market operator's decision problem under net transfer capacity (NTC) market coupling is formulated in Problem (5). The net positions in each scenario are calculated from the bilateral exchanges on each border (Eq. (5b)), which in turn determine the cross-border capacity obligations (Eq. (4h)). The bilateral exchanges are bounded by the pre-

defined maximum entry capacities $\mathcal{E}^{ATC}$ in Eq. (5c).

$$\max_{x^{CO} \in \mathcal{X}^{CO}} \sum_{z \in \mathcal{Z}^{cm}} \lambda_z^{cm} p_z^{CO} \quad (5a)$$

s.t. Eqs. (4b)–(4h)

$$\mathcal{P}^{ATC} := \left\{ p \in \mathbb{R}^{|\mathcal{Z}|} \mid \exists e^{CO} \in \mathcal{E}^{ATC} : \right.$$

$$\left. - \sum_{n \in \mathcal{N}_z} r_{s,n} = \sum_{(z',z) \in \mathcal{B}} e_{z',z}^{CO} - \sum_{(z,z') \in \mathcal{B}} e_{z,z'}^{CO}, \forall z \in \mathcal{Z} \right\} \quad (5b)$$

$$\mathcal{E}^{ATC} = [-ATC^-, ATC^+] \quad (5c)$$

E. Market clearing

The zonal electricity price $\lambda_{z,t}^e$ and the zonal capacity price λ_z^{cm} are dual variables linked to the market-clearing conditions (6) and (7), which couple the agents' decision problems. The energy market clears in every zone and period (Eq. (6)), where $\bar{\lambda}_{z,t}^e$ is the administrative price cap. The complementarity condition describes that when a zonal balance is attained, the market clears between 0 and $\bar{\lambda}_{z,t}^e$. The price is zero with excess supply, or at the cap with excess demand. When multiple zones clear at the cap simultaneously, multiple, equivalent distributions of demand curtailment may exist ($\text{ens}_{z,t}$ in Eq. (2c)). We introduce an ex-post curtailment-allocation rule to circumvent this multiplicity of equilibria, detailed in Appendix II. In every hour in which at least two zones clear at the cap, each zone's demand curtailment is recomputed to the shortfall it faces when it can rely only on generation capacity within the zone and on the cross-border capacity it has procured. Zones with

TABLE I: Investment costs $C_{i,z}^f$ and operational cost parameters $A_{i,z}$ and $B_{i,z}$. Peak generator costs vary by zone.

Technology $i \in \mathcal{I}$	$C_{i,z}^f$ [€/MWh·year]	$A_{i,z}$ [€/MWh ²]	$B_{i,z}$ [€/MWh]
Base	190 000	0.008	20
Mid	110 000	0.032	55
Peak – zone A	60 000	0.04	90
Peak – zone B	70 000	0.06	90
Peak – zone C	80 000	0.08	90

surplus capacity export to other zones in proportion to the deficit, so that the system ENS is conserved (Appendix II).

The capacity market clears annually in every zone participating in the mechanism (Eq. (7)). The capacity demand D_z^{cm} is assumed inelastic and capacity payments λ_z^{cm} are uncapped.

$$0 \leq \lambda_{z,t}^e \leq \bar{\lambda}_{z,t}^e \perp \sum_{i \in \mathcal{I}} g_{i,z,t} + p_{z,t}^E - d_{z,t}^e, \quad \forall z \in \mathcal{Z}, t \in \mathcal{T} \quad (6)$$

$$0 \leq \lambda_z^{\text{cm}} \perp \sum_{i \in \mathcal{I}^{\text{cm}}} y_{i,z}^{\text{cm}} + p_z^{\text{CO}} - D_z^{\text{cm}} \geq 0, \quad \forall z \in \mathcal{Z}^{\text{cm}}. \quad (7)$$

IV. CASE STUDY

We use a stylised three-zone (A, B, C), four-node network from [22] to illustrate the benefits of the proposed capacity-market coupling. Zone A has two nodes ($n1$ and $n2$), while zones B and C have one node, $n3$ and $n4$. Each cross-border line has a capacity of $F_{\ell}^{\text{max}} = 3$ GW, and the intra-zonal line has a capacity of 500 MW. All lines have identical susceptance.

We focus on the long-term equilibrium, assuming perfect coordination between long-term instruments used by member states and the coupled annual capacity market. This allows us to avoid bias in our results w.r.t. the existing capacity mix across the different zones.

We distinguish three types of generators: baseload, mid-merit, and peaking units (Table I). Renewable generation (solar and wind) is modelled as negative demand, with exogenous investment levels. Installed renewable capacities are 15.4 GW (solar) and 13.7 GW (wind) in Zone A, 8.8 GW and 5.3 GW in Zone B, and 18.7 GW and 8.3 GW in Zone C. In Zone A, which has two nodes, renewable capacity is allocated proportionally to nodal demand.

Zonal electricity demand $D_{z,t}^e$, renewable availability factors $AF_{i,z,t}$ and installed renewable capacities $Y_{i,z}$ are sourced from the ENTSO-E Transparency Platform [26] for Germany (Zone A), Belgium (Zone B), and the Netherlands (Zone C) for a winter day (11 November 2024). We normalise the historical demand and renewable capacity data using scaling factors of 5, 1, and 1.3 for zones A, B, and C, to create zones of comparable market size. We then simulate scarcity conditions by capping the renewable availability factor at 10% during the peak periods and multiplying the scaled peak demand by 1.5 to obtain new peak demands of 19 GW, 16 GW, and 18 GW in Zones A, B, and C. The peak events are non-coincident. Our calibration of the peak period ensures that unserved energy occurs when an administrative price cap is introduced, justifying the need for a capacity market.

Electricity demand is represented as a two-block structure comprising an inelastic and a price-elastic component. The inelastic block, $(1 - \epsilon_z)D_{z,t}^e$, is valued at the maximum willingness to pay $WTP_z = 20,000$ €/MWh. The elastic block, $\epsilon_z D_{z,t}^e$ with $\epsilon_z = 0.20$, is derived from a quadratic utility function and yields a linear inverse demand curve that falls from WTP_z at zero elastic consumption to zero €/MWh at $\epsilon_z D_{z,t}^e$, with slope $-WTP_z/(\epsilon_z D_{z,t}^e)$. In designs with an administrative price cap at 4,000 €/MWh, the elastic consumers whose valuation exceeds the cap (16% of $D_{z,t}^e$) cannot express their WTP in the market and bid at the price cap. The consumer's problem (Problem (2)) is accordingly parametrised with $(WTP_z, \epsilon_z) = (20,000, 0.20)$ in markets without a price cap and with the truncation (4,000, 0.04) in price-capped designs (Table II). Both describe the same inverse demand curve, truncated at the cap, with identical slope. Curtailment of demand valued above the price cap is random (Eq. (2c)), so unserved energy is ex-post valued at its average valuation of 18,667 €/MWh. The demand for capacity is inelastic and depends on the capacity market design.

We study seven distinct market design scenarios (Table II). The first two scenarios are Energy-only markets with and without price caps. The remaining scenarios enable the study of how capacity markets with varying degrees of cross-border participation address underinvestment resulting from price caps in the electricity market. Below, we discuss how the capacity demand is computed and summarise the resulting values in Table III:

1) *EOM-ref*: The energy-only market without a price cap. All generators recover their costs from the energy market, assuming no market imperfections beyond the lack of a locational component in the electricity price in zonal markets.

2) *EOM-cap*: An energy-only market with a price cap, capturing the "missing money" problem, which leads to underinvestment in firm capacity and energy not served (ENS).

3) *CM-NoCBP*: A capacity market without cross-border participation. The zonal capacity demand is set to the peak residual demand valued at the price cap in each zone, i.e. the maximum residual demand served at 4,000 €/MWh. Zones do not explicitly or implicitly account for the availability of imports in their national capacity markets, nor carry out a joint scarcity assessment.

4) *CM-ref*: A reference capacity market scenario whose capacity demands result from a joint scarcity assessment (Figure 2 (Right)), but without cross-border participation. The joint scarcity assessment considers three zonal scarcity scenarios, representing each zone's individual peak demand. The set of scarcity scenarios considers the residual demand valued at the price cap of 4,000 €/MWh across all zones at each time step, any zone experiences ENS under *EOM-cap*. The joint scarcity assessment sets the system-wide capacity demand to the maximum coincident peak residual demand. This capacity demand is distributed over the zones proportionally to the average share of the zone's residual demand to the overall system residual demand during scarcity situations. This scenario allows isolating the effect of coordinating capacity requirements from that of cross-border participation. The resulting capacity

TABLE II: Summary of the market designs analysed, indicating whether a capacity market (CM) is implemented, the type of cross-border participation (CBP), and the willingness-to-pay (WTP) or price-cap (€/MWh) in the energy market.

Scenario	CM	CBP	WTP / Price-cap
EOM-ref	—	—	20,000
EOM-cap	—	—	4,000
CM-NoCBP	✓	—	4,000
CM-FBMC	✓	Explicit - Flow-based	4,000
CM-NTC	✓	Explicit - MEC	4,000
CM-Implicit	✓	Implicit	4,000

demand forms the basis for all scenarios considering cross-border participation.

5) *CM-Implicit*: A capacity market with implicit cross-border participation. Each zone's capacity demand is that determined in the joint scarcity assessment minus the average imports it receives in the scarcity scenarios under scenario *CM-ref*.

6) *CM-NTC*: A capacity market with explicit cross-border participation limited by a pre-defined maximum entry capacity (MEC) on each border. We use capacity demands from scenario *CM-ref*. For comparability, we determine best-case MECs using the investment decisions from the *CM-FBMC* scenario, following the procedure outlined in Section III-D2. Thus, results from this scenario should be viewed as a best-case NTC, since we exclude some of the investment distortions that would arise from an endogenously determined MEC. As a consequence, results differ from *CM-FBMC* only in the treatment of cross-border capacity trade.

7) *CM-FBMC*: The proposed flow-based coupling of capacity markets. We use capacity demands obtained in scenarios *CM-ref*. The participation of foreign capacity in each zone's capacity market is constrained by the permissible flows across the system under predefined scarcity scenarios (Figure 2). Scarcity events depict the time steps in which ENS occurs under *EOM-cap*. We represent projected scarcity events as (i) "simultaneous scarcity", where all zones demand 95 per cent of their peak residual demand simultaneously, and (ii) "zonal scarcity", where the zone experiencing critical scarcity demands 100 per cent of its peak residual demand, while neighbouring zones each simultaneously demand 90 per cent of their peak residual demand. The scarcity scenarios are identical to those used in the joint scarcity assessment.

TABLE III: Residual peak demand valued at the price cap (MW) in each zone during observed scarcity scenarios and zonal capacity demands (MW) per capacity market design.

(a) Residual peak demand valued at the price cap (MW)

Scenario	Zone A	Zone B	Zone C	Total
Scarcity at A	16,870	13,293	14,722	44,885
Scarcity at B	16,006	14,829	14,770	45,605
Scarcity at C	15,046	13,293	16,450	44,789

(b) Capacity demand (MW)

Scenario	Zone A	Zone B	Zone C	Total
CM-NoCBP	16,870	14,829	16,450	48,149
CM-ref, CM-FBMC, CM-NTC	16,156	13,957	15,493	45,606
CM-Implicit	14,778	13,084	14,238	42,100

All code and input data are available online [27].

Below, we present the resulting capacity mixes (Section IV-A). Next, we discuss the implications for cross-border capacity trade and congestion rent (Section IV-B). Lastly, we reflect on the system costs and the costs to consumers (Section IV-C).

A. Investment decisions and capacity mix

In the reference Energy-Only Market (*EOM-ref*), investment is driven by scarcity pricing. In this idealised *EOM-ref*, prices may rise to the WTP of 20,000 €/MWh, at which the entire price-responsive share of demand (20% of the reference level) would voluntarily curtail. However, in equilibrium scarcity prices reach at most 16,608 €/MWh. Realised peak demand falls to 17.5, 13.7 and 15 GW in zones A, B and C, from reference peaks of 19, 16 and 18 GW. Implementing a price cap (*EOM-cap*) truncates prices at 4,000 €/MWh, with consequences on both consumer demand and generation capacity investments. The forgone scarcity rents cause a decline in investment, primarily in peaking units (Fig. 3). Total capacity falls by 13.5 GW relative to the reference energy-only market. On the consumer side, the volume of elastic demand that can express its willingness to pay in the market decreases from 20% to 4% of the reference level. The combination of less firm capacity and less price-responsive demand visible in the market results in Energy Not Served (ENS) of 38.9 GWh, 28.9 GWh, and 37.3 GWh in zones A, B, and C.

A capacity market without cross-border participation (*CM-NoCBP*) successfully eliminates ENS; however, it does so through autarky. In line with observations in the literature [28], the system sees a surge in low-capex, high-opex peaking units (pink bars), increasing total capacity by 8 GW w.r.t. *EOM-ref*, indicating overprocurement. We observe a capacity displacement of 1.4 GW away from Zone A w.r.t. *EOM-ref*, as the isolation prevents Zone A from capitalising on its export potential. To mitigate this over-procurement and capacity displacement, we carry out a joint scarcity assessment in *CM-ref*, and further introduce cross-border participation under three designs: *CM-implicit*, *CM-NTC*, and *CM-FBMC*.

Coordinating capacity requirements through the joint scarcity assessment eliminates unserved energy with less installed capacity than in *CM-NoCBP*. Zone B, for example, procures 872 MW less than its own peak residual demand and covers the gap with 874 MW of imports at its peak. The flows presumed via the joint assessment, therefore, materialise in the energy market, though *CM-ref* neither remunerates nor secures them. Coordination, however, does not relocate investment. Without cross-border participation, the capacity displaced from Zone A w.r.t. *EOM-ref* grows to 2.1 GW, above the 1.4 GW under *CM-NoCBP*, while Zones B and C add 1.9 and 5.7 GW (Fig. 3). In total, *CM-ref* installs 2.5 GW below *CM-NoCBP*, with the reduction spread across all zones (0.7, 0.9 and 0.9 GW in Zones A, B and C).

Implicit cross-border participation *CM-implicit* deducts the 3.5 GW of imports that arise in each zone's scarcity period under *CM-ref* (1.4, 0.9 and 1.3 GW in Zones A, B and

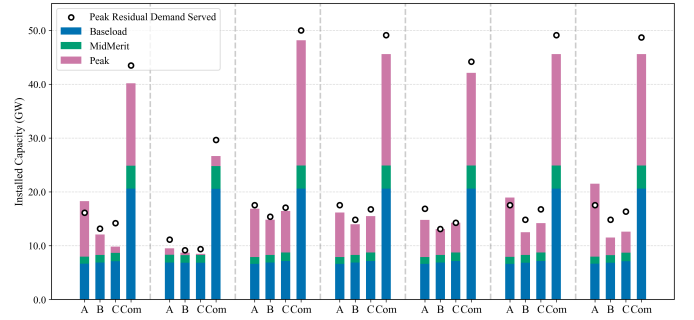
C) without remunerating them. However, the unremunerated import capacity does not materialise because, without explicit remuneration, the foreign capacity expected to support these imports is not built. Total installed capacity falls to 42.1 GW, 1.9 GW above *EOM-ref* and 6.0 GW below *CM-NoCBP*. Consequently, unserved energy re-emerges across all zones (12.4 GWh system-wide). Zone A bears the largest capacity displacement w.r.t *EOM-ref* with 3.5 GW less capacity investment, as capacity it otherwise would build for export is left unremunerated.

Explicit cross-border participation under both *CM-NTC* and *CM-FBMC* starts from the same joint scarcity assessment as *CM-ref*, thus total installed capacity stays at 45.6 GW, 2.5 GW below *CM-NoCBP*, and unserved energy remains negligible, at less than 10^{-4} % of served annual demand in the system. As foreign capacity obligations are remunerated in the explicit designs, Zone A with lower cost of new entry exploits this opportunity and builds 2.8 GW (*CM-NTC*) and 5.3 GW (*CM-FBMC*) above its own capacity demand to serve capacity obligations remunerated by Zones B and C. This is also observed in the obligations it exports (Fig. 4). Comparing both explicit designs, *CM-FBMC* relocates more capacity, shifting 2.6 GW more of investment into the lower-cost Zone A relative to *CM-NTC*. Recall that we base the maximum entry capacities used in *CM-NTC* on the available capacity in the *CM-FBMC*. As such, the result excludes the investment distortions that would result from using the NTC approach and can be interpreted as an upper bound for *CM-NTC*.

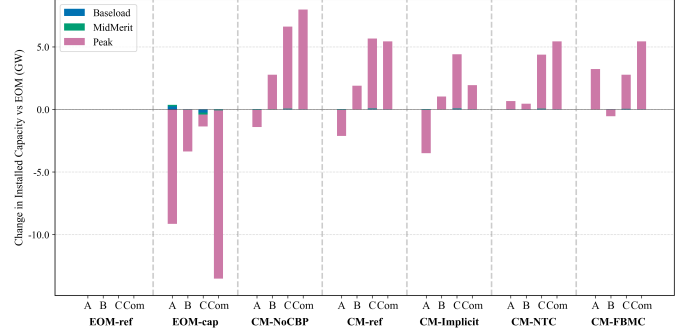
Notably, all capacity market designs result in a total capacity that exceeds the *EOM-ref*. While *EOM-cap* under-invests by 13.5 GW, capacity mechanisms add 1.9–8 GW above *EOM-ref* level. This arises from feedback between electricity prices and the elastic electricity demand. In *EOM-ref*, high scarcity prices reduce electricity consumption through self-rationing by consumers whose WTP is below the market-clearing price, and consequently, the peak residual demand served is reduced to 44 GW. The price cap truncates electricity prices (Section IV-B), removing the efficient self-rationing of consumers with demand valued above the price cap (4,000 €/MWh). To avoid involuntary curtailment of demand, more firm capacity is required. As a result, the peak residual demand served rises from 44 GW in *EOM-ref* to 49–50 GW in the scenarios considering a capacity market.

B. Cross-border trades and congestion revenues

Cross-border trade of capacity obligations is 2.6 GW higher under *CM-FBMC* than under *CM-NTC* (Figure 4). In *CM-NTC*, bilateral transfer limits determined based on investment levels that materialise in *CM-FBMC* are $A \leftrightarrow B$: [740.3, −620] MW, $A \leftrightarrow C$: [574.3, −454] MW, and $B \leftrightarrow C$: [694, −694] MW. Zone A is the sole net exporter in both designs. Under *CM-NTC* it sells 2.8 GW of obligations, of which Zones B and C procure 1.5 and 1.3 GW. Under *CM-FBMC* it sells 5.3 GW, and both neighbours roughly double their foreign procurement, to 2.4 GW (Zone B) and 2.9 GW (Zone C), reducing within-zone investment by the



(a) Capacity mix by zone and market design. "Com" indicates the aggregated result across all zones.



(b) Change in installed capacity w.r.t. *EOM-ref*.

Fig. 3: Capacity markets restore investment levels in peaking units but risk over-procurement without coordinated cross-border participation; *EOM-cap* and capacity markets without explicit cross-border capacity remuneration (*CM-NoCBP*, *CM-ref*, *CM-implicit*) displace capacity away from cheaper Zone A relative to *EOM-ref*, while the explicit designs relocate capacity to Zone A.

same amounts. The higher cross-border capacity trade is shown to be physically deliverable during scarcity with Zone C importing 3.7 GWh/h, against 2.5 GWh/h under *CM-NTC*.

We show the network operator's congestion rents in the energy and capacity market in Table IV. In the reference Energy-only market (*EOM-ref*), high price differentials during scarcity drive congestion rents of 73.1 M€. During congestion, prices peak at 14,100 €/MWh in Zone A and 16,608 €/MWh in Zone C. These spreads generate high revenue. Implementing a price cap (*EOM-cap*) drastically reduces these rents to 2.6 M€. The cap artificially reduces flows (due to under-investment across all zones) and the price spread between zones, as all zones hit the 4,000 €/MWh cap simultaneously.

The capacity market designs without explicit cross-border participation further erase these rents, albeit for different reasons. In *CM-NoCBP*, overprocurement suppresses scarcity pricing. With prices peaking at 560–650 €/MWh in all zones, spreads are minimal and congestion rents collapse to 0.5 M€. In *CM-ref* and *CM-implicit*, scarcity prices reach the cap; however, similar to *EOM-cap*, prices are near the cap in all zones at the same periods, lowering price spreads and congestion rents fall to 0.1 and 0.5 M€.

Explicitly remunerating cross-border participation restores the value of transmission through "capacity congestion rent".

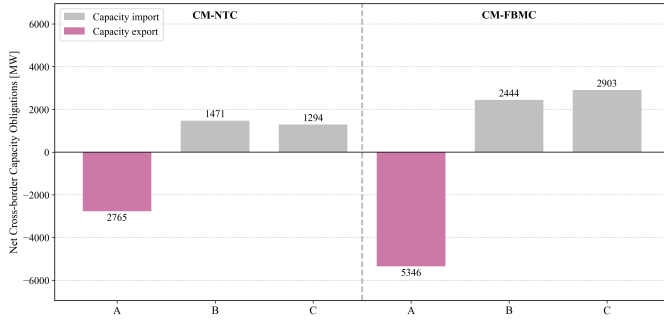


Fig. 4: Net cross-border capacity trade by zone.

TABLE IV: The network operator's congestion rents (M€/yr) across market designs.

Market	EOM-ref	EOM-cap	CM-NoCBP	CM-ref	CM-Implicit	CM-NTC	CM-FBMC
Energy	73.07	2.64	0.50	0.07	0.53	0.47	32.22
Capacity	0.00	0.00	0.00	0.00	0.00	39.08	46.46
Total revenues	73.07	2.64	0.50	0.07	0.53	39.55	78.68

CM-NTC generates 39.6 M€, most of it in the capacity market, restoring more than half of the *EOM-ref* level. *CM-FBMC* yields 78.7 M€, narrowly surpassing *EOM-ref*. Unlike *CM-NTC*, it earns a third of this in the energy market, as greater scarcity exports from Zone A congest interconnectors, causing price spreads to persist during scarcity.

C. System costs, social welfare and costs to consumers

1) *System Costs*: By design, the energy-only market *EOM-ref* presents the least-cost market design, with a total system cost of 15,398 M€/yr (Table V).¹ The introduction of a price cap (*EOM-cap*) suppresses investment costs by 870 M€/yr due to the "missing money" problem, resulting in a surge in the cost of unserved energy (1,963 M€/yr). Consequently, the *EOM-cap* is the least efficient design, incurring a cost increase of 7% relative to the reference case.

Capacity markets reduce ENS costs but increase investment costs. When compared to *EOM-ref*, *CM-NoCBP* increases investment costs by 640 M€/yr, increasing total system cost by 4.2%. Coordinating the capacity requirement in *CM-ref* reduces the required capacity investments by 2.5 GW and lowers annualised investment costs by 181 M€/yr, limiting the total system-cost increase relative to *EOM-ref* to 3.1% (470 M€/yr). *CM-Implicit* reduces investment costs by a further 245 M€/yr, but this is largely offset by 232 M€/yr in unserved-energy costs, leaving total system costs only marginally below *CM-ref* and 2.9% above *EOM-ref*. Unserved energy under *CM-Implicit* is 12.4 GWh per year, equivalent to roughly 0.005% of annual demand (Table V). *CM-implicit* is sensitive to assumptions on the availability of imports during scarcity. Higher import assumptions would increase the volume of unserved energy, and because ENS costs rise faster

TABLE V: Served demand (GWh/yr), unserved energy (ENS – GWh/yr), system costs (M€/yr) and cost increase w.r.t. the reference EOM case (*EOM-ref*).

Market Design	Demand Served	ENS Volume	Gen. Cost	Inv. Cost	ENS Cost	Total Cost	Increase (%)
EOM-ref	271,630	0	10,036	5,361	0	15,397	–
EOM-cap	271,532	105.14	10,015	4,491	1,963	16,468	6.96
CM-NoCBP	271,664	0	10,048	6,001	0	16,049	4.23
CM-ref	271,659	0.01	10,046	5,820	0.1	15,866	3.05
CM-Implicit	271,641	12.42	10,038	5,575	232	15,846	2.91
CM-NTC	271,659	<0.01	10,046	5,780	0	15,826	2.78
CM-FBMC	271,659	0.25	10,048	5,738	4.7	15,791	2.56

than investment savings, the total system cost would increase. Without import deductions, the design collapses to *CM-ref*.

Explicit cross-border participation lowers total costs by relocating peaking capacity from zones with a high investment cost to lower-cost zones while keeping ENS below 0.25 GWh/yr. *CM-NTC* reduces investment costs by 40 M€/yr relative to *CM-ref* (0.7% of 5 820 M€/yr), and by 82 M€/yr (1.4%) under *CM-FBMC*. It removes the unserved energy of *CM-Implicit*, with ENS falling to 253 MWh/yr (below 10^{-4} % of served annual demand) in *CM-FBMC* and to 0.5 MWh/yr in *CM-NTC*. *CM-FBMC* is the least-cost capacity-market design, 258 M€/yr below *CM-NoCBP*, 55 M€/yr below *CM-Implicit*, and 35 M€/yr (or 0.22%) below a best-case NTC (*CM-NTC*), at a total annualised cost of 15,791 M€/yr.

2) *Social welfare*: We decompose the change in welfare w.r.t *EOM-ref* in changes in consumer surplus, producer surplus and congestion rents (Figure 5a).² *EOM-ref* leaves market-clearing price uncapped, enabling efficient self-rationing by consumers with a WTP below the clearing price, and providing a reference value for surplus generated in the market. The largest welfare loss is observed in *EOM-cap* (–864 M€/yr), the lowest in *CM-FBMC* (–171 M€/yr), consistent with what is obtained in (Section IV-C1). Under the price cap, consumers bear the loss (–982 M€/yr, driven by the unserved energy), while producers gain 189 M€/yr as inframarginal rents rise, particularly for renewables and baseload generators. Capacity markets reverse this trend. Producer surplus decreases as capacity revenues compensate investment cost but lower inframarginal rents. However, it significantly improves consumer surplus. Only *CM-NoCBP* (–125 M€/yr, due to over-procurement) and *CM-Implicit* (–17 M€/yr, energy-not-served) leave consumers worse off.

We distinguish two sources of welfare gains. First, coordinating the capacity demand through a joint scarcity assessment (*CM-NoCBP* to *CM-ref*) improves welfare by 172 M€/yr, with most accruing to consumers. Secondly, explicit cross-border participation adds 41 M€/yr under *CM-NTC* and 75 M€/yr under *CM-FBMC*, which materialises almost entirely as congestion revenue. *CM-FBMC* is the only design that fully restores the value of interconnection compared to *EOM-ref*.

¹Total system costs equal the sum of generation costs, investment costs and cost of lost load. The cost of lost load is the volume of ENS valued at 18,667 €/MWh; the average value of the demand that is randomly curtailed at the price cap of 4,000 €/MWh.

²To compute the change in consumer surplus, we re-evaluate energy market outcomes using the uncapped demand curve. The obtained consumer surplus is corrected with the cost of energy not served and capacity payments.

3) *Cost to consumers*: Figure 5b summarises the cost of electricity for consumers normalised by the demand served (€/MWh), decomposed into three components: the energy cost, cost of energy not served and capacity cost.

In *EOM-ref*, average costs reach 99.7 €/MWh in Zone B and 100.9 €/MWh in Zone C. Zone A sees a lower cost of 98.1 €/MWh. The system-wide average is 99.5 €/MWh. Neither a capacity market nor unserved energy exists in this reference case. Introducing a price cap increases consumer costs across all zones because unserved energy emerges. System-wide costs rise from 99.5 €/MWh in *EOM-ref* to 103.9 €/MWh in *EOM-cap*. The cap limits scarcity prices but triggers underinvestment and ENS costs. These effects are shown in the hatched areas of Figure 5b.

Capacity markets provide reliability at a premium. *CM-NoCBP* eliminates ENS; however, capacity payments represent 12.3% of the total cost to consumers. Capacity market prices in Zones A, B, and C clear at or below the costs of a new build: 59,777 €/MW/yr, 69,623 €/MW/yr, and 80,000 €/MW/yr. The system-level average cost to consumers falls to 100.8 €/MWh, 3.1 €/MWh below the *EOM-cap* and 1.3 €/MWh above the reference Energy-only market. Coordinating capacity demands through a joint scarcity assessment in *CM-ref* lowers capacity prices by roughly 10,000 €/MW/yr in every zone (to 48,904, 58,965 and 69,886 €/MW/yr) and reduces capacity payments to 9.9% of the cost. Consumers see a net saving of 0.6 €/MWh.

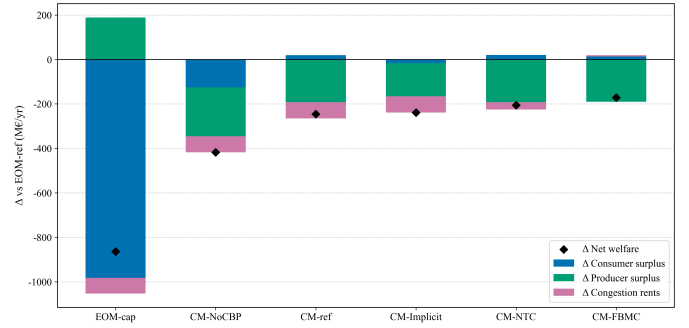
CM-Implicit goes further. Zones count on imports from foreign capacity without remunerating them. Capacity payments fall to 7.5% of the cost (38,254–59,246 €/MW/yr), but the imports do not materialise during scarcity events. Unserved energy adds 0.9 €/MWh and higher energy costs a further 1.6 €/MWh, leaving consumers worse off than under *CM-ref*.

Under *CM-FBMC*, consumers face the same average system cost as in *CM-NTC*. Explicit cross-border participation (*CM-NTC* and *CM-FBMC*) leaves the system-wide cost at the *CM-ref* level, but redistributes it across zones. Zone A's consumers face a higher capacity cost due to demand from neighbouring zones while Zones B and C pay 0.1 €/MWh less. Between both designs, *CM-FBMC* lowers capacity prices in Zones B and C by 390 and 4,076 €/MW/yr and raises Zone A's by 3,876 €/MW/yr.

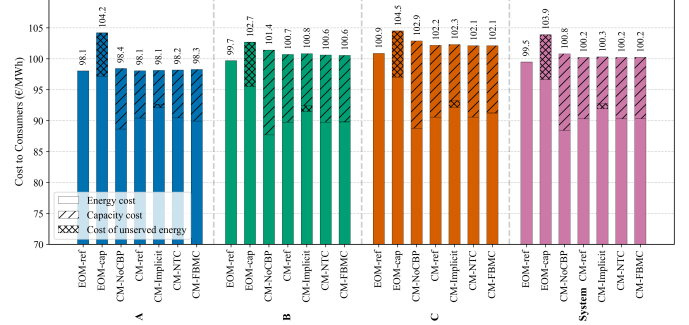
V. DISCUSSION

We propose a flow-based coupling of European capacity markets (*CM-FBMC*), representing a significant departure from current arrangements. Below, we reflect on its implications and on the challenges it may raise.

One of the most significant hurdles to regional capacity markets is presented by the unique needs of Member States. Political reluctance to relinquish control over national resource adequacy and capacity mix hinders progress toward regional integration of capacity markets. Our proposed approach avoids this conflict. It does so by separating long-term investment needs and short-term adequacy. Member States still determine long-term (multi-year) instruments to de-risk investment in capital-intensive assets. The coupled annual capacity markets



(a) Welfare decomposition showing the change w.r.t *EOM-ref* in consumer and producer surplus, congestion rents and the net welfare across market designs



(b) Average consumer cost by zone, decomposed into energy, capacity, and ENS (€/MWh). "System" represents the average cost of all zones and enables comparison of total system costs.

Fig. 5: We show the impact of cross-border capacity participation and coupling design on (a) Welfare decomposition [M€/yr] and (b) total average cost [€/MWh] in Zones A–C.

then provide an opportunity to re-trade capacity obligations held by governments and facilitate investment in fast-to-deploy assets. Each MS may define its own long-term contract structure and parameters of the capacity demand curve (e.g. the demand curve may be defined "centrally" or "decentrally") – only the participation in annual capacity auctions, the traded contract (e.g., reliability options), and coincidental scarcity assessments need to be harmonised.

Even as we accommodate national preferences, the proposed mechanism continues to guide efficient regional investment. It reduces overprocurement by adopting a regional adequacy assessment that exploits the degree of non-simultaneity in scarcity events. A common view of scarcity scenarios across the system incentivises investment levels only up to the peak simultaneous scarcity across zones, rather than the sum of individual zonal peaks. As we demonstrate in our case study IV-A, efficiency gains in *CM-FBMC* are driven both by the mechanism's ability to exploit the degree of non-simultaneity in scarcity events within the network and by relocating investment to cheaper locations.

Two factors may erode these efficiency gains. First, overly conservative regional scarcity expectations at Y-1 may drive inaccurate capacity procurement. Second, several frictions may impede the efficient transmission of price signals between annual, coupled capacity markets and the long-term, national instruments. For example, member states may strategically overprocure firm capacity to guarantee national autonomy or

underprocure to free-ride on neighbouring resources. The coupled annual auction can disincentivise such behaviour. Member states are mandated to re-market all procured capacity under the national mechanisms into the coupled annual auction. Consequently, the cost of overprocurement is made explicit as the spread between the cost of the long-term contracts and the annual clearing price. Consider a Member State "A" that overprocures multi-year, long-term capacity contracts. Neighbouring zones then benefit from purchasing reliability obligations from "A". If transmission constraints in the coupled capacity auction become binding, capacity prices in zone "A" drop below the cost of some of the long-term contracts held by Member State "A". This makes the cost of overprocurement explicit. Member states may, however, choose to absorb the cost, for example, as the price for autonomy, and if all member states systematically overprocure in the first layer, the outcome becomes similar to that of the *CM-NoCBP*.

Compared to current approaches to cross-border participation in capacity markets, our design resolves the "circular dependency" inherent in both *CM-NTC* and *CM-Implicit* approaches. Both rely on ex-ante parameterisations – either in setting Maximum Entry Capacities (*CM-NTC*) or to reduce local capacity targets (*CM-Implicit*). The implicit approach is particularly prone to a "chicken-and-egg" dilemma: it deducts expected imports from local demand without providing remuneration to incentivise investment in the capacity that enables these imports. As such, the assumed surplus may be crowded out in the long run. Our results show this fragility through the re-emergence of unserved energy in the *CM-Implicit* scenario, confirming that reliance on non-remunerated cross-border import may undermine resource adequacy. An increase in the valuation of reliability would sharply penalise the unserved energy risk inherent in the implicit approach, reducing its economic efficiency.

The *CM-FBMC* addresses this issue structurally. By using a detailed grid model to clear capacity offers, it ensures that cleared capacity is deliverable w.r.t. network constraints under a set of defined scarcity scenarios. Flow-based domains internalise loop flows, critical network elements, and contingencies, thereby capturing grid physics more accurately than NTC-based "maximum-entry-capacity" caps. FBMC thus requires less conservative assumptions, unleashing more cross-border trade of firm capacity obligations. Additionally, the flow-based market coupling of annual capacity markets provides explicit price signals for reliability. Zones with adequacy issues will clear their capacity markets at higher prices, providing incentives for cross-border trade and investment in non-capital-intensive assets.

As the proposed design is yet to be implemented in any known real-world setup, we elaborate on the limitations of our work and potential implementation challenges. Our equilibrium model assumes perfect competition and risk-neutral agents in both electricity and capacity markets. Extending the analysis to investigate the effect of risk aversion and market power on cross-border capacity market design constitutes a relevant avenue for future research. Additionally, we certify capacity deliverability under a stylised finite scenario set $\mathcal{S}$. Operationalising the market design would require a

robust set of scarcity scenarios. As TSOs already generate year-ahead scarcity scenarios and flow-based domains for the European Resource Adequacy Assessment (ERAA), extending these processes to produce harmonised, year-ahead anticipated scarcity scenarios (flow-based domains, availability of generation assets and demand) is the next logical step. Such a role may be assumed by Europe's Regional Coordination Centres (RCC). Note, however, that our network representation only considers state- N and ignores contingencies during scarcity events. Real-world implementation would account for grid uncertainties through the $N - k$ security criterion. The combination of a scarcity scenario set and $N - k$ contingencies may considerably increase the computational effort required for market clearing. Lastly, because flow-based market coupling has been criticised for opacity, its application in capacity markets must prioritise transparency.

VI. CONCLUSION

Capacity markets are becoming a structural element of the European electricity market, but lack harmonisation and cross-border participation necessary for regional efficiency. Our study addresses this issue by introducing a coordinating layer for national, long-term capacity remuneration mechanisms. We propose linking annual, zonal capacity auctions using the flow-based market coupling methodology, in which long-term contracts held by governments can be re-marketed and new assets with short lead times can be attracted. As such, the coupled annual capacity market is compatible with long-term instruments issued by Member States to mitigate investment risk.

By endogenously modelling network flows in anticipated scarcity scenarios, our approach eliminates the circular dependencies between import assumptions and available capacity in implicit cross-border participation, and between maximum entry capacities and available capacity in current explicit cross-border participation methods, and guarantees the delivery of contracted capacity during those scarcity events.

Our modelling results show that this design more accurately values the contribution of interconnection and relocates capacity to cheaper locations under deliverability constraints. Moreover, our design addresses challenges posed by fragmented national capacity mechanisms while respecting the heterogeneous adequacy requirements of Member States and their sovereign right to steer their capacity mix.

Future work should focus on detailing the design of the annual auctions and the capacity product to be traded. Key features include the degree of harmonization required across member states and the valuation of resources that have non-symmetric contributions to meeting demand during scarcity situations across zones, such as energy-constrained assets.

APPENDIX I

The equilibrium is computed using an exchange-form consensus ADMM implemented in Julia, in which each agent's convex quadratic subproblem is solved in parallel with Gurobi. Each iteration solves Problems (1)–(4) in parallel, given the prices, then updates $\lambda_{z,t}^e$ and λ_z^{cm} as duals of the zonal

imbalances. The algorithm terminates when, for every zone and for both markets (electricity and capacity), the primal and dual residuals fall below an absolute tolerance $\epsilon = 0.1$ (in MW for primal residuals; in €/MWh and €/MW-year for EOM and CM dual residuals). The penalty parameter ρ is adapted per zone and market via residual balancing following Boyd et al [29]. Converged runs typically require 460-2,500 iterations (13-38 seconds of wall time), with each iteration taking residuals to decay steadily (Table VI). All runs are cold-started, with prices, quantities, and penalty terms initialised at zero.

Because each agent solves a convex quadratic problem, a converged ADMM solution satisfies the conditions for competitive equilibrium. As degeneracy exists in some variables (e.g., within-zone investment allocation, network operator capacity allocation, and unserved energy when price cap hits multiple price zones simultaneously), solutions may be non-unique. However, these multiple equilibria differ only in nodal quantities (except for unserved energy, which we treat ex-post, see Appendix II); aggregate installed capacity and zonal outcomes remain consistent across repeated runs.

TABLE VI: Final ADMM residuals and iteration counts across market designs. Reported residuals are the maximum value over all zones. The convergence tolerance is $\epsilon = 0.1$

Design	Iterations	$\max r_E^p$	$\max r_E^d$	$\max r_{CM}^p$	$\max r_{CM}^d$
EOM-ref	1,504	0.012	0.100	0	0
EOM-cap	462	0.017	0.100	0	0
CM-NoCBP	498	0.047	0.099	0.031	0.077
CM-ref	845	0.084	0.038	0.092	0.024
CM-Implicit	2,223	0.027	0.100	0.017	0.029
CM-NTC	830	0.038	0.097	0.057	0.099
CM-FBMC	733	0.031	0.047	0.025	0.088

APPENDIX II

During joint-scarcity hours, defined as hours in which at least two zones experience $\text{ens}_{z,t} > 0$, these zones face the common price cap, so market clearing may not uniquely determine the zonal allocation of ENS. We resolve this multiplicity ex-post using Eq. (8), where $\text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}$ is the shortfall corrected for the zone's net position in the capacity market p_z^{CO} . A positive net position indicates an importing zone.

$$\text{ens}_{z,t} = s_t \cdot \max \{0, \text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}\}, \quad (8a)$$

$$s_t = \frac{\sum_z \text{ens}_{z,t}}{\sum_z \max \{0, \text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}\}}, \quad (8b)$$

$$\tilde{p}_{z,t}^E = p_{z,t}^E + \text{ens}_{z,t} - \text{ens}_{z,t} \cdot s_t. \quad (8c)$$

Since $\sum_z p_{z,t}^E = \sum_z p_z^{\text{CO}} = 0$, $s_t \leq 1$, and total ENS is conserved by construction. Surplus zones receive no curtailment, while deficit zones share total ENS in proportion to their shortfalls. If no zone is in surplus, $s_t = 1$ and $\tilde{p}_{z,t}^E = p_z^{\text{CO}}$. The reallocation affects only the zonal distribution of ENS and net positions; all other outcomes remain unchanged.

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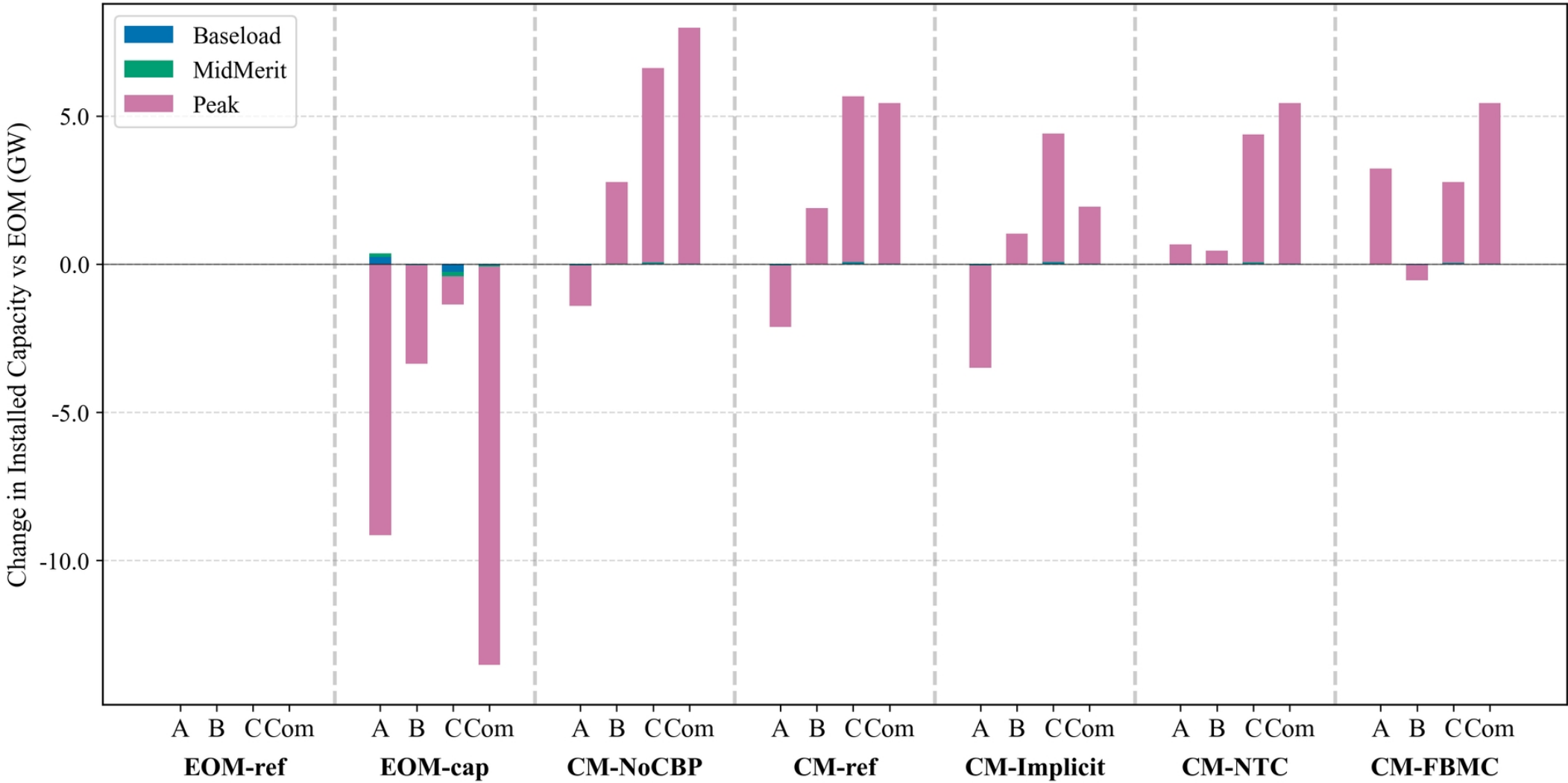
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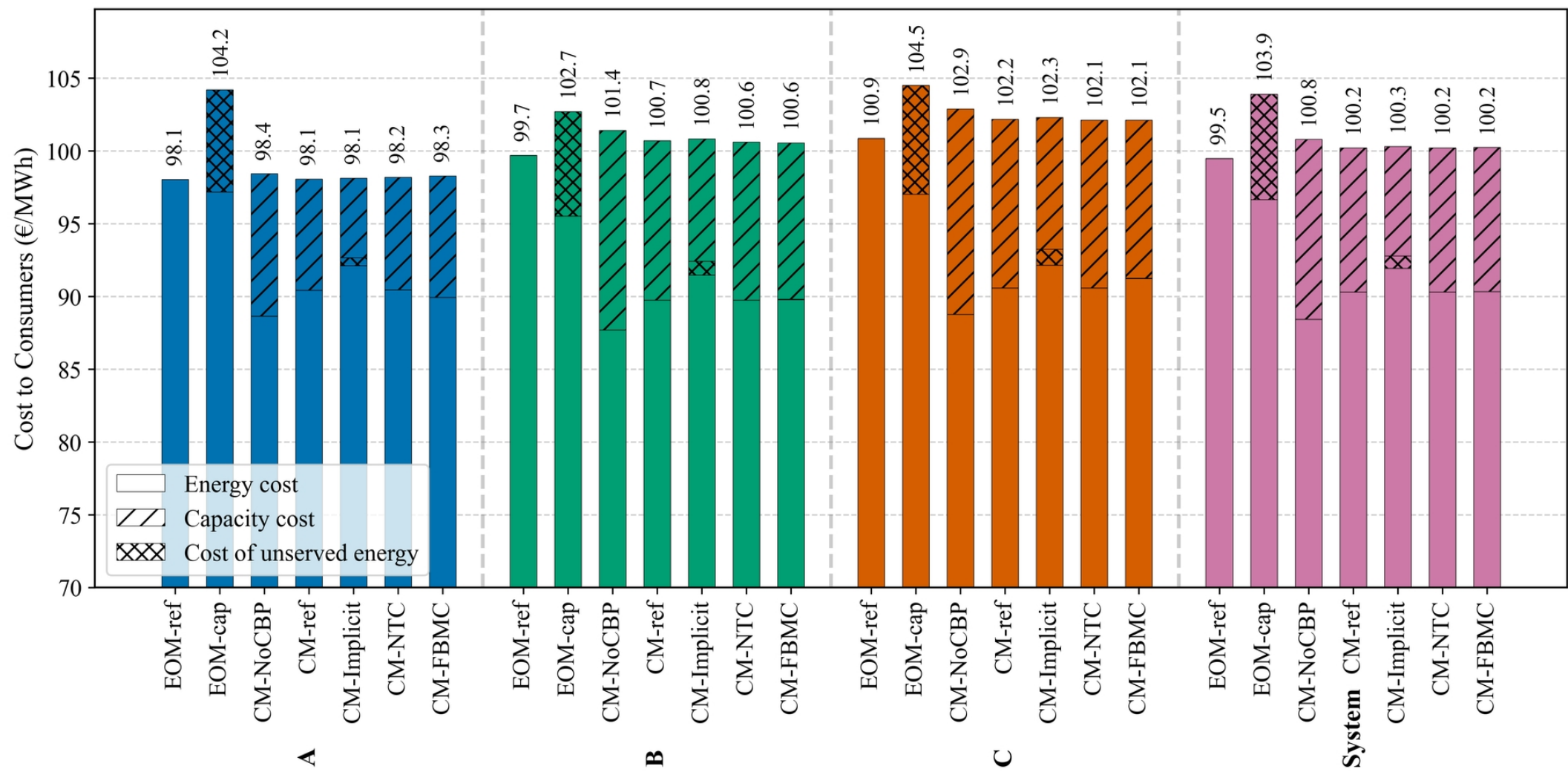
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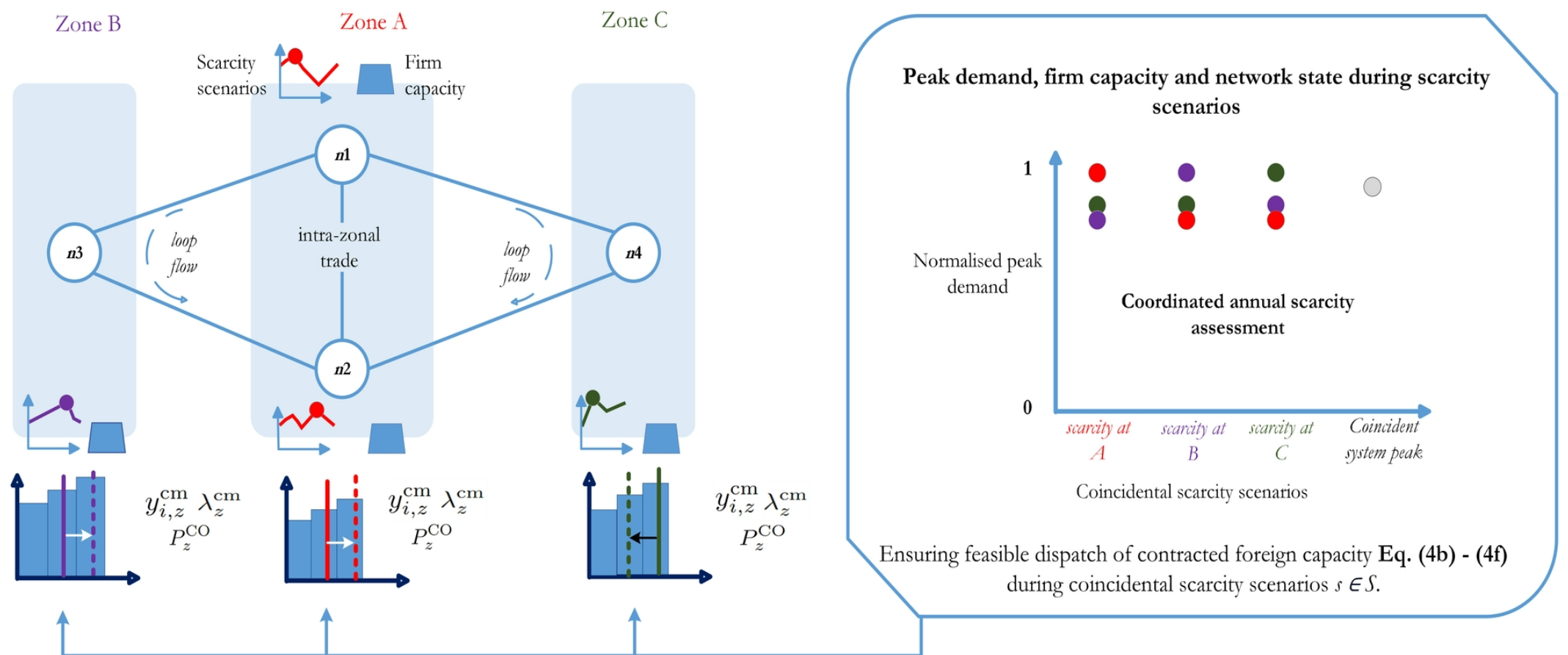
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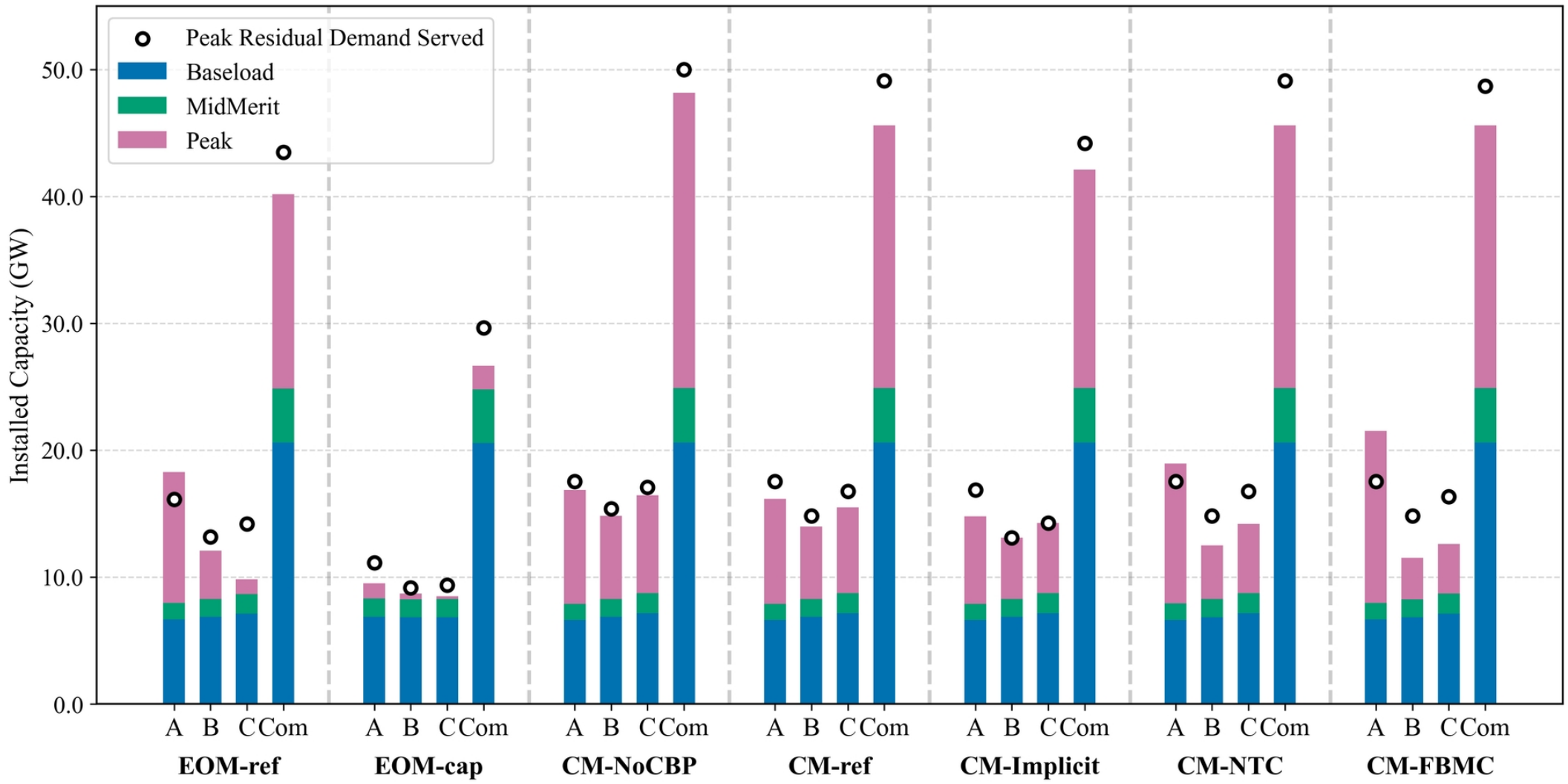
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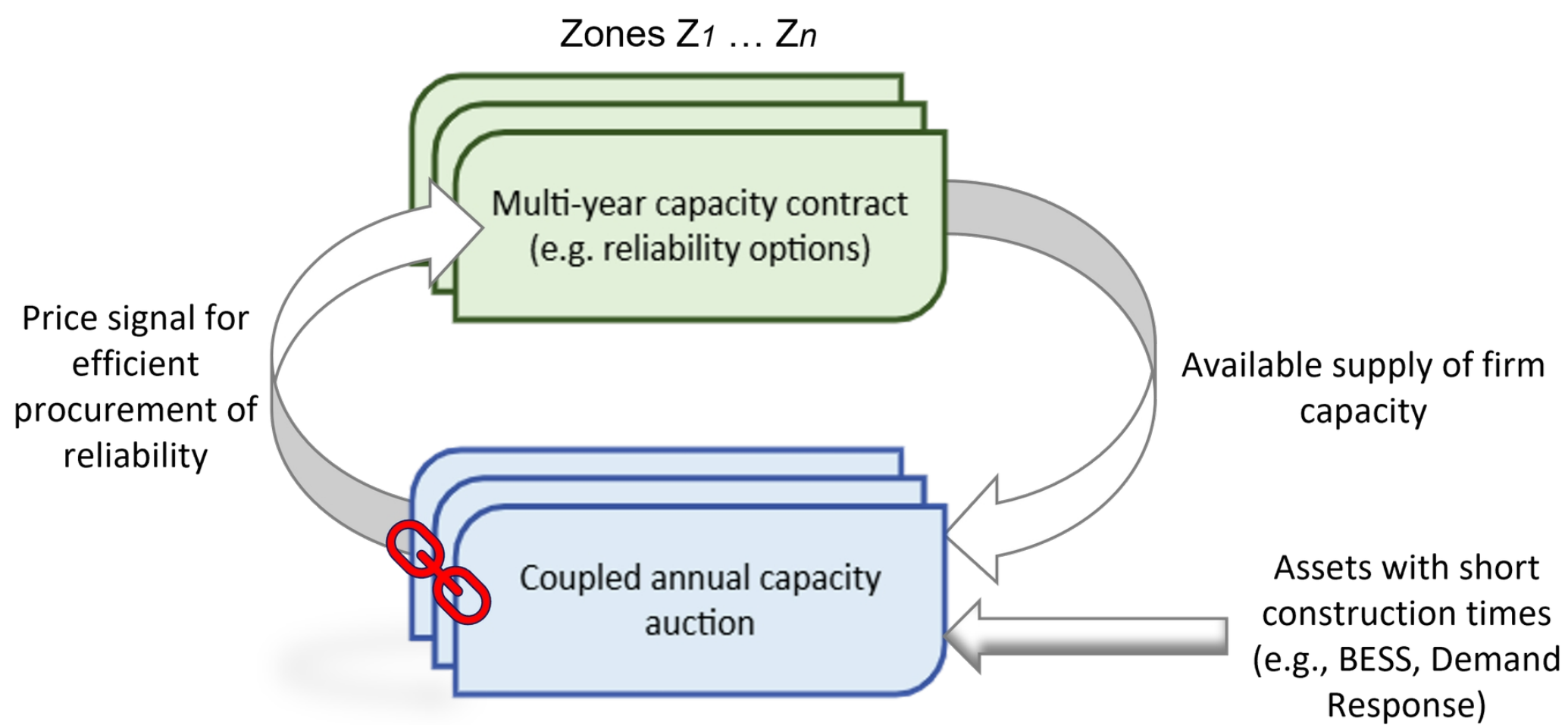
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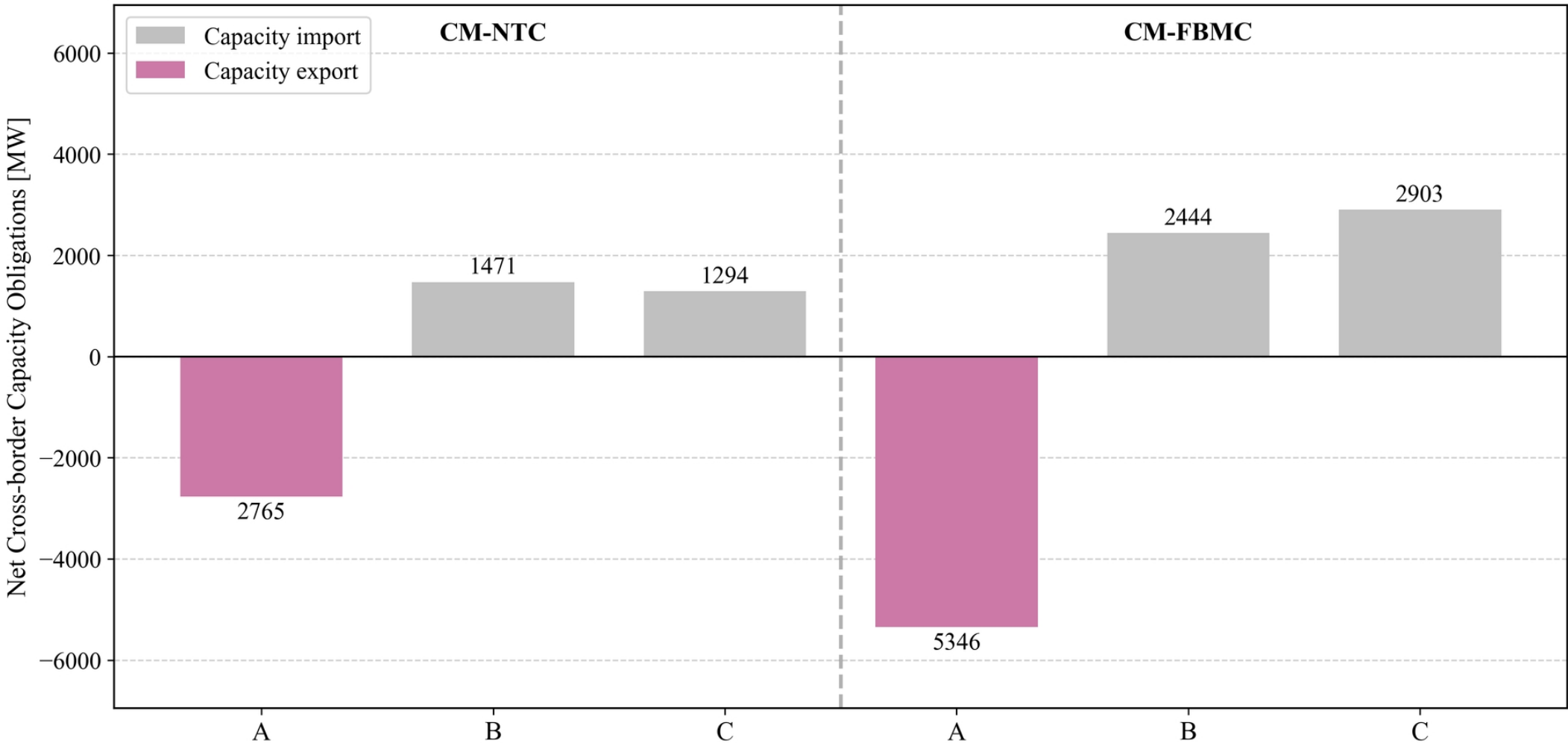


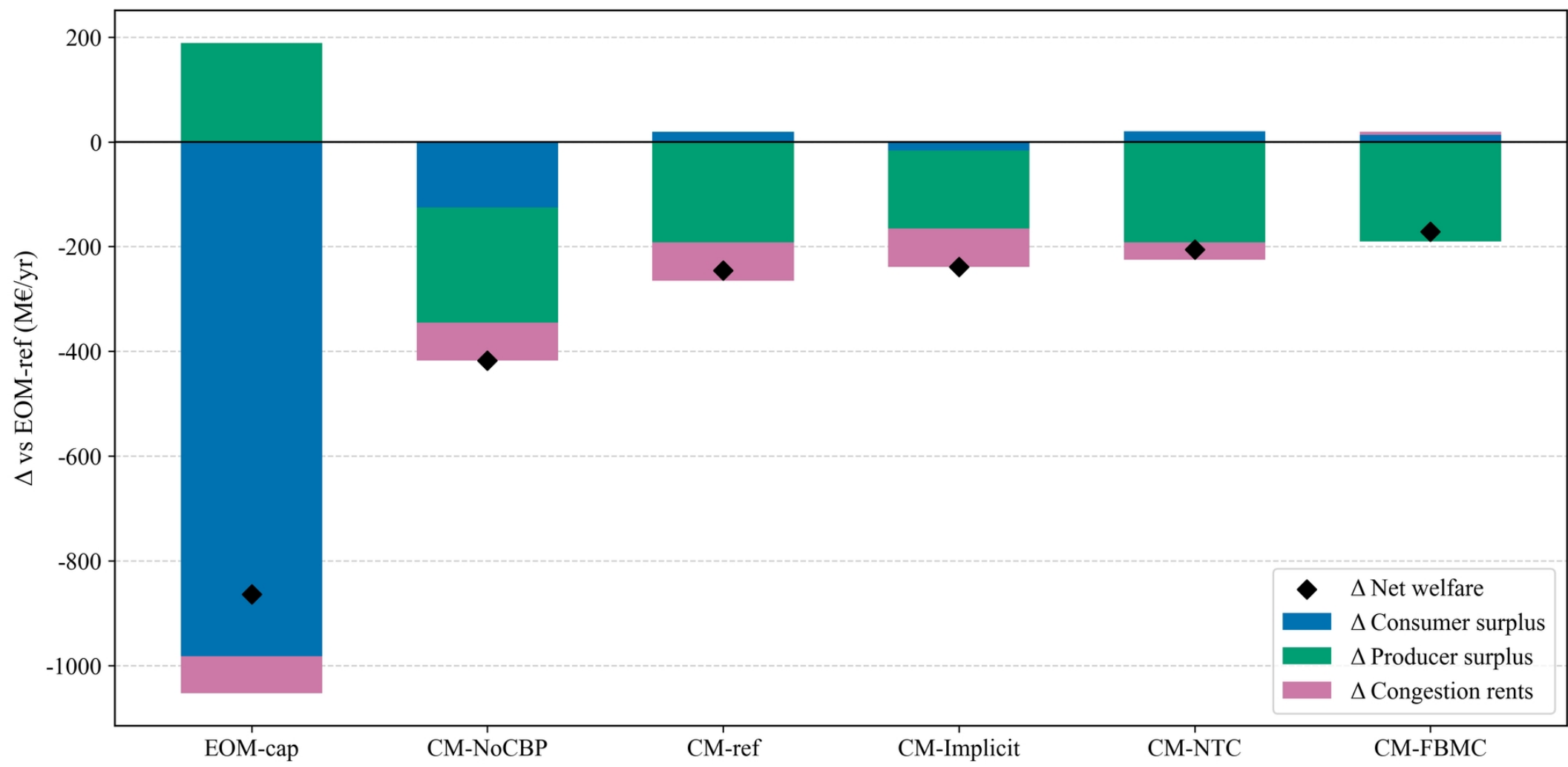












Coupling Europe's Capacity Markets

Kamaldeen Adekola, Laurens de Vries, Kenneth Bruninx, *Member, IEEE*

Abstract—European Member States are increasingly introducing national capacity mechanisms (CMs) to manage growing adequacy risks. However, isolated national CMs are inefficient in highly interconnected electricity systems, such as the European system. While progress has been made in facilitating cross-border participation by generation capacity in CMs, existing arrangements are prone to under- or over-investment and do not properly value the contribution of interconnection capacity to Member States' adequacy targets. ~~In this paper, we propose a novel conceptual design for a coupled European capacity market that utilises the logic of flow-based market coupling. We argue that national capacity markets, offering long-term contracts, need a second tier: an annual capacity auction coupled across zones, which sets zonal prices for firm, deliverable capacity. We design such an auction by extending the logic of flow-based market coupling to capacity markets. In a comparative analysis of different market design scenarios in an illustrative multi-zone case study, using a bespoke long-run equilibrium problem, we show that the proposed flow-based coupling of capacity markets reduces system costs by harnessing available capacity in neighbouring market zones while ensuring deliverability with respect to network constraints in all scarcity situations. auction removes the circular dependency between imports and available capacity in current cross-border participation schemes, locates investment where it is cheapest while keeping it deliverable under network constraints in defined scarcity situations, and makes the value of additional interconnection capacity explicit.~~

Index Terms—Flow-based Market Coupling, Capacity Market, Market Coupling, Cross-border Participation

I. INTRODUCTION

THE reliable and cost-effective supply of electricity is fundamental to any modern economy. After the liberalisation of the electricity sector in Europe, energy-only markets were expected to drive private investments [1]. However, the resulting electricity price signals fail to provide sufficient incentives for adequate investment in generation capacity due to market imperfections such as price caps (the "missing money" problem) and market incompleteness (the "missing (risk) markets" problem) [2], [3]. As a result, many governments have introduced capacity mechanisms to offer additional financial incentives for investments in generation capacity [4].

Previously viewed as tools of last resort, capacity mechanisms have increasingly become a common element in European electricity markets [5]. Member states have opted to implement capacity mechanisms as national measures, exercising their sovereign right to steer their energy mix [5]. Recent developments indicate a convergence toward single-buyer, market-wide capacity markets [6], [7]. In these markets, a central authority defines the capacity requirement based on the target reliability standard. Providers submit bids to supply capacity, and the market clears at the intersection of supply offers and the administratively set demand curve [6].

However, capacity mechanisms across Europe still (i) feature non-harmonised design parameters and (ii) allow for

limited cross-border participation, despite this being mandated by law [5]. Contract durations vary significantly, typically between 1 and 15 years. Auctions occur at different lead times, ranging from 1 to 5 years [6]. Despite reliability options becoming the preferred contract type, the specific contract forms, remuneration and penalties for non-delivery vary [7]. Member states conduct their auctions independently, and cross-border participation is limited to implicit participation—reducing capacity procurement based on expected but unremunerated imports, and explicit participation, which limits foreign resources' contribution to a static "maximum entry capacity" (MEC) at each border [7]. Both methods necessitate overly conservative cross-border assumptions that stifle regional investment efficiency [8].

Because the EU electricity market is highly interconnected, distorted incentives for generation and interconnection investment may arise from such national capacity mechanisms [9]. Furthermore, the current mechanisms for cross-border market participation fail to properly value interconnection and its availability during periods of scarcity [10]. Consequently, this results in a high risk of either over- or under-procuring capacity, depending on the Member States' ability to assess the availability of and willingness to rely on imports during periods of scarcity. Moreover, a circular dependency between available capacity and expected imports makes this assessment inherently challenging [8]. The limitations of current patchwork approaches to mitigate the distorted investment incentives arising from national capacity mechanisms have been acknowledged in the literature [10], [11].

Researchers increasingly call for a more harmonised and regional capacity market. Bucksteeg et al. highlight the potential cost savings of regional-level coordination of capacity mechanisms [12]. Menegatti and Meeus propose regionalisation of capacity markets in three sequential steps to minimise overprocurement of resources [8]. Höschle et al. [13] call for a fully integrated, market-based capacity mechanism that clears de-rated capacity, including generation, demand, and storage resources across borders. Capacity-price differentials would drive trade, while congestion rents would signal the need for new interconnectors. However, the existing literature falls short of providing a design for implementing regionalisation of resource adequacy, recognising the political reality that Member States prioritise sovereignty over their national adequacy and capacity mix.

This leads to a clear research question: How should capacity mechanisms in Europe ~~be designed~~ evolve to facilitate the use of cross-border resources? ~~To this end, we propose a two-tiered mechanism in which Member States would grant long-term contracts to new capacity, which is subsequently re-marketed as reliability options in annual, coupled capacity markets. We propose an additional tier: an annual capacity auction, coupled~~

across zones. It complements the multi-year national capacity auctions in which Member States grant long-term contracts, such as reliability options, to de-risk investment. Member States, as holders of these long-term contracts, re-market the contracted volumes as annual products in the coupled auctions (Section II).

The Belgian and Irish capacity mechanisms offer a precedent for this "split" auction [14], [15]. In the Belgian capacity mechanism, a "Y-4" auction secures the bulk of capacity four years in advance of delivery, allowing derisking capital-intensive investments with longer construction times. This is complemented by a "Y-1" auction held one year prior to delivery, which adjusts volumes based on updated demand forecasts and facilitates participation by existing capacity and fast-to-construct assets (such as battery storage).

Similarly, the German BMWK proposes a Combined Capacity Market [16]. It features an "investment layer" (a centralised capacity market) that provides long-term security for capital-intensive, controllable capacities, and an "adequacy layer" (a decentralised capacity market) where market participants manage residual peak loads through demand-side flexibility or by acquiring capacity certificates. We extend these two-tiered approaches into a European capacity mechanism **by coupling their annual layer across borders**.

The long-term contracts allow Member States to steer the development of their own energy mix, whereas the annual coupled auctions capitalise on the benefits of a highly interconnected system by facilitating cross-border participation of resources. A key challenge in these annual auctions is the inclusion of intra-zonal and inter-zonal network limitations that may impede delivery of contracted capacity.

We propose leveraging flow-based market coupling (FBMC) logic, currently used to couple Europe's day-ahead and intra-day markets. This allows us to account for network feasibility of flows during scarcity events in annual capacity market auctions (i.e. implicit transmission capacity allocation). Furthermore, we avoid the issues associated with defining Maximum Entry Capacities (MECs), such as simplifying assumptions on the capacity and availability of intra-zonal lines and the availability of generation capacity in neighbouring systems during scarcity situations, which can lead to conservative estimates. As more zones adopt capacity mechanisms, the use of MECs becomes more cumbersome and less accurate [8].

To illustrate the functionality and benefits of this new mechanism, which leverages FBMC logic to couple Europe's capacity markets, we develop a long-run equilibrium model and apply it in a stylised case study. In summary, our contribution is threefold:

Firstly, we ~~develop a conceptual design of a coupled capacity mechanism~~. We propose an annual coupled capacity auction that is compatible with long-term contracts issued by member states to domestic resources to incentivise or de-risk investments. Cross-border participation in the annual capacity auction is enabled via FBMC logic. Combined, this preserves national sovereignty over security of supply and investment decisions while reducing inefficiencies associated with national capacity mechanisms.

Secondly, we develop a long-run equilibrium model of

the energy and capacity markets, cast as a non-cooperative, perfectly competitive game, combining the work of Höschle et al. [13] (cross-border capacity markets) and Aravena et al. [17] (FMBC modelling). We replace the static "maximum entry capacity" approximation used in previous studies [11], [13] with an endogenously defined flow-based domain. By embedding flow-based constraints in the capacity market clearing, we ensure the deliverability of the cleared capacity bids w.r.t. network constraints during scarcity events.

Finally, we present findings from a stylised case study that allows us to disentangle the impact of various approaches to cross-border participation in capacity markets in a long-run context. We demonstrate how the proposed mechanism facilitates larger cross-border capacity trades, correctly values interconnection capacity and reduces investment costs in generation capacity while maintaining resource adequacy by reallocating investments to more favourable locations within the network.

The remainder of this paper is organised as follows. Section II introduces a conceptual framework for coupling European capacity mechanisms. We present the mathematical formalisation in Section III. Section IV illustrates the proposed design with a numerical example. Section V reflects on the broader implications. We offer concluding remarks in Section VI.

II. COUPLING CAPACITY MARKETS: PROPOSED DESIGN

Based on de Vries et al. [18] and ENTSO-E [7], we highlight the following set of non-exhaustive design criteria for coupling capacity mechanisms:

- Satisfying both short-term and long-term adequacy needs
- Ensuring the deliverability w.r.t. network constraints of capacity offers during scarcity
- Enabling locational signals in capacity remuneration
- Recognising the heterogeneity of national resource adequacy requirements
- Reducing the need for the harmonisation of design parameters across member states

With the above in mind, ~~we propose a two-tiered approach for future capacity mechanisms~~ we propose an annual, coupled European capacity auction as a coordinating layer for national capacity mechanisms awarding long-term contracts (Figure 1). ~~This two-tiered approach recognises that while investment eyes are multi-year, adequacy and network conditions are non-static.~~ Our primary contribution is the design of the second layer, but we reflect on the requirements it places on the first layer. On the first tier is the "Investment Layer", a long-term national capacity mechanism (e.g., in the form of a centralised, market-wide auction that uses reliability options as contracts). These reliability options should span a duration long enough — typically 10-15 years — to de-risk capital-intensive dispatchable resources, and auctions should be conducted well in advance of delivery (Y-4 or Y-5) to accommodate construction lead times. Reliability options are preferred as they effectively reduce risk for both generators and consumers. They include a "pay-back" obligation when market prices exceed a predetermined strike price. This creates incentives for resources to remain available during scarcity

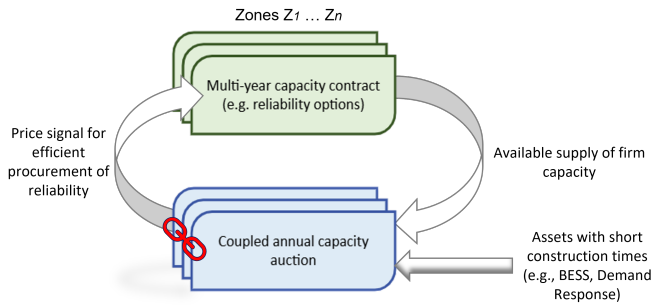


Fig. 1: The proposed capacity market design features two layers: a national capacity market for long-term contracts and a coupled annual capacity mechanism allowing secondary trade of long-term contracts. The proposed annual coupled capacity auction and its interface with national capacity contracts. Member States grant multi-year contracts (e.g., reliability options) to de-risk investment and re-market the contracted volumes as annual products in the coupled auctions.

periods [19]. The Investment Layer remains national in scope as Member States (MS) often have distinct energy policy objectives, such as technology restrictions or carbon targets, and also apply different adequacy standards. Thus, a national mechanism avoids the bottleneck of lengthy negotiations over a harmonised set of design criteria that would be necessary for a cross-border multi-year auction.

However, coordination of the investment tier with the second tier is essential. Member States, acting as the primary contract holders, are obliged to bid the total volume of their multi-year contracted capacity into the second-tier Coupled Annual Capacity Market as annual products. This ensures that long-term investments are fully reflected in the coupled annual capacity market, preventing artificial scarcity in the second tier.

This second tier consists of an annual, coupled capacity market. Firm capacity across all zones participates in a single annual uniform-price auction that implicitly accounts for network flows during anticipated scarcity events. By implicitly allocating transmission capacity in the clearing algorithm (see Section III-D1), we ensure that the cleared capacity can dispatch to meet demand during scarcity events. The uniform-price auction establishes a marginal clearing price for each zone in the coupled region, which reflects the value of one additional megawatt of firm capacity in that zone.

The annual capacity mechanism allows Member States to adjust for over- or under-procurement of long-term reliability options through cross-border trade. Countries that have procured excess options can offer them in the annual capacity auction, which is cleared subject to network constraints. The short-term auction then sends price signals for efficient procurement of reliability options in the long-term auction.

A critical input to the annual, coupled capacity markets is the set of scarcity events, which determine the demand, available generation, and the state of the network throughout the system during critical periods. These scarcity events can either be directly reflected in a stochastic market clearing – similar to how network contingencies are reflected in FBMC – or used to inform a coordinated scarcity assessment. In

the latter approach, Regional Coordination Centres (RCCs) perform a Monte Carlo simulation to determine the minimum firm capacity in each zone that satisfies both simultaneous and isolated scarcity events within the coupled region. This is a natural extension of ENTSO-E’s adequacy assessments [20], and in line with ACER’s guidelines that advocate accounting for critical network elements in adequacy assessments [21].

To reflect the network feasibility of flows from contracted capacity to serve demand during scarcity events, we extend the logic behind flow-based market coupling (FBMC) to an annual, coupled, European cross-border capacity market for annual capacity contracts. Rather than computing border-to-border maximum entry capacities (MECs), which suffer from circular dependency with neighbours’ available capacity, FBMC explicitly models flows on critical network elements resulting from cross-border trade [22]. It enables the coupling of European capacity markets within a common grid model, capturing the physical flows induced by zonal net positions, such as loop flows and internal bottlenecks. In doing so, it allows the deliverability of both foreign and domestic capacity to be assessed under defined scarcity conditions (Section II).

The short lead time and duration of the awarded capacity contracts efficiently signal short-term adequacy needs and promote the participation of flexible assets with short construction times, such as storage and demand-side response. Additionally, it mitigates the private financial risk associated with cross-border trade of long-term capacity obligations. Generation companies show limited appetite for long-term cross-border trade, as securing transmission rights years in advance is often impossible due to the limited liquidity. By implicitly allocating transmission capacity in an annual capacity market clearing, we remove the burden on generators to secure transmission rights. Instead, the risk of regional price differences is shifted to the state, which acts through the TSOs, who then capture a “capacity congestion rent” – the capacity price spread multiplied by the volume of reliability obligations exchanged across the zones. The collected rent can be reinvested in maintaining and further developing interconnection infrastructure or reducing network tariffs for consumers.

The proposed design respects the heterogeneity of adequacy requirements within the coupled region. Member States can impose their own reliability standards by defining the expected scarcity events for which annual capacity obligations must be available. They also determine their reliance on neighbouring countries through the interaction between the two layers of the proposed mechanism. For example, a state favouring national autonomy can prioritise multi-year capacity contracts. Thus, we strike a balance between national autonomy and regional integration. Consequently, the degree of regional efficiency will be determined by the level of coordination between the first and second layers in Figure 1.

Furthermore, the proposed design requires limited harmonisation of design choices and capacity market parameters across member states. For example, the responsibility for purchasing the required firm capacity demand in the annual auctions may be centralised, passed to load-serving entities and large electricity consumers in a decentralised capacity market, or a combination of both approaches. The multi-year contracts

$$0 \leq d_{z,t}^{\text{ela}} \leq \epsilon_z D_{z,t}^e, \quad d_{z,t}^{\text{inel}}, \text{ens}_{z,t} \geq 0, \quad \forall t \in \mathcal{T} \quad (2d)$$

C. Transmission system operator - energy market

The energy market operator coordinates the clearing of the coupled electrical energy markets. Based on agents' offers $g_{i,z,t}$ and bids $d_{z,t}^e$ received, it determines the zonal energy prices $\lambda_{z,t}^e$ and network flows $p_{z,t}^E$ for each time period t and zone z . The energy markets are coupled using flow-based market coupling, captured via the exact projection method of Aravena et al. [17]. The transmission system operator allocates cross-border transmission capacity in the coupled zonal electricity markets. Acting as a price-taking agent, it takes electricity prices $\lambda_{z,t}^e$ and determines the zonal net positions $p_{z,t}^E$ that maximise congestion rent [25]. Network feasibility is imposed through flow-based market coupling, which we capture via the exact projection method of Aravena et al. [17]. The set of decision vectors for the energy market operator is denoted by $x_E = (p_{z,t}^E, f_{t,\ell}, \hat{g}_{n,t}, \hat{y}_n) \in \mathcal{X}_E, x_E = (p_{z,t}^E, f_{t,\ell}, \hat{g}_{i,n,t}, \hat{y}_{i,n}) \in \mathcal{X}_E$.

The energy market clearing transmission system operator's decision problem is presented in Problem (3). The market clearing problem is solved independently for every time step $t \in \mathcal{T}$. Eqs. (3c)–(3e) ensure that, for all periods t , there exists an auxiliary nodal dispatch $\hat{g}_{n,t} \leq \sum \hat{g}_{i,n,t}$ that meets nodal demand $d_{n,t}^e$. Zonal electricity demand is allocated to a nodal demand variable $d_{n,t}^e$ through coefficients $S_{n,z} \in [0, 1]$, which satisfy $\sum_{n \in \mathcal{N}_z} S_{n,z} = 1$ for each zone z . $\hat{y}_{i,n}$ is the nodal allocation of installed capacity at zone z . The nodal allocation of existing capacity is fixed by the same coefficients used to allocate demand, $\hat{y}_{i,n} = S_{n,z} y_{i,z}$, while new capacity is sited freely subject to Eq. (3e).

The transmission network consists of a set of nodes $\mathcal{N}$ and transmission lines $\mathcal{L}$. The subset of nodes belonging to zone $z \in \mathcal{Z}$ is denoted by $\mathcal{N}_z$. Each line $\ell \in \mathcal{L}$, has a thermal capacity $F_\ell^{\max}$. The nodal injection at node n is denoted by $r_{n,t}$, and the corresponding line flows on ℓ are given by a DC load-flow approximation using the power transfer distribution factors PTDF $_{\ell,n}$.

The exact projection method [17] enforces network feasibility of flows by requiring each admissible zonal net position $p_{z,t}^E$ to lie within the flow-based domain. The domain is constructed by aggregating the nodal injections $r_{n,t}$ that satisfy the power balance (Eq. (3g)) and thermal limits (Eq. (3i)). By mapping the nodal constraints into the zonal space via (Eq. (3f)), we project exactly the nodal feasible set onto the zonal net positions.

$$\max_{x_E \in \mathcal{X}_E} \sum_{z \in \mathcal{Z}} \lambda_{z,t}^e p_{z,t}^E \quad (3a)$$

$$\text{s.t. } d_{n,t}^e = S_{n,z} d_{z,t}^e, \quad \forall n \in \mathcal{N}_z, z \in \mathcal{Z}, t \in \mathcal{T} \quad (3b)$$

$$\sum_{i \in \mathcal{I}} \hat{g}_{i,n,t} - d_{n,t}^e = r_{n,t}, \quad \forall n \in \mathcal{N}, t \in \mathcal{T} \quad (3c)$$

$$\hat{g}_{i,n,t} \leq A F_{i,z,t} \hat{y}_{i,n}, \quad \forall i \in \mathcal{I}, n \in \mathcal{N}_z, z \in \mathcal{Z}, t \in \mathcal{T} \quad (3d)$$

$$\sum_{n \in \mathcal{N}_z} \hat{y}_{i,n} = y_{i,z}, \quad \forall i \in \mathcal{I}, z \in \mathcal{Z} \quad (3e)$$

$$p_{z,t}^E = - \sum_{n \in \mathcal{N}_z} r_{n,t}, \quad \forall z \in \mathcal{Z}, t \in \mathcal{T} \quad (3f)$$

$$\sum_{n \in \mathcal{N}} r_{n,t} = 0, \quad \forall t \in \mathcal{T} \quad (3g)$$

$$f_{t,\ell} = \sum_{n \in \mathcal{N}} \text{PTDF}_{\ell,n} \cdot r_{n,t}, \quad \forall \ell \in \mathcal{L}, t \in \mathcal{T} \quad (3h)$$

$$-F_\ell^{\max} \leq f_{t,\ell} \leq F_\ell^{\max}, \quad \forall \ell \in \mathcal{L}, t \in \mathcal{T} \quad (3i)$$

D. Transmission system operator - capacity market

The capacity market operator coordinates the clearing of cross-border capacity markets. The transmission system operator maximizes the capacity congestion rent in coupled capacity auctions, while certifying the deliverability of cross-border capacity obligations w.r.t. network constraints in anticipated scarcity conditions. Scarcity scenarios $s \in \mathcal{S}$ represent stressed system states (e.g., simultaneous peak load or low renewable output) that define the critical periods against which deliverability of procured local and foreign capacity w.r.t. network constraints must be demonstrated. The administratively set zonal demand for firm capacity is denoted by D_z^{cm} . The set of decision variables of the capacity market operator is $x_{\text{CO}} = (p_z^{\text{CO}}, \hat{g}_{s,n}^{\text{sc}}, f_{s,\ell}^{\text{cm}}, \hat{y}_n^{\text{cm}}) \in \mathcal{X}_{\text{CO}}$, where p_z^{CO} denotes the net cross-border capacity obligation at zone z , $\hat{g}_{s,n}^{\text{sc}}$ the auxiliary dispatch at node n in scarcity scenario s , $f_{s,\ell}^{\text{cm}}$ the anticipated network flows during defined scarcity period, and $\hat{y}_n^{\text{cm}}$ is the nodal allocation of the total capacity obligation Y_z^{cm} by generators in zone z . This nodal allocation cannot exceed the total available capacity in the node $\hat{y}_{i,n}$ (Eqs. (4c) and (4d)). This nodal allocation is determined in Problem (3) and, hence, is a parameter in Problems (4) and (5).

We compare two approaches to explicit cross-border capacity trade, differing mainly in how the feasible domain for cross-border trade is defined: (i) a flow-based domain computed endogenously at clearing, which captures exactly the set of net positions deliverable across all scarcity scenarios; and (ii) an ex-ante defined Net Transfer Capacity (NTC) domain, which imposes bilateral maximum entry capacities (MECs) on each border.

1) *Flow-based coupling of capacity markets*: Our proposed approach consists (i) a coupled capacity market clearing and (ii) a simultaneous joint scarcity assessment (Figure 2). The flow-based coupling clearing of the capacity markets under flow-based coupling is formalised in Problem (4). In the proposed coupled capacity market, accepted capacity offers Y_z^{cm} in zone z must be deliverable to the relevant load centres under a set of anticipated scarcity scenarios $s \in \mathcal{S}$. The capacity market coupling allows net cross-border obligations p_z^{CO} only if they are certified to be deliverable to the relevant load centres in each anticipated scarcity scenario $s \in \mathcal{S}$. It does so by limiting the transmission capacity made available for cross-zonal capacity exchange, ensuring the feasibility of network flows during all scarcity situations.

Thus, we require the existence of a feasible nodal dispatch $\hat{g}_{s,n}^{\text{sc}}$ of the committed firm capacity that meets the energy demand at every node n and satisfies all network constraints (including intra-zonal bottlenecks) in every scarcity scenario. As a result, any MW of capacity cleared in the auction can contribute to the nodal capacity requirement $D_{s,n}^{\text{cm}}$ when scarcity materialises. The auction yields for each zone z ,

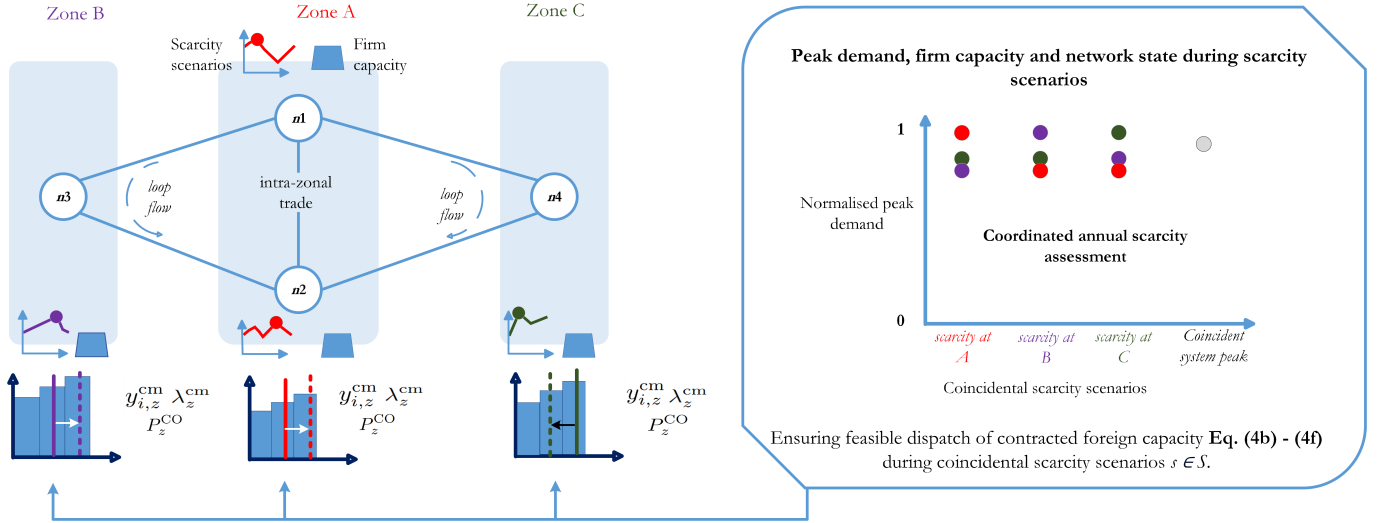


Fig. 2: Left: The annual, coupled capacity auction (zones A, B, and C) clears capacity offers $y_{i,z}^{cm}$, produces capacity market prices λ_z^{cm} , and net cross-border capacity obligations p_z^{CO} subject to the flow-based network constraints. Firm capacity may contribute to adequacy in different zones depending on the scarcity conditions and transmission constraints. Right: A joint assessment of the scarcity events across zones determines the maximum coincidental system capacity requirement, lowering zonal capacity demands compared to isolated national assessments. "Scarcity at A" represents a scarcity situation in zone A and the state of other zones within the coupled region during A's critical scarcity event.

a single annual capacity price λ_z^{cm} , the domestic capacity obligation Y_z^{cm} and the net cross-border capacity obligation p_z^{CO} that meets all nodal adequacy requirements $D_{s,n}^{cm}$ across the defined scarcity scenarios. In each scenario s , the zones in $\mathcal{Z}_s^{sc}$ are placed at their scarcity demand, above the capacity demand $D_{s,n}^{cm} = S_{n,z} D_z^{cm}$ allocated to them by the joint assessment, while the remaining zones are placed at their coincident demand (Figure 2, right). The transmission system operator determines how much cross-zonal capacity exchange can be supported while remaining deliverable across the prescribed scarcity scenarios. To this end, we required that there must exist a nodal dispatch $\hat{g}_{s,n}^{sc}$ that serves demand while respecting the network constraints, including intra-zonal lines at state-N thermal limits in each scenario. The resulting zonal net positions must remain consistent with the cross-zonal capacity obligations. In particular, a zone that has sold capacity abroad must be able to sustain the corresponding export position when that capacity is called during the importing zone's scarcity event. The auction yields, for each zone z , a single annual capacity price λ_z^{cm} , the domestic capacity obligation Y_z^{cm} , and the net cross-border capacity obligation p_z^{CO} .

$$\max_{x_{CO} \in \mathcal{X}_{CO}} \sum_{z \in \mathcal{Z}^{cm}} \lambda_z^{cm} p_z^{CO} \quad (4a)$$

$$\text{s.t. } \hat{g}_{s,n}^{sc} \leq \hat{y}_n^{cm}, \quad \forall s \in \mathcal{S}, n \in \mathcal{N} \quad (4b)$$

$$\hat{y}_n^{cm} \leq \sum_{i \in \mathcal{I}^{cm}} \hat{y}_{i,n}, \quad \forall n \in \mathcal{N}_z, z \in \mathcal{Z} \quad (4c)$$

$$\sum_{n \in \mathcal{N}_z} \hat{y}_n^{cm} = Y_z^{cm}, \quad \forall z \in \mathcal{Z} \quad (4d)$$

$$\hat{g}_{s,n}^{sc} - D_{s,n}^{cm} = r_{s,n}, \quad \forall s \in \mathcal{S}, n \in \mathcal{N}_z \quad (4e)$$

$$-\sum_{n \in \mathcal{N}_z} r_{s,n} \leq p_z^{CO}, \quad \forall s \in \mathcal{S}, z \in \mathcal{Z} \setminus \mathcal{Z}_s^{sc} \quad (4f)$$

$$\sum_{n \in \mathcal{N}} r_{s,n} = 0, \quad \forall s \in \mathcal{S}, \quad (4g)$$

$$\sum_{z \in \mathcal{Z}} p_z^{CO} = 0, \quad (4h)$$

$$-F_\ell^{\max} \leq f_{s,\ell}^{cm} \leq F_\ell^{\max}, \quad \forall s \in \mathcal{S}, \ell \in \mathcal{L} \quad (4i)$$

$$f_{s,\ell}^{cm} = \sum_{n \in \mathcal{N}} \text{PTDF}_{\ell,n} \cdot r_{s,n}, \quad \forall s \in \mathcal{S}, \ell \in \mathcal{L} \quad (4j)$$

2) *Cross-border capacity market using NTC*: The cross-border capacity market using NTC serves as **our best-case proxy** for what is observed in European capacity markets [5]. In line with current practice in explicit cross-border capacity participation, we bound bilateral capacity trade within an ex-ante NTC domain $\mathcal{P}^{\text{ATC}}$. This domain specifies the maximum entry capacity at each border before the capacity market auction. Since our model excludes pre-allocated capacity, the NTC is equivalent to the available transfer capacity (ATC).

For comparability with the flow-based approach, we compute the largest NTC domain that fits entirely within the flow-based feasibility polytope. We obtain this domain by solving a separate optimisation problem following Aravena et al. [17]. The resulting set, $\mathcal{E}^{\text{ATC}}$, gives the maximum **feasible net transfer capacity on each border while still respecting all network constraints** net transfer capacity on each border that still yields feasible network flows during the scarcity scenarios considered in the determination of the flow-based domain (Section III-D1).

Note that in reality, the NTC domain influences agents' investment decisions, and these decisions in turn reshape the domain. To isolate the effect of the chosen cross-border

capacity calculation method (NTC versus flow-based), we fix the investment levels used to construct the NTC domain to those obtained under the flow-based approach. Consequently, we isolate the inefficiency of using NTC rather than flow-based for capacity market clearing. Put differently, we do not capture how investing in the wrong location lowers permissible cross-border exchanges. It is thus a theoretical upper bound to the performance of NTC-coupled markets w.r.t flow-based market coupling.

The capacity market operator's decision problem under net transfer capacity (NTC) market coupling is formulated in Problem (5). ~~The corresponding zonal net capacity obligation p_z^{CO} is given in Eq. It is calculated as the net traded capacity in each zone, as defined in Eq. (5b). The net positions in each scenario are calculated from the bilateral exchanges on each border (Eq. (5b)), which in turn determine the cross-border capacity obligations (Eq. (4h)). The NTC is bilateral exchanges are bounded by the predefined maximum entry capacities $\mathcal{E}^{\text{ATC}}$ in Eq. (5c).~~

$$\max_{x_{\text{CO}} \in \mathcal{X}_{\text{CO}}} \sum_{z \in \mathcal{Z}^{\text{cm}}} \lambda_z^{\text{cm}} p_z^{\text{CO}} \quad (5a)$$

s.t. Eqs. (4b)–(4h)

$$\mathcal{P}^{\text{ATC}} := \left\{ p \in \mathbb{R}^{|\mathcal{Z}|} \mid \exists e^{\text{CO}} \in \mathcal{E}^{\text{ATC}}: \right. \\ \left. - \sum_{n \in \mathcal{N}_z} r_{s,n} = \sum_{(z',z) \in \mathcal{B}} e_{z',z}^{\text{CO}} - \sum_{(z,z') \in \mathcal{B}} e_{z,z'}^{\text{CO}}, \forall z \in \mathcal{Z} \right\} \quad (5b)$$

$$\mathcal{E}^{\text{ATC}} = [-\text{ATC}^-, \text{ATC}^+] \quad (5c)$$

E. Market clearing

The zonal electricity price $\lambda_{z,t}^e$ and the zonal capacity price λ_z^{cm} are dual variables linked to the market-clearing conditions (6) and (7), which couple the agents' decision problems. The energy market clears in every zone and period (Eq. (6)), where $\bar{\lambda}_{z,t}^e$ is the administrative price cap. The complementarity condition describes that when a zonal balance is attained, the market clears between 0 and $\bar{\lambda}_{z,t}^e$. The price is zero with excess supply, or at the cap with excess demand. When multiple zones clear at the cap simultaneously, multiple, equivalent distributions of demand curtailment may exist ($\text{ens}_{z,t}$ in Eq. (2c)). We introduce an ex-post curtailment-allocation rule to circumvent this multiplicity of equilibria, detailed in Appendix II. In every hour in which at least two zones clear at the cap, each zone's demand curtailment is recomputed to the shortfall it faces when it can rely only on generation capacity within the zone and on the cross-border capacity it has procured. Zones with surplus capacity export to other zones in proportion to the deficit, so that the system ENS is conserved (Appendix II).

The capacity market clears annually in every zone participating in the mechanism (Eq. (7)). The capacity demand D_z^{cm} is assumed inelastic and capacity payments λ_z^{cm} are uncapped.

$$0 \leq \lambda_{z,t}^e \leq \bar{\lambda}_{z,t}^e \perp \sum_{i \in \mathcal{I}} g_{i,z,t} + p_{z,t}^{\text{E}} - d_{z,t}^e, \quad \forall z \in \mathcal{Z}, t \in \mathcal{T} \quad (6)$$

TABLE I: Investment costs $C_{i,z}^f$ and operational cost parameters $A_{i,z}$ and $B_{i,z}$. Peak generator costs vary by zone.

Technology $i \in \mathcal{I}$	$C_{i,z}^f$ [€/MW.year]	$A_{i,z}$ [€/MWh] ² [€/MWh] ²	$B_{i,z}$ [€/MWh]
Base	190 000	0.008	20
Mid	110 000	0.032	55
Peak – zone A	60 000	0.04	90
Peak – zone B	70 000	0.06	90
Peak – zone C	80 000	0.08	90

$$0 \leq \lambda_z^{\text{cm}} \perp \sum_{i \in \mathcal{I}^{\text{cm}}} y_{i,z}^{\text{cm}} + p_z^{\text{CO}} - D_z^{\text{cm}} \geq 0, \quad \forall z \in \mathcal{Z}^{\text{cm}}. \quad (7)$$

IV. CASE STUDY

We use a stylised three-zone (A, B, C), four-node network from [22] to illustrate the benefits of the proposed capacity-market coupling. Zone A has two nodes ($n1$ and $n2$), while zones B and C have one node, $n3$ and $n4$. Each cross-border line has a capacity of $F_{\ell}^{\text{max}} = 3$ GW, and the intra-zonal line has a capacity of 500 MW. All lines have identical susceptance.

We focus on the long-term equilibrium, assuming perfect coordination between long-term instruments used by member states and the coupled annual capacity market. This allows us to avoid bias in our results w.r.t. the existing capacity mix across the different zones.

We distinguish three types of generators: baseload, mid-merit, and peaking units (Table I). Renewable generation (solar and wind) is modelled as negative demand, with exogenous investment levels. Installed renewable capacities are 15.4 GW (solar) and 13.7 GW (wind) in Zone A, 8.8 GW and 5.3 GW in Zone B, and 18.7 GW and 8.3 GW in Zone C. In Zone A, which has two nodes, renewable capacity is allocated proportionally to nodal demand.

Zonal electricity demand $D_{z,t}^e$, renewable availability factors $AF_{i,z,t}$ and installed renewable capacities $Y_{i,z}$ are sourced from the ENTSO-E Transparency Platform [26] for Germany (Zone A), Belgium (Zone B), and the Netherlands (Zone C) for a winter day (11 November 2024). We normalise the historical demand and renewable capacity data using scaling factors of 5, 1, and 1.3 for zones A, B, and C, to create zones of comparable market size. We then simulate scarcity conditions by capping the renewable availability factor at 10% during the peak periods and multiplying the scaled peak demand by 1.5 to obtain new peak demands of 19 GW, 16 GW, and 18 GW in Zones A, B, and C. The peak events are non-coincident. Our calibration of the peak period ensures that unserved energy occurs when an administrative price cap is introduced, justifying the need for a capacity market.

~~Energy demand is represented as a two-block structure comprising a fixed (inelastic) and a flexible (elastic) component. The elastic block is derived from a quadratic utility function with a maximum willingness to pay of $\text{WTP}_z = 20,000$ €/MWh, and calibrated so that the elastic component of demand ϵ_z is 20% of the reference demand $D_{z,t}^e$ at the maximum willingness to pay WTP_z and decreases to 4% at the price cap of 4000 €/MWh. The slope of the inverse~~

demand curve is constant at $-WTP_z/(\epsilon_z D_{z,t}^e)$, producing a smooth downward-sloping price-quantity curve, indicating a diminishing marginal utility among flexible consumers. The demand for capacity is inelastic and dependent on the capacity market design. Electricity demand is represented as a two-block structure comprising an inelastic and a price-elastic component. The inelastic block, $(1 - \epsilon_z)D_{z,t}^e$, is valued at the maximum willingness to pay $WTP_z = 20,000$ €/MWh. The elastic block, $\epsilon_z D_{z,t}^e$ with $\epsilon_z = 0.20$, is derived from a quadratic utility function and yields a linear inverse demand curve that falls from WTP_z at zero elastic consumption to zero €/MWh at $\epsilon_z D_{z,t}^e$, with slope $-WTP_z/(\epsilon_z D_{z,t}^e)$. In designs with an administrative price cap at 4,000 €/MWh, the elastic consumers whose valuation exceeds the cap (16% of $D_{z,t}^e$) cannot express their WTP in the market and bid at the price cap. The consumer's problem (Problem (2)) is accordingly parametrised with $(WTP_z, \epsilon_z) = (20,000, 0.20)$ in markets without a price cap and with the truncation (4,000, 0.04) in price-capped designs (Table II). Both describe the same inverse demand curve, truncated at the cap, with identical slope. Curtailment of demand valued above the price cap is random (Eq. (2c)), so unserved energy is ex-post valued at its average valuation of 18,667 €/MWh. The demand for capacity is inelastic and depends on the capacity market design.

We study **sixteen** distinct market design scenarios (Table II). The first two scenarios are Energy-only markets with and without price caps. The remaining scenarios enable the study of how capacity markets with varying degrees of cross-border participation address underinvestment resulting from price caps in the electricity market. **Below, we discuss how the capacity demand is computed and summarise the resulting values in Table III:**

- 1) *EOM-ref*: The energy-only market without a price cap. All generators recover their costs from the energy market, assuming no market imperfections beyond the lack of a locational component in the electricity price in zonal markets.
- 2) *EOM-cap*: An energy-only market with a price cap, capturing the "missing money" problem, which leads to underinvestment in firm capacity and energy not served (ENS).
- 3) *CM-NoCBP*: A capacity market without cross-border participation. The zonal capacity demand is set to the peak residual demand valued at the price cap in each zone, i.e. the maximum residual demand served at 4,000 €/MWh. Zones do not explicitly or implicitly account for the availability of imports in their national capacity markets, nor carry out a joint scarcity assessment.
- 4) *CM-ref*: A reference capacity market scenario whose capacity demands result from a joint scarcity assessment (Figure 2 (Right)), but without cross-border participation. The joint scarcity assessment considers three zonal scarcity scenarios, representing each zone's individual peak demand. The set of scarcity scenarios considers the residual demand valued at the price cap of 4,000 €/MWh across all zones at each time step, any zone experiences ENS under *EOM-cap*. The joint scarcity assessment sets the system-wide capacity demand to the maximum coincident peak residual demand. This capacity

TABLE II: Summary of the market designs analysed, indicating whether a capacity market (CM) is implemented, the type of cross-border participation (CBP), and the willingness-to-pay (WTP) or price-cap (€/MWh) in the energy market.

Scenario	CM	CBP	WTP / Price-cap
EOM-ref	—	—	20,000
EOM-cap	—	—	4,000
CM-NoCBP	✓	—	4,000
CM-FBMC	✓	Explicit - Flow-based	4,000
CM-NTC	✓	Explicit - MEC	4,000
CM-Implicit	✓	Implicit	4,000

demand is distributed over the zones proportionally to the average share of the zone's residual demand to the overall system residual demand during scarcity situations. This scenario allows isolating the effect of coordinating capacity requirements from that of cross-border participation. The resulting capacity demand forms the basis for all scenarios considering cross-border participation.

5) *CM-Implicit*: A capacity market with implicit cross-border participation. ~~To determine capacity demand, expected scarcity imports from neighbouring zones are subtracted from the peak residual demand in each zone. This creates a circular dependency, as imports depend on available capacity. We use the investment outcome of the CM-FBMC to determine available imports, thereby enabling comparability with other scenarios.~~ Each zone's capacity demand is that determined in the joint scarcity assessment minus the average imports it receives in the scarcity scenarios under scenario *CM-ref*.

6) *CM-NTC*: A capacity market with explicit cross-border participation limited by a pre-defined maximum entry capacity (MEC) on each border. **We use capacity demands from scenario CM-ref.** For comparability, we ~~set these~~ **determine best-case** MECs using the investment decisions from the *CM-FBMC* scenario, following the procedure outlined in Section III-D2. **Thus, results from this scenario should be viewed as a best-case NTC, since we exclude some of the investment distortions that would arise from an endogenously determined MEC.** As a consequence, results ~~reflect only the difference~~ **differ from CM-FBMC only** in the treatment of cross-border capacity trade.

7) *CM-FBMC*: ~~A~~ **The proposed** flow-based coupling of the ~~capacity market~~ **capacity** markets. We use capacity demands obtained in scenarios *CM-ref*. The participation of foreign capacity in each zone's capacity market is constrained by the permissible flows across the system under predefined scarcity scenarios (Figure 2). ~~We use capacity demands obtained in scenarios CM-ref.~~ Scarcity events depict the time steps in which ENS occurs under *EOM-cap*. We represent projected scarcity events as (i) "simultaneous scarcity", where all zones demand 95 percent of their peak residual demand simultaneously, and (ii) "zonal scarcity", where the zone experiencing critical scarcity demands 100 per cent of its peak residual demand, while neighbouring zones each simultaneously demand 90 per cent of their peak residual demand. The scarcity scenarios are identical to those used in the joint scarcity assessment.

All code and input data are available online [27].

TABLE III: Residual peak demand valued at the price cap (MW) in each zone during observed scarcity scenarios and zonal capacity demands (MW) per capacity market design.

(a) Residual peak demand valued at the price cap (MW)				
Scenario	Zone A	Zone B	Zone C	Total
Scarcity at A	16,870	13,293	14,722	44,885
Scarcity at B	16,006	14,829	14,770	45,605
Scarcity at C	15,046	13,293	16,450	44,789

(b) Capacity demand (MW)				
Scenario	Zone A	Zone B	Zone C	Total
CM-NoCBP	16,870	14,829	16,450	48,149
CM-ref, CM-FBMC, CM-NTC	16,156	13,957	15,493	45,606
CM-Implicit	14,778	13,084	14,238	42,100

Below, we present the resulting capacity mixes (Section IV-A). Next, we discuss the implications for cross-border capacity trade and congestion rent (Section IV-B). Lastly, we reflect on the system costs and the costs to consumers (Section IV-C).

A. Investment decisions and capacity mix

In the reference Energy-Only Market (*EOM-ref*), investment is driven by scarcity pricing. In this idealised *EOM-ref*, prices may rise to the WTP of 20,000 €/MWh, at which the entire price-responsive share of demand (20% of the reference level) would voluntarily curtail. However, in equilibrium scarcity prices reach at most 16,608 €/MWh. Realised peak demand falls to 17.5, 13.7 and 15 GW in zones A, B and C, from reference peaks of 19, 16 and 18 GW. Implementing a price cap (*EOM-cap*) limits these scarcity rents truncates prices at 4,000 €/MWh, with consequences on both consumer demand and generation capacity investments. The forgone scarcity rents ~~This results in~~ cause a decline in investment, primarily in peaking units (Fig. 3). Total capacity falls by 13.5 GW relative to the reference energy-only market. On the consumer side, the volume of elastic demand that can express its willingness to pay in the market decreases from 20% to 4% of the reference level. The combination of less firm capacity and less price-responsive demand visible in the market results in Energy Not Served (ENS) ~~reaches 41.2 GWh, 30 GWh, and 33 GWh of 38.9 GWh, 28.9 GWh, and 37.3 GWh in zones A, B, and C.~~

A capacity market without cross-border participation (*CM-NoCBP*) successfully eliminates ENS; however, it does so through autarky. In line with observations in the literature [28], the system sees a surge in low-capex, high-opex peaking units (pink bars), increasing total capacity by ~~10 GW~~ 8 GW w.r.t. *EOM-ref*, indicating overprocurement. We observe a capacity displacement of ~~600 MW~~ 1.4 GW away from Zone A w.r.t. *EOM-ref*, as the isolation prevents Zone A from capitalising on its export potential. To mitigate this over-procurement and capacity displacement, we carry out a joint scarcity assessment in *CM-ref*, and further introduce cross-border participation under three designs: *CM-implicit*, *CM-NTC*, and *CM-FBMC*.

Coordinating capacity requirements through the joint scarcity assessment eliminates unserved energy with less in-

stalled capacity than in *CM-NoCBP*. Zone B, for example, procures 872 MW less than its own peak residual demand and covers the gap with 874 MW of imports at its peak. The flows presumed via the joint assessment, therefore, materialise in the energy market, though *CM-ref* neither remunerates nor secures them. Coordination, however, does not relocate investment. Without cross-border participation, the capacity displaced from Zone A w.r.t. *EOM-ref* grows to 2.1 GW, above the 1.4 GW under *CM-NoCBP*, while Zones B and C add 1.9 and 5.7 GW (Fig. 3). In total, *CM-ref* installs 2.5 GW below *CM-NoCBP*, with the reduction spread across all zones (0.7, 0.9 and 0.9 GW in Zones A, B and C).

~~By deducting expected foreign imports from demand without remuneration, the implicit cross-border participation *CM-implicit* minimises investment, adding only 4 GW w.r.t. *EOM-ref* (a 6 GW reduction compared to the *CM-NoCBP*), with some unserved energy re-emerging (4.2 GWh across the system). The latter is to be expected, as capacity demands were set assuming the investment levels in neighbouring zones from the *CM-FBMC* scenario. However, without explicit remuneration, these investments do not materialise, resulting in the unavailability of the expected imports. A capacity displacement of 600 MW from Zone A w.r.t. *EOM-ref* is observed, similar to *CM-NoCBP* scenario. Implicit cross-border participation *CM-implicit* deducts the 3.5 GW of imports that arise in each zone's scarcity period under *CM-ref* (1.4, 0.9 and 1.3 GW in Zones A, B and C) without remunerating them. However, the unremunerated import capacity does not materialise because, without explicit remuneration, the foreign capacity expected to support these imports is not built. Total installed capacity falls to 42.1 GW, 1.9 GW above *EOM-ref* and 6.0 GW below *CM-NoCBP*. Consequently, unserved energy re-emerges across all zones (12.4 GWh system-wide). Zone A bears the largest capacity displacement w.r.t. *EOM-ref* with 3.5 GW less capacity investment, as capacity it otherwise would build for export is left unremunerated.~~

Explicit cross-border participation under both *CM-NTC* and *CM-FBMC* utilises coordinated scarcity assessment, thus ~~reducing system-wide over-procurement w.r.t. *CM-NoCBP* by 2.5 GW while serving the same peak residual demand starts from the same joint scarcity assessment as *CM-ref*, thus total installed capacity stays at 45.6 GW, 2.5 GW below *CM-NoCBP*, and unserved energy remains negligible, at less than 10^{-4} % of served annual demand in the system. Unlike *CM-implicit* and *CM-NoCBP*, no displacement of capacity away from Zone A w.r.t. *EOM-ref* is observed. Both *CM-NTC* and *CM-FBMC* shift capacity investment toward lower-cost and system-friendly locations to varying degrees. When compared to *CM-NTC*, *CM-FBMC* better optimises location, shifting 1.3 GW of investment away from the costlier Zone C to the lower-cost Zone A. As foreign capacity obligations are remunerated in the explicit designs, Zone A with lower cost of new entry exploits this opportunity and builds 2.8 GW (*CM-NTC*) and 5.3 GW (*CM-FBMC*) above its own capacity demand to serve capacity obligations remunerated by Zones B and C. This is also observed in the obligations it exports (Fig. 4). Comparing both explicit designs, *CM-FBMC* relocates more capacity, shifting 2.6 GW more of investment~~

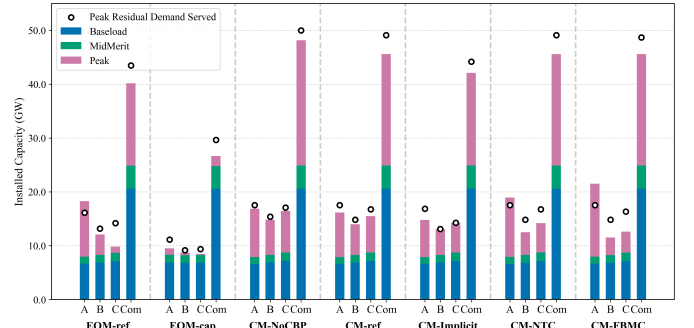
into the lower-cost Zone A relative to *CM-NTC*. Recall that we base the maximum entry capacities used in *CM-NTC* on the available capacity in the *CM-FBMC*. As such, the result excludes the investment distortions that would result from using the NTC approach and can be interpreted as an upper bound for *CM-NTC*.

Notably, all capacity market designs result in a total capacity that exceeds the *EOM-ref*. While *EOM-cap* under-invests by 13.5 GW, capacity mechanisms add 4–10 GW 1.9–8 GW above *EOM-ref* level. This arises from feedback between electricity prices and the elastic electricity demand. In *EOM-ref*, high scarcity prices reduce electricity consumption through self-rationing by consumers whose WTP is below the market-clearing price, and consequently, the peak residual demand served is reduced to 44 GW. Capacity markets suppress these price spikes (Section IV-B) and the associated demand elasticity, with the peak residual demand served increasing to 48–50 GW. The price cap truncates electricity prices (Section IV-B), removing the efficient self-rationing of consumers with demand valued above the price cap (4,000 €/MWh). Consequently, To avoid involuntary curtailment of demand, more firm capacity is required to guarantee the same level of reliability. As a result, the peak residual demand served rises from 44 GW in *EOM-ref* to 49–50 GW in the scenarios considering a capacity market.

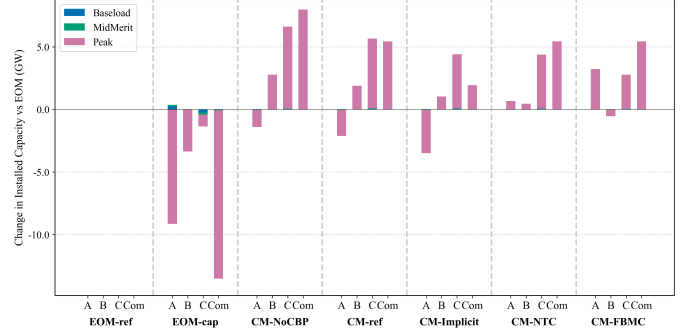
B. Cross-border trades and congestion revenues

The total cross-border capacity trade is 1.4 GW lower under *CM-NTC* than under *CM-FBMC* (Figure 4). In the NTC design, Zone A is the sole net exporter and provides 3.5 GW of capacity obligations to its neighbours. Zones B and C are net importers, 369 MW and 3 GW. Under *CM-FBMC*, Zone A exports 4.8 GW. This allows Zone C to reduce local investment. Zone C instead imports 4.4 GW of obligations from Zone A—an increase of 1.3 GW over the NTC case. Zone B remains a net importer with 408 MW. Higher trade under *CM-FBMC* does not compromise physical deliverability. For example, Zone C successfully imports 5 GWh during scarcity. Cross-border trade of capacity obligations is 2.6 GW higher under *CM-FBMC* than under *CM-NTC* (Figure 4). In *CM-NTC*, bilateral transfer limits determined based on investment levels that materialise in *CM-FBMC* are $A \leftrightarrow B$: [740.3, -620] MW, $A \leftrightarrow C$: [574.3, -454] MW, and $B \leftrightarrow C$: [694, -694] MW. Zone A is the sole net exporter in both designs. Under *CM-NTC* it sells 2.8 GW of obligations, of which Zones B and C procure 1.5 and 1.3 GW. Under *CM-FBMC* it sells 5.3 GW, and both neighbours roughly double their foreign procurement, to 2.4 GW (Zone B) and 2.9 GW (Zone C), reducing within-zone investment by the same amounts. The higher cross-border capacity trade is shown to be physically deliverable during scarcity with Zone C importing 3.7 GWh/h, against 2.5 GWh/h under *CM-NTC*.

We show the network operator's congestion rents in the energy and capacity market in Table IV. In the reference Energy-only market (*EOM-ref*), high price differentials during scarcity drive congestion rents of 72.4 M€ 73.1 M€. During congestion, prices peak at 14,057 €/MWh 14,100 €/MWh in



(a) Capacity mix by zone and market design. "Com" indicates the aggregated result across all zones.



(b) Change in installed capacity w.r.t. *EOM-ref*.

Fig. 3: Capacity markets restore investment levels in peaking units but risk over-procurement without coordinated cross-border participation; our proposed flow-based coupling (*CM-FBMC*) limits total build compared to *CM-NTC*. *CM-NoCBP*, *CM-FBMC* and *CM-NTC* reallocate investments across zones compared to *CM-implicit* *EOM-cap* and capacity markets without explicit cross-border capacity remuneration (*CM-NoCBP*, *CM-ref*, *CM-implicit*) displace capacity away from cheaper Zone A relative to *EOM-ref*, while the explicit designs relocate capacity to Zone A.

Zone A and 16,608 €/MWh in Zone C. These spreads generate high revenue. Implementing a price cap (*EOM-cap*) drastically reduces these rents to 1.1 M€ 2.6 M€. The cap artificially reduces flows (due to under-investment across all zones) and the price spread between zones, as all zones hit the 4,000 €/MWh cap simultaneously. *CM-implicit* follows the same trend; revenue there falls to 0.1 M€. Finally, in *CM-NoCBP*, over-procurement of capacity leads to lower flows and high price convergence across zones. Congestion revenues collapse to 0.5 M€. The capacity market designs without explicit cross-border participation further erase these rents, albeit for different reasons. In *CM-NoCBP*, overprocurement suppresses scarcity pricing. With prices peaking at 560–650 €/MWh in all zones, spreads are minimal and congestion rents collapse to 0.5 M€. In *CM-ref* and *CM-implicit*, scarcity prices reach the cap; however, similar to *EOM-cap*, prices are near the cap in all zones at the same periods, lowering price spreads and congestion rents fall to 0.1 and 0.5 M€.

Explicitly remunerating cross-border participation restores the value of transmission through "capacity congestion rent". *CM-NTC* generates 64.6 M€, partially restoring the levels seen

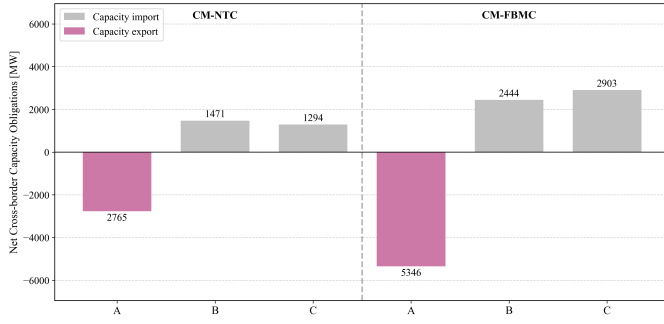


Fig. 4: Net cross-border capacity trade by zone.

TABLE IV: The network operator's congestion rents (M€/yr) across market designs.

Market	EOM-ref	EOM-cap	CM-NoCBP	CM-ref	CM-Implicit	CM-NTC	CM-FBMC
Energy	73.07	2.64	0.50	0.07	0.53	0.47	32.22
Capacity	0.00	0.00	0.00	0.00	0.00	39.08	46.46
Total revenues	73.07	2.64	0.50	0.07	0.53	39.55	78.68

in *EOM-ref*. *CM-FBMC* yields congestion rents of 87.1 M€, surpassing the reference market level *CM-NTC* generates 39.6 M€, most of it in the capacity market, restoring more than half of the *EOM-ref* level. *CM-FBMC* yields 78.7 M€, narrowly surpassing *EOM-ref*. Unlike *CM-NTC*, it earns a third of this in the energy market, as greater scarcity exports from Zone A congest interconnectors, causing price spreads to persist during scarcity.

C. System costs, social welfare and costs to consumers

1) *System Costs*: By design, the energy-only market *EOM-ref* presents the least-cost market design, with a total system cost of 15,398 M€/yr (Table V).¹ The introduction of a price cap (*EOM-cap*) suppresses investment costs by 870 M€/yr due to the "missing money" problem, resulting in a surge in the cost of unserved energy (2,103 M€)(1,963 M€/yr). Consequently, the *EOM-cap* is the least efficient design, incurring a cost increase of nearly 87% relative to the reference case.

Capacity markets eliminate ENS costs but increase investment costs. When compared to *EOM-ref*, *CM-NoCBP* increases investment costs by over 780 M€, or 5.2% 640 M€/yr, increasing total system cost by 4.2%. The *CM-NTC* design incurs a lower investment cost than the isolated case, but still 3.62% above the reference case. *CM-FBMC* brings the total costs down to 15,929 M€. This reduces the cost increase over *EOM-ref* to 3.45%. The *CM-Implicit* achieves the lowest cost among the capacity mechanisms (a 2.6% increase in cost). Recall, however, that ENS emerges, unlike the explicit designs, which guarantee adequacy. The *CM-Implicit* costs are thus highly sensitive to the cost of unserved energy. Coordinating the capacity requirement in *CM-ref* reduces the required capacity investments by 2.5 GW and lowers annualised investment costs by 181 M€/yr, limiting the total system-cost

¹Total system costs equal the sum of generation costs, investment costs and cost of lost load. The cost of lost load is the volume of ENS valued at 18,667 €/MWh; the average value of the demand that is randomly curtailed at the price cap of 4,000 €/MWh.

TABLE V: Served demand (GWh/yr), unserved energy (ENS – GWh/yr), system costs (M€/yr) and cost increase w.r.t. the reference *EOM-ref*.

Market Design	Demand Served	ENS Volume	Gen. Cost	Inv. Cost	ENS Cost	Total Cost	Increase (%)
EOM-ref	271,630	0	10,036	5,361	0	15,397	–
EOM-cap	271,532	105.14	10,015	4,491	1,963	16,468	6.96
CM-NoCBP	271,664	0	10,048	6,001	0	16,049	4.23
CM-ref	271,659	0.01	10,046	5,820	0.1	15,866	3.05
CM-Implicit	271,641	12.42	10,038	5,575	232	15,846	2.91
CM-NTC	271,659	<0.01	10,046	5,780	0	15,826	2.78
CM-FBMC	271,659	0.25	10,048	5,738	4.7	15,791	2.56

increase relative to *EOM-ref* to 3.1% (470 M€/yr). *CM-Implicit* reduces investment costs by a further 245 M€/yr, but this is largely offset by 232 M€/yr in unserved-energy costs, leaving total system costs only marginally below *CM-ref* and 2.9% above *EOM-ref*. Unserved energy under *CM-Implicit* is 12.4 GWh per year, equivalent to roughly 0.005% of annual demand (Table V). *CM-implicit* is sensitive to assumptions on the availability of imports during scarcity. Higher import assumptions would increase the volume of unserved energy, and because ENS costs rise faster than investment savings, the total system cost would increase. Without import deductions, the design collapses to *CM-ref*.

Explicit cross-border participation lowers total costs by relocating peaking capacity from zones with a high investment cost to lower-cost zones while keeping ENS below 0.25 GWh/yr. *CM-NTC* reduces investment costs by 40 M€/yr relative to *CM-ref* (0.7% of 5 820 M€/yr), and by 82 M€/yr (1.4%) under *CM-FBMC*. It removes the unserved energy of *CM-Implicit*, with ENS falling to 253 MWh/yr (below 10^{-4} % of served annual demand) in *CM-FBMC* and to 0.5 MWh/yr in *CM-NTC*. *CM-FBMC* is the least-cost capacity-market design, 258 M€/yr below *CM-NoCBP*, 55 M€/yr below *CM-Implicit*, and 35 M€/yr (or 0.22%) below a best-case NTC (*CM-NTC*), at a total annualised cost of 15,791 M€/yr.

2) *Social welfare*: We decompose the change in welfare w.r.t *EOM-ref* in changes in consumer surplus, producer surplus and congestion rents (Figure 5a).² *EOM-ref* leaves market-clearing price uncapped, enabling efficient self-rationing by consumers with a WTP below the clearing price, and providing a reference value for surplus generated in the market. The largest welfare loss is observed in *EOM-cap* (–864 M€/yr), the lowest in *CM-FBMC* (–171 M€/yr), consistent with what is obtained in (Section IV-C1). Under the price cap, consumers bear the loss (–982 M€/yr, driven by the unserved energy), while producers gain 189 M€/yr as inframarginal rents rise, particularly for renewables and baseload generators. Capacity markets reverse this trend. Producer surplus decreases as capacity revenues compensate investment cost but lower inframarginal rents. However, it significantly improves consumer surplus. Only *CM-NoCBP* (–125 M€/yr, due to over-procurement) and *CM-Implicit* (–17 M€/yr, energy-not-served) leave consumers worse off.

²To compute the change in consumer surplus, we re-evaluate energy market outcomes using the uncapped demand curve. The obtained consumer surplus is corrected with the cost of energy not served and capacity payments.

We distinguish two sources of welfare gains. First, coordinating the capacity demand through a joint scarcity assessment (*CM-NoCBP* to *CM-ref*) improves welfare by 172 M€/yr, with most accruing to consumers. Secondly, explicit cross-border participation adds 41 M€/yr under *CM-NTC* and 75 M€/yr under *CM-FBMC*, which materialises almost entirely as congestion revenue. *CM-FBMC* is the only design that fully restores the value of interconnection compared to *EOM-ref*.

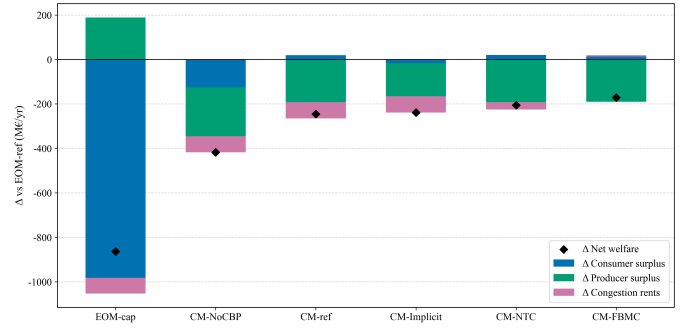
3) *Cost to consumers*: Figure 5b summarises the cost of electricity for consumers normalised by the demand served (€/MWh), decomposed into three components: the energy cost, cost of energy not served and capacity cost.

In *EOM-ref*, average costs reach 99.7 €/MWh in Zone B and 100.9 €/MWh in Zone C. Zone A sees a lower cost of 98.1 €/MWh. The system-wide average is 99.5 €/MWh. Neither a capacity market nor unserved energy exists in this reference case. Introducing a price cap increases consumer costs across all zones because unserved energy emerges. System-wide costs rise from 99.5 €/MWh in *EOM-ref* to 104.4 €/MWh in *EOM-cap*. The cap limits scarcity prices but triggers underinvestment and ENS costs. These effects are shown in the hatched areas of Figure 5b.

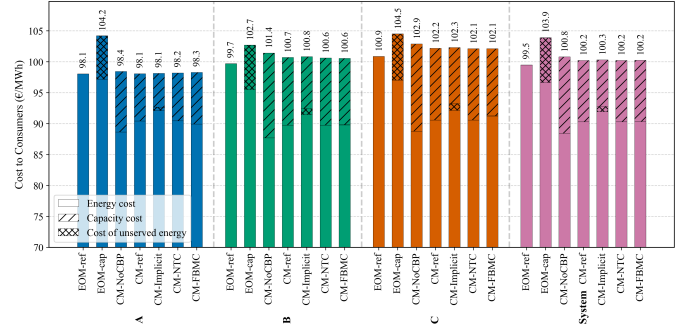
Capacity markets provide reliability at a premium. *CM-NoCBP* eliminates ENS; however, capacity payments represent 12.8% of the total cost. Capacity market prices in Zones A, B, and C clear at the costs of new entry: 60,000 €/MW, 70,000 €/MW, and 80,000 €/MW. The system-level average cost falls to 101.3 €/MWh, below the *EOM-cap* but still above the reference Energy-only market 12.3% of the total cost to consumers. Capacity market prices in Zones A, B, and C clear at or below the costs of a new build: 59,777 €/MW/yr, 69,623 €/MW/yr, and 80,000 €/MW/yr. The system-level average cost to consumers falls to 100.8 €/MWh, 3.1 €/MWh below the *EOM-cap* and 1.3 €/MWh above the reference Energy-only market. Coordinating capacity demands through a joint scarcity assessment in *CM-ref* lowers capacity prices by roughly 10,000 €/MW/yr in every zone (to 48,904, 58,965 and 69,886 €/MW/yr) and reduces capacity payments to 9.9% of the cost. Consumers see a net saving of 0.6 €/MWh.

In *CM-Implicit*, zones account for foreign support without paying for it explicitly, thus reducing system-wide average costs to 100.0 €/MWh. Zone C sees its average cost drop to 99.9 €/MWh, 1 €/MWh below *EOM-ref*. Capacity costs fall to 8.9% of the total, with capacity clearing prices ranging from 45,956 €/MW to 65,323 €/MW, reflecting lower demand. *CM-Implicit* goes further. Zones count on imports from foreign capacity without remunerating them. Capacity payments fall to 7.5% of the cost (38,254–59,246 €/MW/yr), but the imports do not materialise during scarcity events. Unserved energy adds 0.9 €/MWh and higher energy costs a further 1.6 €/MWh, leaving consumers worse off than under *CM-ref*.

Under *CM-FBMC*, consumers face the same average system cost as in *CM-NTC*. However, the flow-based approach reduces capacity prices in Zones B and C by 1,019 €/MW and 2,275 €/MW compared to the NTC case. As a result, capacity prices as a percentage of total costs decline from 12.2% to 12%. Explicit cross-border participation (*CM-NTC* and *CM-*



(a) Welfare decomposition showing the change w.r.t *EOM-ref* in consumer and producer surplus, congestion rents and the net welfare across market designs



(b) Average consumer cost by zone, decomposed into energy, capacity, and ENS (€/MWh). “System” represents the average cost of all zones and enables comparison of total system costs.

Fig. 5: We show the impact of cross-border capacity participation and coupling design on (a) Welfare decomposition [M€/yr] and (b) total average cost [€/MWh] in Zones A–C.

FBMC) leaves the system-wide cost at the *CM-ref* level, but redistributes it across zones. Zone A’s consumers face a higher capacity cost due to demand from neighbouring zones while Zones B and C pay 0.1 €/MWh less. Between both designs, *CM-FBMC* lowers capacity prices in Zones B and C by 390 and 4,076 €/MW/yr and raises Zone A’s by 3,876 €/MW/yr.

V. DISCUSSION

We propose a flow-based coupling of European capacity markets (*CM-FBMC*), representing a significant departure from current arrangements. Below, we reflect on its implications and on the challenges it may raise.

One of the most significant hurdles to regional capacity markets is presented by the unique needs of Member States. Political reluctance to relinquish control over national resource adequacy and capacity mix hinders progress toward regional integration of capacity markets. Our proposed approach avoids this conflict. It does so by separating long-term investment needs and short-term adequacy. Member States still determine long-term (multi-year) instruments to de-risk investment in capital-intensive assets. The coupled annual capacity markets then provide an opportunity to re-trade capacity obligations held by governments and facilitate investment in fast-to-deploy assets. Each MS may define its own long-term contract structure and parameters of the capacity demand curve (e.g. the demand curve may be defined “centrally” or “decentrally”) – only the participation in annual capacity auctions, the traded

contract (e.g., reliability options), and coincidental scarcity assessments need to be harmonised.

Even as we accommodate national preferences, the proposed mechanism continues to guide efficient regional investment. It reduces overprocurement by adopting a regional adequacy assessment that exploits the degree of non-simultaneity in scarcity events. A common view of scarcity scenarios across the system incentivises investment levels only up to the peak simultaneous scarcity across zones, rather than the sum of individual zonal peaks. As we demonstrate in our case study IV-A, efficiency gains in *CM-FBMC* are driven both by the mechanism's ability to exploit the degree of non-simultaneity in scarcity events within the network and by relocating investment to cheaper locations. ~~Two factors may diminish these efficiency gains. III-defined scarcity expectations can lead to inaccurate capacity procurement. Furthermore, if MS ignore price signals and promotes investment in the wrong locations (i.e., non-deliverable locations), trade in the annual, coupled capacity market remains limited.~~

Two factors may erode these efficiency gains. First, overly conservative regional scarcity expectations at Y-1 may drive inaccurate capacity procurement. Second, several frictions may impede the efficient transmission of price signals between annual, coupled capacity markets and the long-term, national instruments. For example, member states may strategically overprocure firm capacity to guarantee national autonomy or underprocure to free-ride on neighbouring resources. The coupled annual auction can disincentivise such behaviour. Member states are mandated to re-market all procured capacity under the national mechanisms into the coupled annual auction. Consequently, the cost of overprocurement is made explicit as the spread between the cost of the long-term contracts and the annual clearing price. Consider a Member State "A" that overprocures multi-year, long-term capacity contracts. Neighbouring zones then benefit from purchasing reliability obligations from "A". If transmission constraints in the coupled capacity auction become binding, capacity prices in zone "A" drop below the cost of some of the long-term contracts held by Member State "A". This makes the cost of overprocurement explicit. Member states may, however, choose to absorb the cost, for example, as the price for autonomy, and if all member states systematically overprocure in the first layer, the outcome becomes similar to that of the *CM-NoCBP*.

~~Nonetheless~~ Compared to current approaches to cross-border participation in capacity markets, our design resolves the "circular dependency" inherent in both *CM-NTC* and *CM-Implicit* approaches. Both rely on ex-ante parameterisations – either in setting Maximum Entry Capacities (*CM-NTC*) or to reduce local capacity targets (*CM-Implicit*). The implicit approach is particularly prone to a "chicken-and-egg" dilemma: it deducts expected imports from local demand without providing remuneration to incentivise investment in the capacity that enables these imports. As such, the assumed surplus may be crowded out in the long run. Our results show this fragility through the re-emergence of unserved energy in the *CM-Implicit* scenario, confirming that reliance on non-remunerated cross-border import may undermine resource adequacy. An increase in the valuation of reliability would sharply penalise

the unserved energy risk inherent in the implicit approach, reducing its economic efficiency.

The *CM-FBMC* addresses this issue structurally. By using a detailed grid model to clear capacity offers, it ensures that cleared capacity is deliverable w.r.t. network constraints under a set of defined scarcity scenarios. Flow-based domains internalise loop flows, critical network elements, and contingencies, thereby capturing grid physics more accurately than NTC-based "maximum-entry-capacity" caps. FBMC thus requires less conservative assumptions, unleashing more cross-border trade of firm capacity obligations. Additionally, the flow-based market coupling of annual capacity markets provides explicit price signals for reliability. Zones with adequacy issues will clear their capacity markets at higher prices, providing incentives for cross-border trade and investment in non-capital-intensive assets.

~~Lastly, we discuss limitations of our work, and potential implementation challenges. As TSOs already generate year-ahead scarcity scenarios and flow-based domains for the European Resource Adequacy Assessment (ERAA), extending these processes to produce harmonised, year-ahead anticipated scarcity scenarios (flow-based domains, availability of generation assets and demand) is the next logical step. Such a role may be assumed by Europe's Regional Coordination Centres (RCC). However, this approach could increase computational effort, as it requires accounting for grid uncertainties and the N-1 security criterion when computing flow-based domains. Furthermore, because flow-based market coupling has been criticised for opacity, its application in capacity markets must prioritise transparency. Lastly, we discuss limitations of our work, and potential implementation challenges.~~ As the proposed design is yet to be implemented in any known real-world setup, we elaborate on the limitations of our work and potential implementation challenges. Our equilibrium model assumes perfect competition and risk-neutral agents in both electricity and capacity markets. Extending the analysis to investigate the effect of risk aversion and market power on cross-border capacity market design constitutes a relevant avenue for future research. Additionally, we certify capacity deliverability under a stylised finite scenario set $\mathcal{S}$. Operationalising the market design would require a robust set of scarcity scenarios. As TSOs already generate year-ahead scarcity scenarios and flow-based domains for the European Resource Adequacy Assessment (ERAA), extending these processes to produce harmonised, year-ahead anticipated scarcity scenarios (flow-based domains, availability of generation assets and demand) is the next logical step. Such a role may be assumed by Europe's Regional Coordination Centres (RCC). Note, however, that our network representation only considers state- N and ignores contingencies during scarcity events. Real-world implementation would account for grid uncertainties through the $N - k$ security criterion. The combination of a scarcity scenario set and $N - k$ contingencies may considerably increase the computational effort required for market clearing. Lastly, because flow-based market coupling has been criticised for opacity, its application in capacity markets must prioritise transparency.

VI. CONCLUSION

Capacity markets are becoming a structural element of the European electricity market, but lack harmonisation and cross-border participation necessary for regional efficiency. Our study addresses this issue ~~through a two-tiered procedure~~ by introducing a coordinating layer for national, long-term capacity remuneration mechanisms. We propose linking annual, zonal capacity auctions using the flow-based market coupling methodology, in which long-term contracts held by governments can be re-marketed and new assets with short lead times can be attracted. As such, the coupled annual capacity market ~~then allows for compatibility~~ is compatible with long-term instruments issued by Member States to mitigate investment risk.

By endogenously modelling network flows in anticipated scarcity scenarios, our approach eliminates the circular dependencies between import assumptions and available capacity in implicit cross-border participation, and between maximum entry capacities and available capacity in current explicit cross-border participation methods, and guarantees the delivery of contracted capacity during those scarcity events.

Our modelling results show that this design more accurately values the contribution of interconnection and ~~yields efficiency~~ ~~gains~~ ~~relocates capacity to cheaper locations under deliverability constraints~~. Moreover, our design addresses challenges posed by fragmented national capacity mechanisms while respecting the heterogeneous adequacy requirements of Member States and their sovereign right to steer their capacity mix.

Future work should focus on detailing the design of the annual auctions and the capacity product to be traded. Key features include the degree of harmonization required across member states and the valuation of resources that have non-symmetric contributions to meeting demand during scarcity situations across zones, such as energy-constrained assets.

APPENDIX I

The equilibrium is computed using an exchange-form consensus ADMM implemented in Julia, in which each agent's convex quadratic subproblem is solved in parallel with Gurobi. Each iteration solves Problems (1)–(4) in parallel, given the prices, then updates $\lambda_{z,t}^e$ and λ_z^{cm} as duals of the zonal imbalances. The algorithm terminates when, for every zone and for both markets (electricity and capacity), the primal and dual residuals fall below an absolute tolerance $\epsilon = 0.1$ (in MW for primal residuals; in €/MWh and €/MW-year for EOM and CM dual residuals). The penalty parameter ρ is adapted per zone and market via residual balancing following Boyd et al [29]. Converged runs typically require 460–2,500 iterations (13–38 seconds of wall time), with each iteration taking residuals to decay steadily (Table VI). All runs are cold-started, with prices, quantities, and penalty terms initialised at zero.

Because each agent solves a convex quadratic problem, a converged ADMM solution satisfies the conditions for competitive equilibrium. As degeneracy exists in some variables (e.g., within-zone investment allocation, network operator capacity allocation, and unserved energy when price cap hits

multiple price zones simultaneously), solutions may be non-unique. However, these multiple equilibria differ only in nodal quantities (except for unserved energy, which we treat ex-post, see Appendix II); aggregate installed capacity and zonal outcomes remain consistent across repeated runs.

TABLE VI: Final ADMM residuals and iteration counts across market designs. Reported residuals are the maximum value over all zones. The convergence tolerance is $\epsilon = 0.1$

Design	Iterations	$\max r_E^p$	$\max r_E^d$	$\max r_{CM}^p$	$\max r_{CM}^d$
EOM-ref	1,504	0.012	0.100	0	0
EOM-cap	462	0.017	0.100	0	0
CM-NoCBP	498	0.047	0.099	0.031	0.077
CM-ref	845	0.084	0.038	0.092	0.024
CM-Implicit	2,223	0.027	0.100	0.017	0.029
CM-NTC	830	0.038	0.097	0.057	0.099
CM-FBMC	733	0.031	0.047	0.025	0.088

APPENDIX II

During joint-scarcity hours, defined as hours in which at least two zones experience $\text{ens}_{z,t} > 0$, these zones face the common price cap, so market clearing may not uniquely determine the zonal allocation of ENS. We resolve this multiplicity ex-post using Eq. (8), where $\text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}$ is the shortfall corrected for the zone's net position in the capacity market p_z^{CO} . A positive net position indicates an importing zone.

$$\text{ens}_{z,t} = s_t \cdot \max \{0, \text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}\}, \quad (8a)$$

$$s_t = \frac{\sum_z \text{ens}_{z,t}}{\sum_z \max \{0, \text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}\}}, \quad (8b)$$

$$\tilde{p}_{z,t}^E = p_{z,t}^E + \text{ens}_{z,t} - \text{ens}_{z,t}. \quad (8c)$$

Since $\sum_z p_{z,t}^E = \sum_z p_z^{\text{CO}} = 0$, $s_t \leq 1$, and total ENS is conserved by construction. Surplus zones receive no curtailment, while deficit zones share total ENS in proportion to their shortfalls. If no zone is in surplus, $s_t = 1$ and $\tilde{p}_{z,t}^E = p_z^{\text{CO}}$. The reallocation affects only the zonal distribution of ENS and net positions; all other outcomes remain unchanged.

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Editor

The reviewers have expressed several concerns in order to improve the quality of the paper, which has been positively assessed. The authors should pay attention to modeling: curtailment methodology, capacity levels, reliability, welfare decomposition, mismatch between design and model, demand curve calibration, numerical results, etc. Other minor issues need to be addressed as well.

Authors' response: Thank you for your detailed review of our work and for managing the review process. We have carefully revised our manuscript in light of the reviewers' comments, which has clearly strengthened the paper. In what follows, we list all comments, our responses, as well as an overview of the changes in the manuscript. All changes in the manuscript are marked in colored text and are reproduced in this document for the reviewers' convenience.

The most important changes include:

- We addressed the mismatch between the proposed design and the presented model (Abstract, Section I and II);
- We introduced a welfare decomposition, comparing designs at similar demand levels (Section IV);
- We included additional details on our numerical results (Appendix I);
- We have included a curtailment methodology (Appendix II)
- We provide a link to a GitHub repository with all our code and input data, ensuring reproducibility.

With these changes, we believe we have demonstrated the methodological soundness and conceptual consistency of our manuscript.

We want to thank you again, and we hope that you find the revised manuscript suitable for publication in the IEEE Transactions on Energy Market, Policy and Regulation.

Yours sincerely,

K. Adekola, on behalf of the authors

Reviewer 1

The paper provides a very useful contribution to the discussion on capacity markets cross-border coordination in the EU. It proposes a concrete, two-tier solution, through capacity markets flow-based coupling. The advantages of the proposed solution is that it is compatible with the allocation of long-term contracts by Member States to steer domestic investments (in the Y-4 auction), and potential differences in the details of national capacity markets designs. I have however several questions on the methodology, assumptions, and practical considerations.

Authors' response: Thank you for your detailed review of our work and your appreciation of our work. Below, we provide a point-by-point response to your comments and an overview of the changes in the manuscript, which have clearly strengthened the paper. All changes in the manuscript are marked in purple.

1. Curtailment allocation in common scarcity:

The authors should precise how curtailment is allocated during common scarcity scenarios. They assume that all bidding zones have the same VOLL or price cap. When the latter is reached in more than one bidding zone (common scarcity periods), then an explicit rule must be applied to decide on the cross-border flows. Indeed, the welfare function in the clearing algorithm is insensitive to delivering in A or B. This however affects the allocation of ens across the region. In theory, the day-ahead algorithm EUPHEMIA applies “curtailment sharing” which equalizes curtailment ratios. In practice, one could assume countries restricting exports to prioritize domestic consumers. Another solution would be to define cross-border flows based on the amount of capacity each country has procured, be it domestic or foreign. This is a crucial assumption as it eventually defines whether countries actually get the capacity they paid for.

Authors' response: Thank you for pointing out this issue. Indeed, without an explicit rule on splitting ENS , multiple equilibria could emerge. We have adopted your third suggestion: an ex-post correction that redefines cross-border flows based on procured capacity.

In joint-scarcity hours (i.e., hours in which at least two zones have $ens_{z,t} > 0$), each zone's curtailment is recomputed to reflect its shortfall after accounting for its own dispatched generation and the foreign capacity it has procured. Under *EOM-cap* and *CM-implicit*, there are no cross-border allocations, so each zone's curtailment equals its generation shortfall. In our updated results, changes in $ens_{z,t}$ occur only in these two scenarios (*pricecap* and *implicit*), as only in these cases, joint-scarcity hours emerge. The rule is applied to all designs, but their results remain unchanged. The changes are reflected in Figure (5b) and in the text reproduced below:

Changes to the paper (p. 6, right column in the revised manuscript): ... where $\bar{\lambda}_{z,t}^e$ is the administrative price cap. The complementarity condition describes that when a zonal balance is attained, the market clears between 0 and $\bar{\lambda}_{z,t}^e$. The price is zero with excess supply, or at the cap with excess demand. **When multiple zones clear at the cap simultaneously, multiple, equivalent distributions of demand curtailment may exist ($\text{ens}_{z,t}$ in Eq. 2c). We introduce an ex-post curtailment-allocation rule to circumvent this multiplicity of equilibria, detailed in Appendix II. In every hour in which at least two zones clear at the cap, each zone's demand curtailment is recomputed to the shortfall it faces when it can rely only on generation capacity within the zone and on the cross-border capacity it has procured. Zones with surplus capacity export to other zones in proportion to the deficit, so that the system ENS is conserved (Appendix II).**

(p. 8, right column in the revised manuscript): On the consumer side, the volume of elastic demand that can express its willingness to pay in the market decreases from 20% to 4% of the reference level. The combination of less firm capacity and less price-responsive demand visible in the market results in Energy Not Served (ENS) **reaches 41.2 GWh, 30 GWh, and 33 GWh of 38.9 GWh, 28.9 GWh, and 37.3 GWh in zones A, B, and C.**

(Appendix II in the revised manuscript): During joint-scarcity hours, defined as hours in which at least two zones experience $\text{ens}_{z,t} > 0$, these zones face the common price cap, so market clearing may not uniquely determine the zonal allocation of ENS. We resolve this multiplicity ex-post using Eq. (1), where $\text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}$ is the shortfall corrected for the zone's net position in the capacity market p_z^{CO} . A positive net position indicates an importing zone.

$$\text{ens}_{z,t} = s_t \cdot \max \{0, \text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}\}, \quad (1a)$$

$$s_t = \frac{\sum_z \text{ens}_{z,t}}{\sum_z \max \{0, \text{ens}_{z,t} + p_{z,t}^E - p_z^{\text{CO}}\}}, \quad (1b)$$

$$\tilde{p}_{z,t}^E = p_{z,t}^E + \text{ens}_{z,t} - \text{ens}_{z,t}. \quad (1c)$$

Since $\sum_z p_{z,t}^E = \sum_z p_z^{\text{CO}} = 0$, $s_t \leq 1$, and total ENS is conserved by construction. Surplus zones receive no curtailment, while deficit zones share total ENS in proportion to their shortfalls. If no zone is in surplus, $s_t = 1$ and $\tilde{p}_{z,t}^E = p_z^{\text{CO}}$. The reallocation affects only the zonal distribution of ENS and net positions; all other outcomes remain unchanged.

2. Why no ENS in EOM-ref? / Do CMs overprocure vs. optimal?

I have questions about the capacity levels and targets. In theory, an optimal capacity market level strikes the exact right balance between the costs of additional ens and the cost of building more generation capacity. Could the authors explain why there is no ens in the energy-only scenario? Is this a result of the assumptions on demand elasticity? Then, the capacity markets could be implemented with the objective of strictly achieving this optimal balance (meeting the national reliability standards). In the current results, the CM-FBMC and NTC scenarios both appear to overprocure, as the implicit scenario eventually comes at a lower cost.

Authors' response: Thank you for the questions. We structure our response in three points:

- Why is there no ENS in EOM-ref? Yes, unserved energy does not arise in *EOM-ref* because demand is reduced along the elastic section of the demand curve. In the idealised *EOM-ref*, prices may rise to the willingness-to-pay level of 20,000 €/MWh, at which the entire price-responsive share of demand (20% of the reference level) would *voluntarily* curtail. In our case study, this point is not reached at equilibrium as scarcity prices peak at 16,608 €/MWh. Thus, all demand reduction falls on consumers with WTP below the clearing price. We have now clearly stated this explanation under EOM-ref in Section (IV-A).
- Calibrating the capacity markets to the optimal balance. The capacity demand is set exogenously based on peak residual demand at the price cap (96% of reference demand), adjusted per design to account for cross-border capacity participation (see below). We do not claim that this level is optimal. It, however, ensures consistency across designs so that the differences between designs reflect the treatment of cross-border participation rather than the calibration of the capacity demand curve.

We agree that a capacity market can be calibrated to procure exactly the volume implied by a reliability standard. Such calibration is, however, not trivial in an interconnected system because the required capacity cannot be determined independently for each zone. A zone's contribution to regional adequacy depends not only on its own peak demand, but also on the coincidence of scarcity across neighbouring zones and on the capacity that can be delivered through the network during those events. Calibrating the capacity demand independently to procure the volume implied by each zone's reliability standard, therefore, overstates the regional requirement when scarcity is non-coincident. Second, under *EOM-ref*, high scarcity prices induce efficient self-rationing by consumers, reducing the peak residual demand that must be served. Once the energy-market price is capped, part of this demand response disappears, so meeting the same reliability objective without involuntary curtailment requires more firm capacity than in the uncapped energy-only equilibrium (see Fig. 1). Indeed, this explains why

all capacity-market scenarios in our case study install more capacity than *EOM-ref*, even when ENS is eliminated. Under the capacity market designs the energy price is capped at 4,000 €/MWh, and consumers with a WTP above the cap—96% of reference demand: the 80% inelastic block valued at 20,000 €/MWh and the 16% of the price-responsive block valued between 4,000 and 20,000 €/MWh—can no longer self-ration in response to market prices. The market clearing hence cannot distinguish a consumer valued at 19,900 €/MWh from one valued at 4,100 €/MWh; both observe the price cap, and any shortfall is rationed irrespective of value. Consequently, a capacity market parametrised at the *EOM-ref* capacity level would *underprocure* capacity, as in the electricity market clearing highly valued demand will be unserved once scarcity pricing is muted. In other words, the system requires more capacity to maintain the same reliability and efficiency.

- Correcting the capacity demand for cross-border participation. To ensure fair comparison among the different cases, we have revised the way we parametrise our capacity demand and introduced a new case to measure the impact of the joint scarcity assessment (see also comment 3 from reviewer 3). Our baseline capacity demand curve is set to the peak residual demand of the inelastic block in the scenarios considering a price cap (i.e., at 96% of reference demand). Note that this leads to a lower capacity demand compared to our initial submission. This baseline is applied only to *CM-NoCBP*. For the other capacity market scenarios, we perform a joint scarcity assessment. The results of the joint scarcity assessment are used to lower the capacity demand in the newly introduced scenario *CM-ref* compared to the *CM-NoCBP* scenario, measuring the impact of the joint scarcity assessment without allowing cross-border participation. We then use the same capacity demand derived from the joint scarcity assessment and apply the respective methodologies for explicit cross-border participation methods *CM-NTC*, *CM-FBMC* and the implicit cross-border participation *CM-Implicit*. Consequently, *CM-Implicit*'s unserved energy reflects its inefficiency, and *CM-Implicit* performs worse than the explicit designs as a result.

In Section (IV) of the revised paper, we have now discussed the parametrisation of capacity demand in greater detail for each of the studied cases.

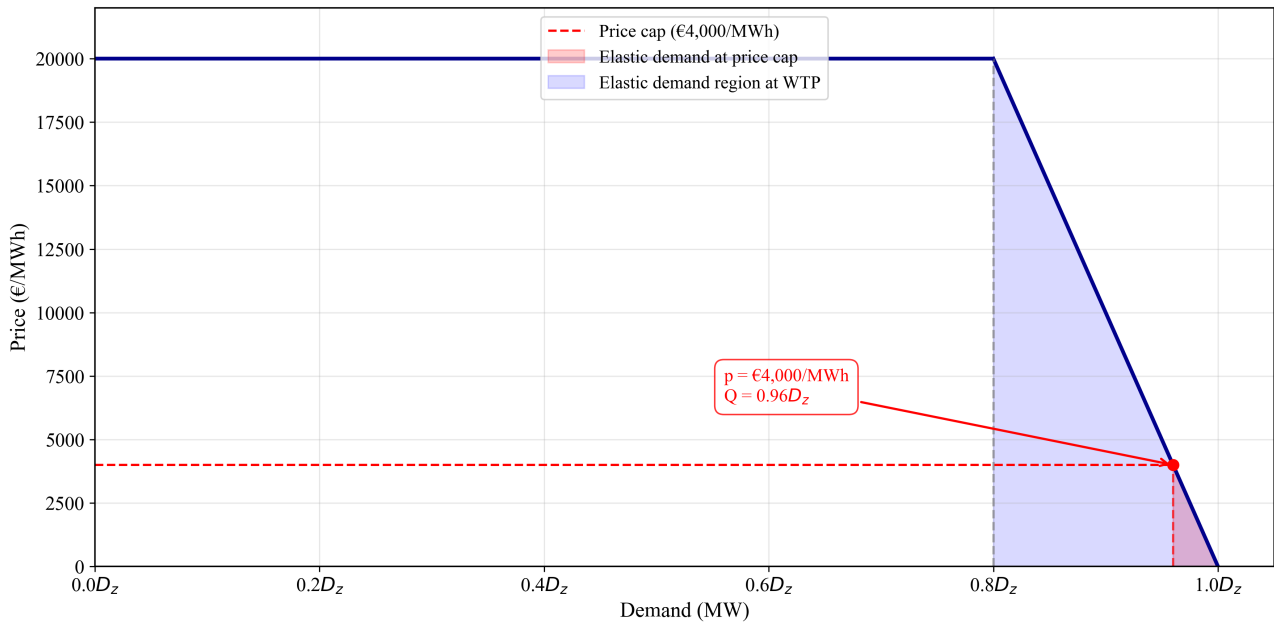


Fig. 1: Inverse demand curve used to represent consumers' behaviour in the model.

Changes to the paper (p. 7, right column in the revised manuscript): We study ~~six~~seven distinct market design scenarios (Table II)....

- 1) *EOM-ref*: The energy-only market without a price cap. All generators recover their costs from the energy market, assuming no market imperfections beyond the lack of a locational component in the electricity price in zonal markets.
- 2) *EOM-cap*: An energy-only market with a price cap, capturing the "missing money" problem, which leads to underinvestment in firm capacity and energy not served (ENS).
- 3) *CM-NoCBP*: A capacity market without cross-border participation. The zonal capacity demand is set to the peak residual demand ~~valued at the price cap in each zone, i.e. the maximum residual demand served at 4,000 €/MWh.~~ Zones do not explicitly or implicitly account for the availability of imports in their national capacity markets, ~~nor carry out a joint scarcity assessment.~~
- 4) *CM-ref*: A reference capacity market scenario whose capacity demands result from a joint scarcity assessment (Figure 2 (Right)), but without cross-border participation. The joint scarcity assessment considers three zonal scarcity scenarios, representing each zone's individual peak demand. The set of scarcity scenarios considers the residual demand valued at the price cap of 4,000 €/MWh across all zones at each time step, any zone experiences ENS under *EOM-cap*. The joint scarcity assessment sets the system-wide capacity demand to the maximum coincident peak residual demand. This capacity demand is distributed over the zones proportionally to the average share of the zone's residual demand to the overall system residual demand during scarcity situations. This scenario allows isolating the effect of coordinating capacity requirements from that of cross-border participation. The resulting capacity demand forms the basis for all scenarios considering cross-border participation.
- 5) *CM-Implicit*: A capacity market with implicit cross-border participation. ~~To determine capacity demand, expected scarcity imports from neighbouring zones are subtracted from the peak residual demand in each zone. This creates a circular dependency, as imports depend on available capacity. We use the investment outcome of the CM-FBMC to determine available imports, thereby enabling comparability with other scenarios.~~ Each zone's capacity demand is that determined in the joint scarcity assessment minus the average imports it receives in the scarcity scenarios under scenario *CM-ref*.
- 6) *CM-NTC*: A capacity market with explicit cross-border participation limited by a pre-defined maximum entry capacity (MEC) on each border. ~~We use capacity demands from scenario CM-ref.~~ For comparability, we ~~set these~~determine best-case MECs using the investment decisions from the *CM-FBMC* scenario, following the procedure outlined in Section III-D2. ~~Thus, results from this scenario should be viewed as a best-case NTC, since we exclude some of the investment distortions that would arise from an endogenously determined MEC.~~ As a consequence, results ~~reflect only the difference~~differ from *CM-FBMC* only in the treatment of cross-border capacity trade.
- 7) *CM-FBMC*: ~~A~~The proposed flow-based coupling of ~~the capacity market~~capacity markets. We use capacity demands obtained in scenarios *CM-ref*. The participation of foreign capacity in each zone's capacity market is constrained by the permissible flows across the system under predefined scarcity scenarios (Figure 2). ~~We use capacity demands obtained in scenarios CM-ref.~~ Scarcity events depict the time steps in which ENS occurs under *EOM-cap*. We represent projected scarcity events as (i) "simultaneous scarcity", where all zones demand 95 percent of their peak residual demand simultaneously, and (ii) "zonal scarcity", where the zone experiencing critical scarcity demands 100 per cent of its peak residual demand, while neighbouring zones each simultaneously demand 90 per cent of their peak residual demand. The scarcity scenarios are identical to those used in the joint scarcity assessment.

Changes to the paper (p. 8, left column in the revised manuscript): In the reference Energy-Only Market (*EOM-ref*), investment is driven by scarcity pricing. In this idealised *EOM-ref*, prices may rise to the WTP of 20,000 €/MWh, at which the entire price-responsive share of demand (20% of the reference level) would voluntarily curtail. However, in equilibrium scarcity prices reach at most 16,608 €/MWh. Realised peak demand falls to 17.5, 13.7 and 15 GW in zones A, B and C, from reference peaks of 19, 16 and 18 GW. Implementing a price cap (*EOM-cap*) ~~limits these scarcity rents~~ truncates prices at 4,000 €/MWh, with consequences on both consumer demand and generation capacity investments. The forgone scarcity rents ~~this results in~~ cause a decline in investment, primarily in peaking units (Fig. 3). Total capacity falls by 13.5 GW relative to the reference energy-only market. On the consumer side, the volume of elastic demand that can express its willingness to pay in the market decreases from 20% to 4% of the reference level. The combination of less firm capacity and less price-responsive demand visible in the market results in Energy Not Served ...

Changes to the paper (p. 9, left column in the revised manuscript): Notably, all capacity market designs result in a total capacity that exceeds the *EOM-ref*. While *EOM-cap* under-invests by 13.5 GW, capacity mechanisms add ~~4-10 GW~~1.9-8 GW above *EOM-ref* level. This arises from feedback between electricity prices and the elastic electricity demand. In *EOM-ref*, high scarcity prices reduce electricity consumption through self-rationing by consumers whose WTP is below the market-clearing price, and consequently, the peak residual demand served is reduced to 44 GW. ~~Capacity markets suppress these price spikes (Section IV-B)) and the associated demand elasticity, with the peak residual demand served increasing to 48-50 GW~~The price cap truncates electricity prices (Section IV-B)), removing the efficient self-rationing of consumers with demand valued above the price cap (4,000 €/MWh). ~~Consequently,~~ To avoid involuntary curtailment of demand, more firm capacity is required ~~to guarantee the same level of reliability.~~ As a result, the peak residual demand served rises from 44 GW in *EOM-ref* to 49-50 GW in the scenarios considering a capacity market.

3. Reliability options — how it was implemented, strike price, reference price?

The authors model reliability options. Could they precise how this was implemented in the model? In particular, what is the strike price assumed, and what is the reference price for a given generator or country? How do these assumptions affect the results? I understand in the Y-1 auction, the reliability options are traded between countries, not market participants? This is an interesting proposal, where financial contracts may provide appropriate incentives for MS to procure the right amount of capacity. It is however unclear to me how would the transfer of these new price signals from/to market participants work in practice.

Authors' response: Thank you for the questions. We will address your remarks point by point below:

- We did not explicitly model reliability options. Cleared capacity in the annual auction is remunerated at an annual payment λ_z^{cm} . In the revised manuscript, we have clarified the scope of the paper in the abstract and in Sections I and II. Please see our response to Comment 5 below for the changes to the manuscript.
- In our proposal, the reliability options would indeed be traded between countries. As the holders of the multi-year contracts, member states would be obligated to re-market the contracted volumes as annual products. Note, however, that the annual auction is not restricted to governments. Uncontracted resources and assets with short lead times could offer their firm capacity directly into the coupled annual auction (Fig. 1).
- The coupled annual auction produces a zonal annual clearing price which reflects the marginal value of an additional MW of firm, deliverable capacity in each zone to regional adequacy as perceived one year ahead of delivery. This provides information to national governments and market participants about an asset's value in terms of its contribution to regional generation system adequacy. We illustrate the informational content of the annual clearing price to market participants under 3 possible outcomes.
 - Structural overcapacity in all zones: Assuming countries still define capacity demands as described above (see our response to comment 2), the zonal capacity price clears below the cost of the long-term contracts held by governments. The cost of overprocurement in long-term contracts is made explicit. No new entry is cleared in the capacity market, and no demand for cross-border capacity obligations, since every zone holds an excess supply of long-term contracts (autarkic scenario).
 - Structural shortage in all zones: New entrants (fast-to-build assets such as storage and peaking units) not previously contracted under national reliability contracts are selected in the auction. Zonal clearing prices in the annual capacity auction rise to at least the net cost of new entry in each zone, triggering new build in every zone. Cross-border trade of capacity obligations becomes significant (subject to interconnection limits), reducing the amount of capacity that must be built in the most expensive zones and consequently lowering the total cost of capacity procurement in the system. This scenario corresponds to our greenfield modelling assumption and results shown in Section IV-C.
 - Mixed: zones holding deliverable surplus capacity become net exporters and clear above the capacity price that would originate from a national auction without cross-border participation, as their capacity contributes to adequacy elsewhere. Zones whose capacity demand is met fully by procurement of foreign capacity clear below their zonal cost of new entry, as no new capacity is needed to meet the reliability target.

The annual clearing price therefore performs three functions: it corrects, at annual frequency, long-term positions taken under deep uncertainty; it provides a locational signal, directing investment toward zones where firm, deliverable capacity is cheapest to provide; and it guides the volume of subsequent long-term procurement with regional information.

- The price signal transfers to market participants directly through their participation in annual capacity market clearing, where high prices attract new fast-to-build assets. In the multi-year national markets, market participants receive price signals indirectly via the member states' response to the outcome of the annual auction.

4. NTC benchmark overestimates real MEC

I have concerns that the “NTC” approach used may be an overestimation of the MEC used in the current EU framework. The MEC is the average statistical contribution of imports during domestic scarcity periods based on ERAA (or more often, national) scenarios. It factors in the interconnectivity limits, but most importantly the availability of foreign resources (hence why it is low or zero on many borders currently). In the proposed approach, the authors could model MEC using, for example, the EOM/an other scenario as a proxy for ERAA, setting the MEC as the average imports observed during the domestic scarcity periods. Could the author either align their model with the current MEC methodology, or explain more thoroughly their choice to opt for a different approach?

Authors' response: Thank you for this observation. We agree that our approach overestimates MEC relative to those in practice. However, our intent is not to replicate the current practice, as the degree of “conservatism” would directly affect how much worse the NTC approach is compared to the FBMC scenario. This would make it difficult to attribute

the inefficiencies to conservative assumptions that lead to lower NTC values or to a less permissive representation of the network in the coupled capacity market clearing. To avoid this, we use a "best-case" MEC to avoid arbitrary derating.

In the revised manuscript, we clearly outline that the *CM-NTC* is a best-case maximum entry capacity on each border. The changes in the manuscript are reproduced below.

Changes to the paper (p. 5, right column in the revised manuscript): The cross-border capacity market using NTC serves as ~~our~~ **best-case** proxy for what is observed in European capacity markets [1]... It is thus a theoretical upper bound to the performance of NTC-coupled markets w.r.t flow-based market coupling.

Changes to the paper (p. 6, left column in the revised manuscript): The resulting set, $\mathcal{E}^{\text{ATC}}$, gives the maximum ~~feasible net transfer capacity on each border while still respecting all network constraints~~ net transfer capacity on each border that still yields feasible network flows during the scarcity scenarios considered in the determination of the flow-based domain (Section III-D1).

Changes to the paper (p. 8, left column in the revised manuscript): 6) *CM-NTC*: ...For comparability, we ~~set these~~ **determine best-case** MECs using the investment decisions from the *CM-FBMC* scenario, following the procedure outlined in Section III-D2. Thus, results from this scenario should be viewed as a best-case NTC, since we exclude some of the investment distortions that would arise from an endogenously determined MEC. As a consequence, results ~~reflect only the difference~~ differ from *CM-FBMC* only in the treatment of cross-border capacity trade.

5. Assumption of Perfect price propagation from Y-1 to Y-4 auction

The authors assume a perfect propagation of capacity prices from the Y-1 to the Y-4 auction. Could the authors specify what justifies this assumption? In practice, capacity prices tend to be much lower in Y-1 auctions (see, for example the Belgian capacity market average prices) than in Y-4 auctions.

Authors' response: Thank you for your comment. In response to comment 1 of Reviewer 2, the revised manuscript now presents our proposal as a second layer required to coordinate multi-year national capacity mechanisms rather than a full-fledged two-tier mechanism. Our contribution is the design of the proposed second layer i.e, the design of its auction and its interface with the investment layer.

The assumption of perfect price propagation from the second tier (Y-1) to the first (Y-4) is now presented as a scoping decision/assumption. The assumption is implied by the greenfield modelling choice. The modelling choice enables us to isolate the second layer by treating the first layer as perfectly responsive to the decision made in the annual layer.

For the sake of brevity, we don't list all changes to the paper related to this reframing below, but focus on the statement related to price propagation. The other changes to the paper are listed in the response to Comment 1 of Reviewer 2.

Changes to the paper (p. 4, left column): In Section III and IV, we zoom in on the second layer: a coupled annual capacity market, and ~~abstract from the first layer~~, assuming a perfect propagation of price and investment signals to the first layer. Thus, our results do not capture the effect of distortions that arise within the first layer, such as overprocurement, cost-allocation and free-riding. We reflect on the impact of these limitations on our analysis in Section V. We capture our proposed design in a mixed complementarity problem and apply it to a case study, demonstrating how this can facilitate the coupling of European capacity markets.

6. Overprocurement incentives / how is Y-4 demand defined / do ROs discourage overprocurement?

One main concern with having national capacity markets is that Member States will overprocure domestic capacity to have more control and autonomy, even if it comes at a higher cost from the regional (and eventually national) point of view. How exactly are capacity demands defined in the Y-4 auction? More precisely, how are imports factored in already in the Y-4 auction without coupling? In a pessimistic, but maybe realistic scenario: each country could overprocure resulting in high domestic capacity prices in Y-4. Then, there would be excess of capacity available in the Y-1 auction, prices would be very low (e.g. we saw CM prices reaching zero in France due to overcapacity), and the expected benefits of the coupled auction would be quite limited. Could the authors explain how the proposed approach would limit the risk of overprocurement? Does the re-selling of reliability options bring any additional economic incentives for countries not to overprocure?

Authors' response: Thank you for the comments and questions. We answer them point-by-point below:

- We do not explicitly model Y-4, and we have clarified our modelled scope better in the revised manuscript. In our conceptual design (Section II), we leave the definition of capacity demand in the long-term (first-layer) auction as a national choice, in line with current practice and the idea that this is a national prerogative in the EU.

- Long-term procurement (Y-4) is undertaken amid high structural uncertainty (e.g., in demand growth and transmission capacity); as such, assumptions cross-border availability of capacity in Y-4 would remain highly uncertain. We focus on the coordination at the annual layer, where the degree of structural uncertainty is relatively limited. The annual capacity auction proposed in our paper then serves as a coordinating layer, providing a reference price signal for subsequent rounds of national auctions. Note that our approach removes the circular dependency between assumed available firm capacity and imports across the border in current cross-border participation mechanisms.
- The annual, coupled capacity market can limit the incentive to overprocure, as it increases liquidity in the Y-1 auction. It hence becomes easier to contract capacity in the Y-1 auction. In addition, by mandating that all procured capacity in long-term contracts is remarketed as annual products, the annual clearing provides an explicit reference price signal. If there is region-wide overprocurement, then the annual auction reveals this explicitly. We note that the price signal is informational, and some MS may choose to pay the price for autonomy.

We discuss this issue in Section V of the revised paper:

Changes to the paper (p. 11, right column in the revised manuscript): The coupled annual auction can disincentivise such behaviour. Member states are mandated to re-market all procured capacity under the national mechanisms into the coupled annual auction. Consequently, the cost of overprocurement is made explicit as the spread between the cost of the long-term contracts and the annual clearing price. Consider a Member State "A" that over-procures multi-year, long-term capacity contracts. Neighbouring zones then benefit from purchasing reliability obligations from "A". If transmission constraints in the coupled capacity auction become binding, capacity prices in zone "A" drop below the cost of some of the long-term contracts held by Member State "A". This makes the cost of overprocurement explicit. Member states may, however, choose to absorb the cost, for example, as the price for autonomy, and if all member states systematically overprocure in the first layer, the outcome becomes similar to that of the *CM-NoCBP*.

7. Which design elements must be harmonised?

The authors argue that the coupling mechanism proposed does not imply perfect harmonisation of national capacity market designs. In practice, differences in monitoring, penalty calculation, de-rating methodologies, are inherently factored in in participants' bids. One could expect this would affect the efficiency of a common clearing. Could the author explain which elements of capacity markets would need to be harmonized for an efficient coupling?

Authors' response: Thank you for the question. Indeed, the product traded in the annual coupled auction must be sufficiently standardised so that capacity from different zones is comparable. For example, de-rating, availability, monitoring and verification, non-delivery penalties and settlement rules pertaining to the annual product should be harmonised. In other words, the annual auction should trade a standardised product. In addition, the participating zones require a coordinated scarcity assessment and a common representation of the network, since these determine whether cross-border obligations can be procured to meet demand. These requirements do not imply full harmonisation of the underlying multi-year national capacity mechanisms. National capacity mechanisms may retain heterogeneous features, such as contract types, contract durations, auction lead times, and technology eligibility. A member state may equally use different monitoring rules in its multi-year long-term contracts, but the resulting capacity must be converted into the common annual product (under the harmonised rules) when it is re-marketed in the annual auction. However, the basis risk associated with any mismatch between the national long-term contract and the standardised annual obligation remains with the member state. Ultimately, the proposed mechanism may offer a strong incentive for member states to align the design of their national long-term contracts with the annual product, but this is not essential at the start.

We leave the required degree of harmonization across borders for future work:

Changes to the paper (Conclusion in the revised manuscript): Future work should focus on detailing the design of the annual auctions and the capacity product to be traded. Key features include the degree of harmonization required across member states and the valuation of resources that have non-symmetric contributions to meeting demand during scarcity situations across zones, such as energy-constrained assets.

8. Energy-limited resources / zone-dependent de-rating

The authors could also reflect on the effect of the integration of a large share of non-fossil, energy-limited, resources expected in EU capacity markets. The de-rating of a given energy-limited asset is not the same for each bidding zone. Today, this is observable for short-duration storage, which de-rating is highly different across countries depending on the expected duration of scarcity periods length. Ideally, the coupling algorithm should account for the fact that a given MW in country x does not contribute equally to security of supply in country y. Could the authors eventually reflect on whether and how this would affect the coupled auction clearing?

Authors' response: Thank you for your comment. The inclusion of energy-limited resources is an important future extension of our work, as such assets are expected to make a larger share of the EU's capacity market participation.

In our proposed annual auction, we assume a de-rating factor of one for all assets in all scarcity situations. Indeed, this simplification would need to be relaxed to accommodate energy-limited resources such as short-duration storage. One approach would be to derive technology-specific de-rating factors from the joint scarcity assessment. Because the scarcity scenarios identify both the zone experiencing scarcity and the system conditions in the remaining zones, each technology could be assigned a zone-specific de-rating factor reflecting its expected firm contribution during each zone's scarcity event. For example, suppose a short-duration storage technology is assessed to provide only 50% of its nameplate capacity during *Scarcity at A*, owing to the duration of the scarcity event, but can provide its full capacity during the shorter *Scarcity at B* event. Its de-rating factor would then be 0.5 in the former scenario and 1 in the latter. The same physical asset could therefore have different accredited capacity contributions depending on the scarcity situation in which it is expected to provide support. To avoid double-counting transmission limitations, these de-rating factors would only need to capture resource-specific limitations, such as energy duration. Transmission deliverability should not be embedded in the de-rating factor, since it is already determined endogenously through the flow-based constraints in the coupled auction.

As we haven't tested our hypothesis above, we leave this for future work:

<u>Changes to the paper (Conclusion in the revised manuscript):</u> Future work should focus on detailing the design of the annual auctions and the capacity product to be traded. Key features include the degree of harmonization required across member states and the valuation of resources that have non-symmetric contributions to meeting demand during scarcity situations across zones, such as energy-constrained assets.

Reviewer 2

This manuscript proposes a coordinated cross-border capacity-market design in which national long-term arrangements are preserved, but obligations are subsequently reallocated through an annual coupled auction that accounts for network constraints using a flow-based framework. The paper formulates the setting as a long-run equilibrium with generators, consumers, an energy-market operator, and a capacity-market operator; the flow-based case is represented as a generalized Nash equilibrium and solved as an MCP with ADMM. The topic is important, and the manuscript has clear potential.

Overall, the paper is conceptually clear, and a natural extension of the flow-based market coupling approach used in short-term markets. From a methodological standpoint, the paper relies on existing techniques. The empirical demonstration is understandably stylized, and not broad enough to establish robustness. With that said, I am broadly positive on the paper on the basis of its topical relevance and the clean experimental design. A few more specific comments follow:

Authors' response: Thank you for your detailed review and interest in our work. Below, we provide a point-by-point response to your comments and an overview of the changes in the manuscript, which have clearly strengthened the paper. All changes in the manuscript are marked in green.

1. The paper does not model the full institutional design it advocates

One high-level issue is that the paper does not model the full institutional design it advocates. While the discussion suggests a two-tier framework to reconcile national sovereignty with efficiency, the first tier goes unmodeled and the paper assumes efficient propagation of the prices from the second tier into the decisions made in the first. This seems like a strong claim, with important implications for overall performance. I can understand wanting to avoid modeling an endogenous first stage, but I think this could be reframed. Rather than specifically proposing a two-tier approach, the paper could simply say: for national capacity mechanisms to be efficient, whatever form they take, we need a second tier that effectively sets a spot price for capacity around which longer-term arrangements can settle. I think you want to claim that your approach solves the deliverability problem without necessarily getting into strategic, free-riding, and cost allocation issues that might arise when incorporating the first tier national problems, and that clarity on this point would help clarify the intended contribution.

Authors' response: We thank the reviewer for pointing out the discrepancy. We have adopted the suggestion to scope our proposal more precisely. Instead of proposing a "two-tiered mechanism", the revised manuscript now presents our proposal as a second layer required to coordinate multi-year national capacity mechanisms, as suggested by the reviewer. We present the contribution as the design of the proposed second layer i.e, the design of its auction and its interface with the investment layer. In Section V, we discuss the consequences of imperfect coordination between the two layers. The revision, therefore, pertains mainly to framing and scope and does not affect the numerical results presented. The changes are reflected in the abstract, Sections I, II, and V. The changes made are further detailed in the box below.

Changes to the paper in the revised manuscript (p. 1, abstract): In this paper, we propose a novel conceptual design for a coupled European capacity market that utilises the logic of flow-based market coupling. We argue that national capacity markets, offering long-term contracts, need a second tier: an annual capacity auction coupled across zones, which sets zonal prices for firm, deliverable capacity. We design such an auction by extending the logic of flow-based market coupling to capacity markets.

(p. 1, right column): This leads to a clear research question: How should capacity mechanisms in Europe be designed to facilitate the use of cross-border resources? This leads to a clear research question: How should capacity mechanisms in Europe be designed to facilitate the use of cross-border resources? To this end, we propose a two-tiered mechanism in which Member States would grant long-term contracts to new capacity, which is subsequently re-marketed as reliability options in annual, coupled capacity markets. We propose an additional tier: an annual capacity auction, coupled across zones. It complements the multi-year national capacity auctions in which Member States grant long-term contracts, such as reliability options, to de-risk investment. Member States, as holders of these long-term contracts, re-market the contracted volumes as annual products in the coupled auctions (Section 2).

(p. 2, left column): We extend these two-tiered approaches into a European capacity mechanism by coupling their annual layer across borders.

(p. 2, right column): With the above in mind, we propose a two-tiered approach for future capacity mechanisms: we propose an annual, coupled European capacity auction as a coordinating layer for national capacity mechanisms awarding long-term contracts (Figure 1). This two-tiered approach recognises that while investment cycles are multi-year, adequacy and network conditions are non-static. Our primary contribution is the design of the second layer, but we reflect on the requirements it places on the first layer.

(Fig. 1 caption): The proposed capacity market design features two layers: a national capacity market for long-term contracts and a coupled annual capacity mechanism allowing secondary trade of long-term contracts. The proposed annual coupled capacity auction and its interface with national capacity contracts. Member States grant multi-year contracts (e.g., reliability options) to de-risk investment and re-market the contracted volumes as annual products in the coupled auctions.

(p. 4, left column): In Section III and IV, we zoom in on the second layer: a coupled annual capacity market, and abstract from the first layer, assuming a perfect propagation of price and investment signals to the first layer. Thus, our results do not capture the effect of distortions that arise within the first layer, such as overprocurement, cost-allocation and free-riding. We reflect on the impact of these limitations on our analysis in Section V. We capture our proposed design in a mixed complementarity problem and apply it to a case study, demonstrating how this can facilitate the coupling of European capacity markets.

2. Welfare decomposition

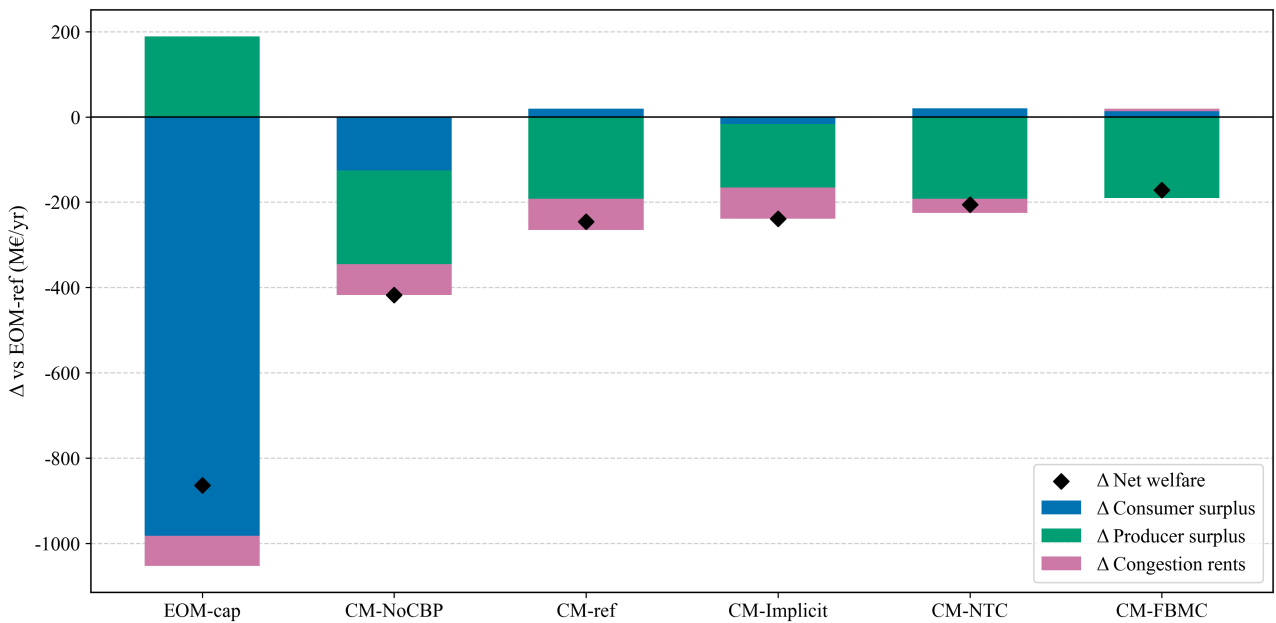
The interpretation of outcomes would benefit from a welfare decomposition. The manuscript seems incongruous in having consumers with elastic demand but then reporting the headline results in terms of total system costs rather than welfare. This is especially important because Total Cost under CM-Implicit is lower than CM-FBMC (as reported in Table IV), raising the question of why the manuscript advocates for CM-FBMC.

Authors' response: Thank you for your comment. We have included a subsection on social welfare (Section IV-C2) and a figure on showing a decomposition of the change in welfare w.r.t *EOM-ref* (Figure 5a). The figure is decomposed into consumer and producer surplus, and the network operator's congestion rents. We show the changes below.

Changes to the paper (p. 10, right column in the revised manuscript and Figure 5a): We decompose the change in welfare w.r.t *EOM-ref* in changes in consumer surplus, producer surplus and congestion rents (Figure 5a).^a *EOM-ref* leaves market-clearing price uncapped, enabling efficient self-rationing by consumers with a WTP below the clearing price, and providing a reference value for surplus generated in the market. The largest welfare loss is observed in *EOM-cap* (−864 M€/yr), the lowest in *CM-FBMC* (−171 M€/yr), consistent with what is obtained in (Section IV-C1). Under the price cap, consumers bear the loss (−982 M€/yr, driven by the unserved energy), while producers gain 189 M€/yr as inframarginal rents rise, particularly for renewables and baseload generators. Capacity markets reverse this trend. Producer surplus decreases as capacity revenues compensate investment cost but lower inframarginal rents. However, it significantly improves consumer surplus. Only *CM-NoCBP* (−125 M€/yr, due to over-procurement) and *CM-Implicit* (−17 M€/yr, energy-not-served) leave consumers worse off.

We distinguish two sources of welfare gains. First, coordinating the capacity demand through a joint scarcity assessment (*CM-NoCBP* to *CM-ref*) improves welfare by 172 M€/yr, with most accruing to consumers. Secondly, explicit cross-border participation adds 41 M€/yr under *CM-NTC* and 75 M€/yr under *CM-FBMC*, which materialises almost entirely as congestion revenue. *CM-FBMC* is the only design that fully restores the value of interconnection compared to *EOM-ref*.

^aTo compute the change in consumer surplus, we re-evaluate energy market outcomes using the uncapped demand curve. The obtained consumer surplus is corrected with the cost of energy not served and capacity payments.



(a) Welfare decomposition showing the change w.r.t *EOM-ref* in consumer and producer surplus, congestion rents and the net welfare across market designs

3. Detail on numerical approach:

The manuscript would benefit from more detail on the numerical approach. First, it wasn't clear to me that all of the proposed designs require an MCP—some comparisons might be made with a equivalent optimization formulation. It would help to clarify where the equilibrium approach is required. Beyond this, additional detail on convergence behavior, stopping criteria, sensitivity to initialization, and whether multiple local solutions are encountered would be useful.

Authors' response: Thank you for your comment. L  t   et al [1] show that the classical equivalence between the central planner problem and the equilibrium problem break down under flow-based market coupling. This is because the network operator's feasible set depends on the generators' investment decisions. In other words, in a central planner problem, generators would be placed optimally *within* zones to manage congestion, reflecting locational incentives that are absent in our problem formulation. The equilibrium is therefore a Generalised Nash equilibrium. We have added this explanation in the Methodology section (Section 3). All results shown converged. We have included numerical details in Appendix I.

Changes to the paper (p. 4, left column in the revised manuscript): L     et al. [1] show that, under flow-based coupling, the classical equivalence between a central planner’s optimisation problem and the game described above breaks down. Embedding the network constraints and generators in a single problem allows optimally siting generators within a market zone to manage congestion. The planner’s solution is that of a market completed by locational payments, which are absent in real-world zonal markets and the game described above [1]. We therefore solve the MCP using a price-search algorithm based on the Alternating Direction Method of Multipliers (ADMM) [2].

Changes to the paper (Appendix I in the revised manuscript): The equilibrium is computed using an exchange-form consensus ADMM implemented in Julia, in which each agent’s convex quadratic subproblem is solved in parallel with Gurobi. Each iteration solves Problems (1)–(4) in parallel, given the prices, then updates $\lambda_{z,t}^e$ and λ_z^{cm} as duals of the zonal imbalances. The algorithm terminates when, for every zone and for both markets (electricity and capacity), the primal and dual residuals fall below an absolute tolerance $\epsilon = 0.1$ (in MW for primal residuals; in €/MWh and €/MW-year for EOM and CM dual residuals). The penalty parameter ρ is adapted per zone and market via residual balancing following Boyd et al [3]. Converged runs typically require 460-2,500 iterations (13-38 seconds of wall time), with each iteration taking residuals to decay steadily (Table). All runs are cold-started, with prices, quantities, and penalty terms initialised at zero.

Because each agent solves a convex quadratic problem, a converged ADMM solution satisfies the conditions for competitive equilibrium. As degeneracy exists in some variables (e.g., within-zone investment allocation, network operator capacity allocation, and unserved energy when price cap hits multiple price zones simultaneously), solutions may be non-unique. However, these multiple equilibria differ only in nodal quantities (except for unserved energy, which we treat ex-post, see Appendix II); aggregate installed capacity and zonal outcomes remain consistent across repeated runs.

Table VI: Final ADMM residuals and iteration counts across market designs. Reported residuals are maximum over zones. The convergence tolerance is $\epsilon = 0.1$

Design	Iterations	$\max r_E^p$	$\max r_E^d$	$\max r_{CM}^p$	$\max r_{CM}^d$
EOM-ref	1,504	0.012	0.100	0	0
EOM-cap	462	0.017	0.100	0	0
CM-NoCBP	498	0.047	0.099	0.031	0.077
CM-ref	845	0.084	0.038	0.092	0.024
CM-Implicit	2,223	0.027	0.100	0.017	0.029
CM-NTC	830	0.038	0.097	0.057	0.099
CM-FBMC	733	0.031	0.047	0.025	0.088

Reviewer 3

This paper proposes a two-tier design for coupling European capacity markets. On the first tier (the Investment Layer), Member States grant long-term (10–15 yr) national reliability-option contracts to de-risk capital-intensive investment while retaining sovereignty over their capacity mix. On the second tier, the State — as primary contract holder — re-markets the contracted volume into an annual, coupled, cross-border capacity auction that clears subject to network feasibility. The central methodological idea is to apply flow-based market coupling (FBMC) logic to the clearing of this annual capacity auction: the deliverability of foreign and domestic firm capacity is assessed against an endogenously computed flow-based domain under a set of defined scarcity scenarios, replacing the static "maximum entry capacity" (MEC) approximation used in prior work. The design is formalised as a long-run equilibrium — a Generalised Nash Equilibrium cast as a mixed complementarity problem and solved with ADMM — and illustrated on a stylised three-zone, four-node case study comparing six market designs (EOM-ref, EOM-cap, CM-NoCBP, CM-Implicit, CM-NTC, CM-FBMC).

The topic is timely, well-motivated and squarely within the journal's scope. The diagnosis of the two current cross-border approaches — implicit deduction without remuneration, and conservative MEC/NTC limits — is accurate, and the core idea of bringing FBMC network feasibility into the capacity clearing is genuinely interesting. The paper is publishable in principle, but in its current form it over-claims relative to what the model represents, embeds the proposed mechanism's own outcomes in its benchmarks, and contains a small set of reproducible numerical and specification errors. All of these are correctable within a (demanding) major revision, without a new study.

Authors' response: Thank you for your detailed review of our work and your appreciation of our work. Below, we provide a point-by-point response to your comments and an overview of the changes in the manuscript, which have clearly strengthened the paper. All changes in the manuscript are marked in red.

1. Mismatch between the proposed design and the model:

Mismatch between the proposed design and the model. Section II builds the design on reliability options with a strike price and a pay-back obligation, and foregrounds storage and demand-side response, but the model implements none of these. Eq. (1a) remunerates capacity with a flat payment $\sum \lambda_z^{cm} \cdot y_{i,z}^{cm}$ — there is no strike price, pay-back, availability obligation or non-delivery penalty anywhere in Problems (1)–(5); deliverability is enforced only by the engineering feasibility constraints (4b)–(4j). Table I contains only three thermal technologies (Base/Mid/Peak); storage and DSR never enter the model, and renewables appear only as exogenous negative demand. The authors do scope the study to the second tier ("assuming a perfect propagation of price signals to the first layer," p.6), so this is a declared limitation rather than a hidden flaw — but the abstract and conclusions are nonetheless stated for the full mechanism. Please add an explicit statement of what the model does and does not represent, and re-word the abstract, Fig. 1 and Section II so that the option pay-back, the strike settlement and storage/DSR are presented as conceptual provisions or future work; every claim that relies on the option's risk-management role should be scoped to the modelled second tier.

Authors' response: Thank you for your feedback. Our intention was indeed to zoom in on the annual layer, but we recognise that we should have been clearer about what we model and what is part of the conceptual design. We have made the following changes:

- We have re-framed the scope of our paper in response to Comment 1 of Reviewer 2. We now state our contribution as the design of the annual coupled auction (the second layer) and its interface with the investment layer. Thus, we have revised the abstract, Section I, Fig. 1 caption, Section II, and Section VI to reflect this. Please refer to Comment 1 of Reviewer 2 for the exact changes we made.
- As requested, in Section II, we have included a paragraph on what the model represents and what we did not model explicitly.

Changes to the paper in the revised manuscript (p. 1, abstract): ~~In this paper, we propose a novel conceptual design for a coupled European capacity market that utilises the logic of flow-based market coupling. We argue that national capacity markets, offering long-term contracts, need a second tier: an annual capacity auction coupled across zones, which sets zonal prices for firm, deliverable capacity. We design such an auction by extending the logic of flow-based market coupling to capacity markets.~~

(p. 1, right column): This leads to a clear research question: How should capacity mechanisms in Europe be designed to facilitate the use of cross-border resources? This leads to a clear research question: How should capacity mechanisms in Europe ~~be designed~~ ~~evolve~~ to facilitate the use of cross-border resources? ~~To this end, we propose a two-tiered mechanism in which Member States would grant long-term contracts to new capacity, which is subsequently re-marketed as reliability options in annual, coupled capacity markets.~~ We propose an additional tier: an annual capacity auction, coupled across zones. It complements the multi-year national capacity auctions in which Member States grant long-term contracts, such as reliability options, to de-risk investment. Member States, as holders of these long-term contracts, re-market the contracted volumes as annual products in the coupled auctions (Section 2).

(p. 2, left column): We extend these two-tiered approaches into a European capacity mechanism ~~by coupling their annual layer across borders.~~

(p. 2, right column): With the above in mind, ~~we propose a two-tiered approach for future capacity mechanisms~~ we propose an annual, coupled European capacity auction as a coordinating layer for national capacity mechanisms awarding long-term contracts (Figure 1). ~~This two-tiered approach recognises that while investment cycles are multi-year, adequacy and network conditions are non-static.~~ Our primary contribution is the design of the second layer, but we reflect on the requirements it places on the first layer.

(Fig. 1 caption): ~~The proposed capacity market design features two layers: a national capacity market for long-term contracts and a coupled annual capacity mechanism allowing secondary trade of long-term contracts.~~ The proposed annual coupled capacity auction and its interface with national capacity contracts. Member States grant multi-year contracts (e.g., reliability options) to de-risk investment and re-market the contracted volumes as annual products in the coupled auctions.

(p. 4, left column): In Section III and IV, we zoom in on the second layer: a coupled annual capacity market, ~~and abstract from the first layer,~~ assuming a perfect propagation of price ~~and investment~~ signals to the first layer. ~~Thus, our results do not capture the effect of distortions that arise within the first layer, such as overprocurement, cost-allocation and free-riding. We reflect on the impact of these limitations on our analysis in Section V.~~ We capture our proposed design in a mixed complementarity problem and apply it to a case study, demonstrating how this can facilitate the coupling of European capacity markets.

2. The headline quantitative advantage is small and not yet shown to be robust:

The headline quantitative advantage is small and not yet shown to be robust. The CM-FBMC vs CM-NTC system-cost difference is about 26 M€ on a system cost of $\approx 15,950$ M€/yr — roughly 0.16%. Generation cost is identical and there is no unserved energy in either design, so the entire saving comes from a marginally lower investment cost under CM-FBMC (5,881 vs 5,908 M€). Three issues make the ranking fragile as presented: (i) CM-NTC fixes investments to the CM-FBMC outcome and is described in the text as “a theoretical upper bound to the performance of NTC-coupled markets,” so 0.16% is measured against a best-case NTC; (ii) the model is a networked bilinear Generalised Nash Equilibrium whose uniqueness is not established — the cited convergence result [24] concerns a risk-averse capacity-market equilibrium and is not shown to transfer to this networked bilinear formulation — and no convergence, KKT-residual or multistart evidence is reported, so an ordering of this magnitude could be sensitive to equilibrium selection; and (iii) the whole case rests on a single engineered winter day with hand-set scaling factors and no sensitivity analysis. Please report the saving with its denominator and percentage and qualify it as relative to a best-case NTC; lead instead with the more robust findings (the ≈ 1.3 GW Zone-C to Zone-A investment reallocation and the elimination of the CM-Implicit unserved energy); provide convergence/KKT-residual evidence and a multistart (uniqueness) check, or state the ordering as indicative; and add a minimal sensitivity sweep (VOLL or reliability standard, the scarcity-coincidence parameters, and independent NTC investment).

Authors’ response: Thank you for the comment. Before we respond to the issues raised above point-by-point, we re-iterate that we have, in response to comment 2 of reviewer 1, changed our calibration of the capacity demand. This changes the numerical results compared to the original submission, but not our qualitative findings. For details, we refer to our response to the comment of Reviewer 1.

- Based on your suggestion, we have rephrased the findings that we highlight in the abstract and conclusion of the revised manuscript.
- We describe savings as a percentage to reflect their denominator. We have also added the volume of ENS as an additional column in Table (IV) for easier reference.

- We have clarified throughout the revised manuscript that *CM-NTC* constitutes a best-case benchmark relative to *CM-FBMC*. We refer to Comment 4 of Reviewer 1 for our detailed motivation for this modelling choice. We also clarify an important distinction. Investments in *CM-NTC* are not fixed to the investment outcome of *CM-FBMC*. Investment remains endogenous in the solution to the *CM-NTC* equilibrium. Only the ex-ante bilateral NTC domain is constructed using the investment realised under *CM-FBMC*. This deliberately favours the NTC benchmark by excluding part of the investment distortion that may result from the interaction between ex-ante bilateral limits and the subsequent investment response.
- We provide numerical convergence evidence in *Appendix I*, including final primal and dual ADMM residuals and iteration counts for all market designs. We do not, however, claim that this establishes the uniqueness of the Generalised Nash Equilibrium. Degeneracy may exist in several disaggregated variables, and we do not provide an analytical uniqueness result or a multistart proof of uniqueness.
- We respectfully do not add the proposed sensitivity sweep. The purpose of the numerical example is a comparative analysis in which the market designs are evaluated under the same exogenous assumptions. The paper does not seek to estimate a general quantitative welfare gain from flow-based coupling. We therefore believe that the proposed parameter sweeps would mainly change the magnitude of the illustrative outcomes rather than the mechanisms we study. We elaborate on each below:
 - *Sensitivity to VoLL or reliability standard*. We understand the reviewer’s concern about whether the small system-cost differences are robust to alternative assumptions about reliability. We address this by qualifying the cost ranking as indicative and no longer presenting the efficiency gain as a headline result. In the revised manuscript, ENS is valued at 18,667 €/MWh, the block-average willingness to pay implied by the calibrated demand curve. Changing this value ex post would only rescale ENS costs and would not affect equilibrium quantities. By contrast, changes to the 4,000 €/MWh price cap, the energy demand curve or the capacity demand would affect investment, ENS, and system costs. However, these parameters are held constant across all market designs precisely to isolate the effect of alternative treatments of cross-border participation. While the mechanisms driving the differences between the designs are clear, we do not claim that the precise ordering of system costs is invariant to alternative price caps or adequacy assumptions. Our objective is instead to compare different treatments of cross-border participation under common assumptions.
 - *Scarcity-coincidence*: We have revised the parameterisation of the scarcity scenarios so that the exogenous coincidence factors used in the initial submission are no longer required. In the revised manuscript, each scarcity scenario is derived directly from coincident observations in the load time series. Specifically, when a zone reaches its peak energy demand, the demand of the other zones is taken from the same period. The resulting set of zonal requirements is then used directly to define the system state against which capacity deliverability is assessed in *CM-NTC* and *CM-FBMC*. Consequently, eliminate the scarcity-coincidence parameter on which the reviewer suggested a sensitivity. The coincidence between zonal scarcity requirements is instead determined by the underlying data. We report the scenarios in Section IV of the revised manuscript. We nevertheless agree that a real-world implementation would require a substantially richer and probabilistically derived set of scarcity scenarios than the illustrative scenarios considered here. We have therefore made this limitation explicit in Section V of the revised manuscript.
 - *Independent NTC investment*: We clarify that investment decisions under *CM-NTC* are already independent and endogenous. What is taken from *CM-FBMC* is only the investment configuration used to construct the ex-ante NTC domain. Constructing this domain endogenously, together with the *CM-NTC* investment decision, is non-trivial as it re-introduces a circular dependency. Bilateral transfer limits (NTC domain) depend on the location and availability of generation capacity, while the location of investment itself depends on those bilateral transfer limits. Resolving this endogenously would therefore require an additional optimisation problem to compute the domain of available transfer capacity. This constitutes a different modelling problem and is outside the scope of the present paper. For reference, *CM-ref* also provides the limiting case in which explicit bilateral exchange disappears: as the NTC limits approach zero, *CM-NTC* approaches *CM-ref*. The comparison between *CM-ref*, best-case *CM-NTC*, and *CM-FBMC*, therefore, already illustrates the progression from no explicit cross-border participation, to bilateral coupling, to flow-based coupling.

Changes to the paper (abstract in the revised manuscript): In a comparative analysis of different market design scenarios in an illustrative multi-zone case study, using a bespoke long-run equilibrium problem, we show that the proposed ~~flow-based coupling of capacity markets reduces system costs by harnessing available capacity in neighbouring market zones while ensuring deliverability with respect to network constraints in all scarcity situations.~~ auction removes the circular dependency between imports and available capacity in current cross-border participation schemes, locates investment where it is cheapest while keeping it deliverable under network constraints in defined scarcity situations, and makes the value of additional interconnection capacity explicit.

Changes to the paper (conclusion in the revised manuscript): Our modelling results show that this design more accurately values the contribution of interconnection and ~~yields efficiency gains~~ ~~relocates capacity to cheaper locations under deliverability constraints~~. Moreover, our design addresses challenges posed by fragmented national capacity mechanisms while respecting the heterogeneous adequacy requirements of Member States and their sovereign right to steer their capacity mix.

Changes to the paper (p. 10, left column in the revised manuscript): Explicit cross-border participation lowers total costs by relocating peaking capacity from zones with a high investment cost to lower-cost zones while keeping ENS below 0.25 GWh/yr. *CM-NTC* reduces investment costs by 40 M€/yr relative to *CM-ref* (0.7 % of 5 820 M€/yr), and by 82 M€/yr (1.4 %) under *CM-FBMC*. It removes the unserved energy of *CM-Implicit*, with ENS falling to 253 MWh/yr (below 10^{-4} % of served annual demand) in *CM-FBMC* and to 0.5 MWh/yr in *CM-NTC*. *CM-FBMC* is the least-cost capacity-market design, 258 M€/yr below *CM-NoCBP*, 55 M€/yr below *CM-Implicit*, and 35 M€/yr (or 0.22%) below a best-case NTC (*CM-NTC*), at a total annualised cost of 15,791 M€/yr.

Changes to the paper (Appendix I in the revised manuscript): Table VI: Final ADMM residuals and iteration counts across market designs. Reported residuals are maximum over zones. The convergence tolerance is $\epsilon = 0.1$

Design	Iterations	$\max r_E^p$	$\max r_E^d$	$\max r_{CM}^p$	$\max r_{CM}^d$
EOM-ref	1,504	0.012	0.100	0	0
EOM-cap	462	0.017	0.100	0	0
CM-NoCBP	498	0.047	0.099	0.031	0.077
CM-ref	845	0.084	0.038	0.092	0.024
CM-Implicit	2,223	0.027	0.100	0.017	0.029
CM-NTC	830	0.038	0.097	0.057	0.099
CM-FBMC	733	0.031	0.047	0.025	0.088

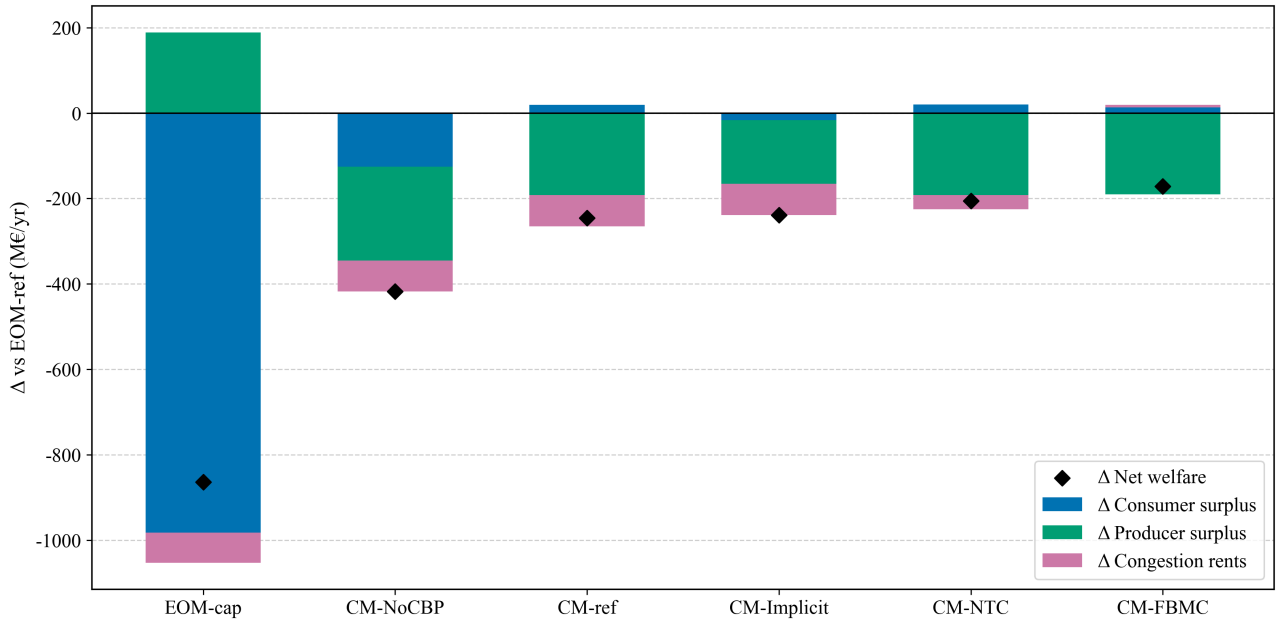
3. Adequacy methodology:

Adequacy methodology. Capacity demand D_z^{cm} is set to peak residual demand and is not tied to any reliability standard (LOLE/EENS or VOLL-CONE), so the procured volume is an assumption rather than a derived requirement. The scarcity scenarios are partly self-referential: "scarcity events depict the time steps in which ENS occurs under EOM-cap" (p.10), i.e. they are defined by the very design the paper sets out to improve, and the 95% / 100% / 90% percentages are stated without source or sensitivity. The deliverability constraints are optimistic: Eq. (3e) carries no availability factor although Eq. (1b) does, and nodal siting $\hat{y}_n$ is a free decision variable of the operator (constrained only by $\sum \hat{y}_n = y_z$), so the operator can relocate capacity within a zone and relax the intra-zonal limits the mechanism is meant to enforce; combined with state-N-only thermal limits (no $N-1$, CNEC, GSK or RAM), the modelled "FBMC logic" is more permissive than operational flow-based coupling (see Felten et al., 2021, Utilities Policy 70:101136, on how flow-based domain approximations affect welfare). Finally, the designs are compared at different served demand (44 GW under EOM-ref vs 48–50 GW under the capacity markets), so the cost ranking is not established at constant welfare. Please tie D_z^{cm} to a reliability standard or state the simplification explicitly; add a sensitivity on the 95/100/90% parameters; add the availability factor to (3e)/(4b) or justify its absence and add a fixed-siting sensitivity; bound the deliverability claim to state-N and the finite scenario set S ; and report a net-welfare or constant-served-demand metric.

Authors' response: Thank you for the thorough assessment and helpful recommendations. We respond to the recommendations as follows:

- We follow the recommendation to explicitly state our simplification of the reliability standard. Setting D_z^{cm} to the peak residual demand is a stylised representation of member-state-defined reliability requirements, which would normally be derived from a reliability standard. We have also refined our definition of D_z^{cm} , which we use consistently across all zones and market designs. This helps us to limit the parameters that drive our results. Please refer to comment 2 of reviewer 1 for a detailed response on our justification for this approach.
- Our result remains dependent on the demand data. Demand data in which zones peak almost simultaneously would reduce the benefit of *CM-ref* relative to *CM-NoCBP*. Rather than a sensitivity, we believe that the inclusion of the *CM-ref* scenario better addresses the reviewer's concern, as it disentangles the benefit of the joint scarcity assessment from that of cross-border trade in capacity obligations.
- We have included the availability factor in Eq. (3d) (formerly Eq. (3e)). This was a typo.

- While the exogenous renewable capacity is fixed ex ante, we allow new capacity to be sited freely to avoid biasing the outcome. Moreover, $\hat{y}_{i,n}$ is neither scenario- or time-indexed and is determined by the TSO's energy-market capacity allocation, so capacity cannot be "resited" on a scenario-by-scenario basis to obtain a deliverability certification.
- We have bound the deliverability claim as requested. Deliverability is now stated with respect to state-N thermal limits and under the finite scenario set $\mathcal{S}$. We have introduced a limitations paragraph in Section (V) that covers the N-1 security.
- We have included a welfare decomposition that shows changes in consumer surplus, producer surplus and congestion rents for the six designs relative to *EOM-ref* (Section IV-C.2 and Fig. 5a).



(a) Welfare decomposition showing the change w.r.t *EOM-ref* in consumer and producer surplus, congestion rents and the net welfare across market designs

Changes to the paper (P. 4, right column revised manuscript): $\hat{g}_n \hat{y}_{i,n}$ is the nodal allocation of installed capacity at zone z . The nodal allocation of existing capacity is fixed by the same coefficients used to allocate demand, $\hat{y}_{i,n} = S_{n,z} y_{i,z}$, while new capacity is sited freely subject to Eq. 3e.

Changes to the paper (P. 5, right column revised manuscript): In each scenario s , the zones in Z_s^{sc} are placed at their scarcity demand, above the capacity demand $D_{s,n}^{\text{cm}} = S_{n,z} D_z^{\text{cm}}$ allocated to them by the joint assessment, while the remaining zones are placed at their coincident demand (Figure 2, right). The transmission system operator determines how much cross-zonal capacity exchange can be supported while remaining deliverable across the prescribed scarcity scenarios. To this end, we required that there must exist a nodal dispatch $\hat{g}_{s,n}^{\text{sc}}$ that serves demand while respecting the network constraints, including intra-zonal lines at state- N thermal limits in each scenario. The resulting zonal net positions must remain consistent with the cross-zonal capacity obligations. In particular, a zone that has sold capacity abroad must be able to sustain the corresponding export position when that capacity is called during the importing zone's scarcity event. The auction yields, for each zone z , a single annual capacity price λ_z^{cm} , the domestic capacity obligation Y_z^{cm} , and the net cross-border capacity obligation p_z^{CO} .

Changes to the paper (P. 12, right column revised manuscript): Note, however, that our network representation only considers state- N and ignores contingencies during scarcity events. Real-world implementation would account for grid uncertainties through the $N - k$ security criterion. The combination of a scarcity scenario set and $N - k$ contingencies may considerably increase the computational effort required for market clearing. Lastly, because flow-based market coupling has been criticised for opacity, its application in capacity markets must prioritise transparency.

Changes to the paper (Section IV-C.2, Fig. 5a – P. 10, right column revised manuscript): We decompose the change in welfare w.r.t *EOM-ref* in changes in consumer surplus, producer surplus and congestion rents (Figure 5a).^a *EOM-ref* leaves market-clearing price uncapped, enabling efficient self-rationing by consumers with a WTP below the clearing price, and providing a reference value for surplus generated in the market. The largest welfare loss is observed in *EOM-cap* (−864 M€/yr), the lowest in *CM-FBMC* (−171 M€/yr), consistent with what is obtained in (Section IV-C1). Under the price cap, consumers bear the loss (−982 M€/yr, driven by the unserved energy), while producers gain 189 M€/yr as inframarginal rents rise, particularly for renewables and baseload generators. Capacity markets reverse this trend. Producer surplus decreases as capacity revenues compensate investment cost but lower inframarginal rents. However, it significantly improves consumer surplus. Only *CM-NoCBP* (−125 M€/yr, due to over-procurement) and *CM-Implicit* (−17 M€/yr, energy-not-served) leave consumers worse off.

^aTo compute the change in consumer surplus, we re-evaluate energy market outcomes using the uncapped demand curve. The obtained consumer surplus is corrected with the cost of energy not served and capacity payments.

4. Reproducibility and the demand-curve calibration:

Reproducibility and the demand-curve calibration. (a) The study cannot be reproduced from the manuscript: the time-aggregation weights and period set, the PTDF/susceptance data and the $S_{n,z}$ nodal-allocation coefficients, the demand/RES scaling and annualisation formulas, the ADMM settings (penalty, tolerances, iterations), and the NTC-domain ATC values are all missing, and there is no data/code-availability statement. (b) The elastic-demand calibration is numerically inconsistent: the text states the elastic share is 20% of reference demand at WTP and "decreases to 4% at the price cap of 4,000 €/MWh," but with the stated linear inverse demand of slope $-WTP/(\epsilon \cdot D)$, the elastic share at the cap is $\epsilon \cdot (WTP - \lambda)/WTP = 0.20 \cdot (16,000/20,000) = 16\%$, not 4% (obtaining 4% at 4,000 €/MWh would require $WTP \approx 5,000$, contradicting the $WTP = 20,000$ used throughout). Please add a reproducibility appendix with all parameters, formulas and an ATC/PTDF table plus a data/code-availability statement, and correct or re-specify the elastic-demand curve so the stated value is consistent with the equations, propagating any change through the results.

Authors' response: Thank you for your suggestions.

- We have added a reproducibility statement with all required data and code. All code, input data and results are available here: https://github.com/kamaladekola/cross_border-CM.
- We apologize for the unclear definition of the demand curve. The stated 4% refers to the residual price-responsive share below the cap, not consumption of the elastic block at the cap. We have one inverse demand curve with an inelastic block $(1 - \epsilon_z) D_{z,t}^e$ valued at $WTP_z = 20,000$ €/MWh and an elastic block $\epsilon_z D_{z,t}^e$, $\epsilon_z = 0.20$, falling linearly from WTP_z to zero with slope $-WTP_z/(\epsilon_z D_{z,t}^e)$. In price-capped designs, 16% of demand valued between the cap and WTP_z cannot express their willingness-to-pay and bids at the cap, so the consumer's problem is parametrised with the truncation (4,000, 0.04). If prices rise to the cap, demand curtailment is random, so unserved energy is valued at the average valuation of the curtailed block, 18,667 €/MWh. The revised text clarifies this. Please refer to Figure 1 below.

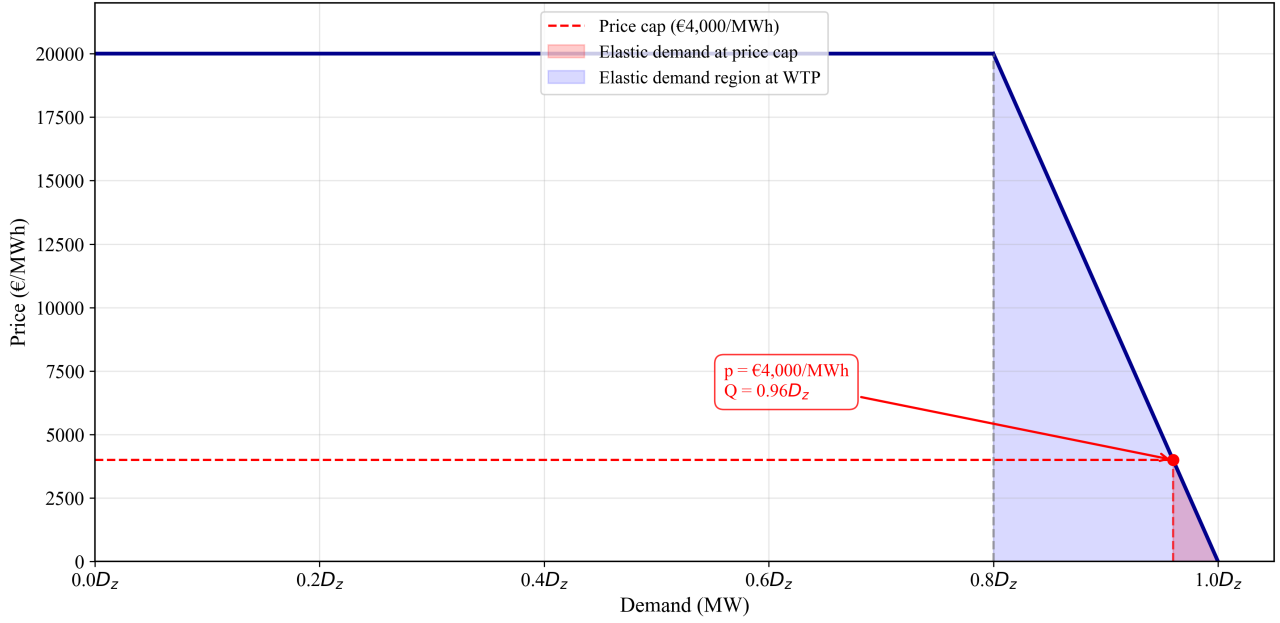


Fig. 4: Inverse demand curve used to represent consumers' behaviour in the model.

Changes to the paper (p. 7, right column in the revised manuscript): Energy demand is represented as a two-block structure comprising a fixed (inelastic) and a flexible (elastic) component. The elastic block is derived from a quadratic utility function with a maximum willingness to pay of $WTP_z = 20,000$ €/MWh, and calibrated so that the elastic component of demand ϵ_z is 20% of the reference demand $D_{z,t}^e$ at the maximum willingness to pay WTP_z and decreases to 4% at the price cap of 4,000 €/MWh. The slope of the inverse demand curve is constant at $-WTP_z/(\epsilon_z \cdot D_{z,t}^e)$, producing a smooth downward-sloping price-quantity curve, indicating a diminishing marginal utility among flexible consumers. The demand for capacity is inelastic and dependent on the capacity market design. Electricity demand is represented as a two-block structure comprising an inelastic and a price-elastic component. The inelastic block, $(1 - \epsilon_z)D_{z,t}^e$, is valued at the maximum willingness to pay $WTP_z = 20,000$ €/MWh. The elastic block, $\epsilon_z D_{z,t}^e$ with $\epsilon_z = 0.20$, is derived from a quadratic utility function and yields a linear inverse demand curve that falls from WTP_z at zero elastic consumption to zero €/MWh at $\epsilon_z D_{z,t}^e$, with slope $-WTP_z/(\epsilon_z D_{z,t}^e)$. In designs with an administrative price cap at 4,000 €/MWh, the elastic consumers whose valuation exceeds the cap (16% of $D_{z,t}^e$) cannot express their WTP in the market and bid at the price cap. The consumer's problem (Problem 2) is accordingly parametrised with $(WTP_z, \epsilon_z) = (20,000, 0.20)$ in markets without a price cap and with the truncation (4,000, 0.04) in price-capped designs (Table II). Both describe the same inverse demand curve, truncated at the cap, with identical slope. Curtailment of demand valued above the price cap is random (Eq. 2c), so unserved energy is ex-post valued at its average valuation of 18,667 €/MWh. The demand for capacity is inelastic and depends on the capacity market design.

(p. 8, right column in revised manuscript): All code and input data are available online [1].

(p. 9, left column in the revised manuscript): **NTC domain.** The bilateral transfer limits determined based on investment levels that materialise in *CM-FBMC* are $A \leftrightarrow B$: [740.3, -620] MW, $A \leftrightarrow C$: [574.3, -454] MW, and $B \leftrightarrow C$: [694, -694] MW. It defines the box for (*CM-NTC*) that sits within the *CM-FBMC* flow-based domain [2].

(p. 8, left column in revised manuscript): Table III: Zonal capacity demand (MW) across market designs.

Scenario	Zone A	Zone B	Zone C	Total
CM-NoCBP	16,870	14,829	16,450	48,149
CM-ref, CM-FBMC, CM-NTC	16,156	13,957	15,493	45,606
CM-Implicit	14,778	13,084	14,238	42,100

5. Formulation and notation:

- State explicitly that the capacity-clearing objective in (4a) is an equilibrium/complementarity condition (with the price λ_z^{cm} as a market-clearing dual) rather than a welfare optimisation; the bilinear price objective is unconventional and should be documented, and the equilibrium concept (variational / normalised) named and linked to what ADMM computes.
- A free index i appears in the objectives (4a)/(5a); the index z is reused as both the outer and the inner summation index; the term $\sum_z p_z^{CO}$ is annihilated by (4h); and (4h) carries a vacuous " $\forall z$ " over an already-summed scalar. Please provide one consolidated definition list mapping the per-agent offers $y_{i,z}^{cm}$, the nodal allocation $\hat{y}_n^{cm}$ and

the zonal obligation Y_z^{cm} , and name all dual variables. Also, λ^{cm} is not listed in the decision vector x_{CO} .

- Availability-factor inconsistency: AF appears in (1b) but is absent from (3e)/(4b).
- In Table I, $A_{i,z}$ is labelled $[\text{€/MWh}]^2$ while the operational cost is written linearly ($C^g = A \cdot g + B$); reconcile the cost-function form with the units and state explicitly that marginal cost is linear (total cost quadratic).

Authors' response: Thank you for your careful reading. We have addressed all your points as follows:

- We have restructured the formulation so that the clearing of the coupled capacity market is no longer expressed through the bilinear objective. We have previously written it in a concise form but we understand that this may have been confusing.
- The restructuring removes the defective objective entirely.
- First, the quoted quotation number has been moved due to the restructuring described above. Eq. (3e) is now Eq. (3d). We have included availability factor (AF) in Eq. (3d) The absence of the availability factor (AF) in Eq. (4b) reflects that only dispatchable capacity is offered in the capacity market, and they are assumed to have $AF_{i,z,t} = 1$.
- The cost function is quadratic with linear marginal cost. The unit in Table 1 has been corrected.

[Changes to the paper \(p. 5, left column in the revised manuscript\):](#)

$$\max_{x_E \in \mathcal{X}_E} \sum_{z \in \mathcal{Z}} \lambda_{z,t}^E p_{z,t}^E \quad (2a)$$

$$\text{s.t. } d_{n,t}^E = S_{n,z} d_{z,t}^E, \quad \forall n \in \mathcal{N}_z, z \in \mathcal{Z}, t \in \mathcal{T} \quad (2b)$$

$$\sum_{i \in \mathcal{I}} \hat{g}_{i,n,t} - d_{n,t}^E = r_{n,t}, \quad \forall n \in \mathcal{N}, t \in \mathcal{T} \quad (2c)$$

$$\hat{g}_{i,n,t} \leq A F_{i,z,t} \hat{y}_{i,n}, \quad \forall i \in \mathcal{I}, n \in \mathcal{N}_z, z \in \mathcal{Z}, t \in \mathcal{T} \quad (2d)$$

$$\sum_{n \in \mathcal{N}_z} \hat{y}_{i,n} = y_{i,z}, \quad \forall i \in \mathcal{I}, z \in \mathcal{Z} \quad (2e)$$

$$p_{z,t}^E = - \sum_{n \in \mathcal{N}_z} r_{n,t}, \quad \forall z \in \mathcal{Z}, t \in \mathcal{T} \quad (2f)$$

$$\sum_{n \in \mathcal{N}} r_{n,t} = 0, \quad \forall t \in \mathcal{T} \quad (2g)$$

$$f_{t,\ell} = \sum_{n \in \mathcal{N}} \text{PTDF}_{\ell,n} \cdot r_{n,t}, \quad \forall \ell \in \mathcal{L}, t \in \mathcal{T} \quad (2h)$$

$$-F_{\ell}^{\max} \leq f_{t,\ell} \leq F_{\ell}^{\max}, \quad \forall \ell \in \mathcal{L}, t \in \mathcal{T} \quad (2i)$$

[Changes to the paper \(p. 5, right column in the revised manuscript\):](#)

$$\max_{x_{CO} \in \mathcal{X}_{CO}} \sum_{z \in \mathcal{Z}^{cm}} \lambda_z^{cm} p_z^{CO} \quad (3a)$$

$$\text{s.t. } \hat{g}_{s,n}^{sc} \leq \hat{y}_n^{cm}, \quad \forall s \in \mathcal{S}, n \in \mathcal{N} \quad (3b)$$

$$\hat{y}_n^{cm} \leq \sum_{i \in \mathcal{I}^{cm}} \hat{y}_{i,n}, \quad \forall n \in \mathcal{N}_z, z \in \mathcal{Z} \quad (3c)$$

$$\sum_{n \in \mathcal{N}_z} \hat{y}_n^{cm} = Y_z^{cm}, \quad \forall z \in \mathcal{Z} \quad (3d)$$

$$\hat{g}_{s,n}^{sc} - D_{s,n}^{cm} = r_{s,n}, \quad \forall s \in \mathcal{S}, n \in \mathcal{N}_z \quad (3e)$$

$$- \sum_{n \in \mathcal{N}_z} r_{s,n} \leq p_z^{CO}, \quad \forall s \in \mathcal{S}, z \in \mathcal{Z} \setminus \mathcal{Z}_s^{sc} \quad (3f)$$

$$\sum_{n \in \mathcal{N}} r_{s,n} = 0, \quad \forall s \in \mathcal{S}, \quad (3g)$$

$$\sum_{z \in \mathcal{Z}} p_z^{CO} = 0, \quad (3h)$$

$$-F_{\ell}^{\max} \leq f_{s,\ell}^{cm} \leq F_{\ell}^{\max}, \quad \forall s \in \mathcal{S}, \ell \in \mathcal{L} \quad (3i)$$

$$f_{s,\ell}^{cm} = \sum_{n \in \mathcal{N}} \text{PTDF}_{\ell,n} \cdot r_{s,n}, \quad \forall s \in \mathcal{S}, \ell \in \mathcal{L} \quad (3j)$$

[Changes to the paper \(p. 7, left column in the revised manuscript\):](#)

Market clearing

The zonal electricity price $\lambda_{z,t}^E$ and the zonal capacity price λ_z^{cm} are dual variables linked to the market-clearing conditions 6 and 7, which couple the agents' decision problems. The energy market clears in every zone and period (Eq. 6), where $\bar{\lambda}_{z,t}^E$ is the administrative price cap. The complementarity condition describes that when a zonal balance is attained, the market clears between 0 and $\bar{\lambda}_{z,t}^E$. The price is zero with excess supply, or at the cap with excess demand. ...

The capacity market clears annually in every zone participating in the mechanism. As the capacity demand D_z^{cm} is assumed inelastic, the capacity market clearing condition is defined by Eq. 7.

$$0 \leq \lambda_{z,t}^E \leq \bar{\lambda}_{z,t}^E \perp \sum_{i \in \mathcal{I}} g_{i,z,t} + p_{z,t}^E - d_{z,t}^E, \quad \forall z \in \mathcal{Z}, t \in \mathcal{T} \quad (4)$$

$$0 \leq \lambda_z^{cm} \perp \sum_{i \in \mathcal{I}^{cm}} y_{i,z}^{cm} + p_z^{CO} - D_z^{cm} \geq 0, \quad \forall z \in \mathcal{Z}^{cm}. \quad (5)$$

[Changes to the paper \(p. 4, right column in the revised manuscript\):](#) ... (i) its dispatch $g_{i,z,t}$ for all time steps $t \in \mathcal{T}$, incurring variable operational cost $C_{i,z}^g(g_{i,z,t}) = \frac{A_{i,z}}{2} g_{i,z,t}^2 + B_{i,z} C_{i,z}^g(g_{i,z,t}) = \frac{A_{i,z}}{2} g_{i,z,t}^2 + B_{i,z} g_{i,z,t}$, where $A_{i,z}$ and $B_{i,z}$ are technology-specific parameters;

[Changes to the paper \(Table I, in the revised manuscript\):](#) $\{\text{€/MWh}\}^2 [\text{€/MWh}]^2$

6. Numerical traceability:

- Table III: the text states the CM-NoCBP energy congestion revenue as 0.5 M€ but the table shows 1.14 M€, and the EOM-cap and CM-NoCBP cells both read 1.14 M€ — this looks like a copy/paste, so please check.
- The EOM-cap unserved energy is reported inconsistently: the text's 104.2 GWh, the Table IV ENS cost and the served-demand reduction imply figures that differ.
- "Zone C successfully imports 5 GWh during scarcity": please clarify the units (energy over what duration) in a capacity-obligation context.

Authors' response: We appreciate the reviewer's thoroughness in helping us identify these slips.

- We ran the simulations again, so Table III has been recomputed for all designs and the quoted results above have been updated. The reported figures have been double-checked for accuracy. We also include a new scenario *CM-ref*
- In our re-run, ENS under *EOM-cap* sums to $38.9 + 28.9 + 37.3 = 105.1$ GWh. The distribution is ex-post curtailment sharing (see our response to Comment 1 of reviewer 1), which, at 18,667 €/MWh, yields exactly the 1,963 M€ ENS cost reported in Table IV. Table IV's served-demand reduction, $271,630 - 271,532 = 98.0$ GWh, is smaller than the 105.2 GWh of ENS by 7.2 GWh. This is due to demand elasticity. *EOM-ref* has no forced curtailment (scarcity prices peak below the willingness to pay, so all curtailment is voluntary, Section IV-A) but more voluntary self-rationing by the elastic block, due to scarcity pricing. *EOM-cap* truncates that price signal, so less of the elastic block self-rations, but 105.2 GWh is instead forcibly curtailed as ENS. The served-demand column, therefore, nets ENS against a reduction in voluntary self-rationing.
- After the rerun, the 5 GWh figure is now 3.7 GWh under *CM-FBMC*. The GWh figures in Section IV-B report the actual energy flows realised during an hour of peak scarcity for which the capacity obligations are certified. We report this to confirm that the capacity obligations materialise as energy flows during the scarcity event. To avoid confusion, we now report it as 3.7 GWh/h.

Changes to the paper (p. 9, right column in the revised manuscript): We show the network operator's congestion rents in the energy and capacity market in Table IV. In the reference Energy-only market (*EOM-ref*), high price differentials during scarcity drive congestion rents of ~~72.4 M€~~ **73.1 M€**. During congestion, prices peak at ~~14,057 €/MWh~~ **14,100 €/MWh** in Zone A and 16,608 €/MWh in Zone C. These spreads generate high revenue. Implementing a price cap (*EOM-cap*) drastically reduces these rents to ~~1.1 M€~~ **2.6 M€**. The cap artificially reduces flows (due to under-investment across all zones) and the price spread between zones, as all zones hit the 4,000 €/MWh cap simultaneously. ~~*CM-Implicit* follows the same trend; revenue there falls to 0.1 M€.~~ Finally, in ~~*CM-NoCBP*, over-procurement of capacity leads to lower flows and high price convergence across zones. Congestion revenues collapse to 0.5 M€~~ The capacity market designs without explicit cross-border participation further erase these rents, albeit for different reasons. In *CM-NoCBP*, overprocurement suppresses scarcity pricing. With prices peaking at 560–650 €/MWh in all zones, spreads are minimal and congestion rents collapse to 0.5 M€. In *CM-ref* and *CM-Implicit*, scarcity prices reach the cap; however, similar to *EOM-cap*, prices are near the cap in all zones at the same periods, lowering price spreads and congestion rents fall to 0.1 and 0.5 M€. Explicitly remunerating cross-border participation restores the value of transmission through "capacity congestion rent". ~~*CM-NTC* generates 64.6 M€, partially restoring the levels seen in *EOM-ref*. *CM-FBMC* yields congestion rents of 87.1 M€, surpassing the reference market level. *CM-NTC* generates 39.6 M€, most of it in the capacity market, restoring more than half of the *EOM-ref* level. *CM-FBMC* yields 78.7 M€, narrowly surpassing *EOM-ref*. Unlike *CM-NTC*, it earns a third of this in the energy market, as greater scarcity exports from Zone A congest interconnectors, causing price spreads to persist during scarcity.~~

Changes to the paper (p. 9, right column in the revised manuscript): Zone A is the sole net exporter in both designs. Under *CM-NTC* it sells 2.8 GW of obligations, of which Zones B and C procure 1.5 and 1.3 GW. Under *CM-FBMC* it sells 5.3 GW, and both neighbours roughly double their foreign procurement, to 2.4 GW (Zone B) and 2.9 GW (Zone C), reducing within-zone investment by the same amounts. ~~The higher cross-border capacity trade is shown to be physically deliverable during scarcity with Zone C importing 3.7 GWh/h, against 2.5 GWh/h under *CM-NTC*.~~

Changes to the paper (Table IV, p. 10, left column in the revised manuscript):

Market	<i>EOM-ref</i>	<i>EOM-cap</i>	<i>CM-NoCBP</i>	<i>CM-ref</i>	<i>CM-Implicit</i>	<i>CM-NTC</i>	<i>CM-FBMC</i>
Energy	73.07	2.64	0.50	0.07	0.53	0.47	32.22
Capacity	0.00	0.00	0.00	0.00	0.00	39.08	46.46
Total	73.07	2.64	0.50	0.07	0.53	39.55	78.68

7. Other minor comments:

- Add a brief limitations paragraph: results hold under perfect competition and risk neutrality; risk aversion would

change investment and the value of reliability options, and market power in a thin cross-border capacity market (a single net-exporter zone) is unmodelled.

- Capacity congestion rent: please specify the settlement chain (generator $\rightarrow$ State $\rightarrow$ TSO $\rightarrow$ consumer) and who bears the spread risk when a State over- or under-procures.

Authors' response: Thank you for your comment. We have added a limitations section to our discussion. We briefly discuss the settlement chain below and leave a detailed analysis for future work. In the first layer, generators receive remuneration specified in their multi-year national capacity contracts from the Member State (or its designated counterparty). The Member State then re-marks the corresponding annual capacity obligations in the coupled auction. Uncontracted resources may participate directly in the annual auction. In the coupled annual auction, buyers and sellers are settled at their respective zonal capacity prices. For q MW transferred from exporting zone A to importing zone B , consumers in zone B pay $\lambda_B^{cm} q$, while the seller in zone A (re-marketed capacity, or directly participating generator) receives $\lambda_A^{cm} q$. The participating TSOs receives the difference:

$$(\lambda_B^{cm} - \lambda_A^{cm})q,$$

which constitutes the capacity congestion rent. This revenue may be used for network investment or returned to consumers through lower network tariffs. The detailed allocation of capacity congestion revenues across participating TSOs is left for future work.

The financial risk associated with over- or under-procurement in the first layer remains with the Member State. If a Member State over-procures, it has already committed to the long-term contract cost, but it can recover part of that cost by re-marketing the excess capacity in the annual coupled auction, to the extent that this capacity contributes to regional adequacy and clears in the market. If a Member State under-procures, it must acquire the shortfall in capacity at the zonal capacity price in the annual auction. This may nevertheless be cheaper than procuring the same capacity domestically in the first layer if lower-cost capacity (e.g., a neighbour that has overprocured or has a lower cost of new entry) is available elsewhere in the coupled market.

Changes to the paper (p. 12, left column in the revised manuscript): ~~Lastly, we discuss limitations of our work, and potential implementation challenges. As TSOs already generate year-ahead scarcity scenarios and flow-based domains for the European Resource Adequacy Assessment (ERAA), extending these processes to produce harmonised, year-ahead anticipated scarcity scenarios (flow-based domains, availability of generation assets and demand) is the next logical step. Such a role may be assumed by Europe's Regional Coordination Centres (RCC). However, this approach could increase computational effort, as it requires accounting for grid uncertainties and the N-1 security criterion when computing flow-based domains. Furthermore, because flow-based market coupling has been criticised for opacity, its application in capacity markets must prioritise transparency. Lastly, we discuss limitations of our work, and potential implementation challenges.~~ Since the proposed design is yet to be implemented in any known real-world setup, we elaborate on the limitations of our work and potential implementation challenges. Our equilibrium model assumes perfect competition and risk-neutral agents in both electricity and capacity markets. Extending the analysis to investigate the effect of risk aversion and market power on cross-border capacity market design constitutes a relevant avenue for future research. Additionally, we certify capacity deliverability under a stylised finite scenario set $\mathcal{S}$. Operationalising the market design would require a robust set of scarcity scenarios. As TSOs already generate year-ahead scarcity scenarios and flow-based domains for the European Resource Adequacy Assessment (ERAA), extending these processes to produce harmonised, year-ahead anticipated scarcity scenarios (flow-based domains, availability of generation assets and demand) is the next logical step. Such a role may be assumed by Europe's Regional Coordination Centres (RCC)

Reviewer 4

This paper proposes a two-tiered capacity remuneration mechanism featuring a long-term national capacity market and a shorter term coupled European mechanism. I like the idea, which I think addresses a major obstacle to EU wide capacity mechanisms while improving the coordination of capacity remuneration between member states. I have a few comments, in particular regarding the presentation of the models which I think could be made clearer/more accurate. My comments/questions are as follows:

Authors' response: Thank you for your detailed review of our work and your appreciation of our work. Below, we provide a point-by-point response to your comments and an overview of the changes in the manuscript, which have clearly strengthened the paper. All changes in the manuscript are marked in orange.

1. Model (1):

- Cost C : do you mean $\frac{A_{i,z}}{2}g^2 + B_{i,z}g$? Because I assume the operational cost should be quadratic so that the marginal cost is linear
- Why is $Y_{i,z}$ a variable, if it is the existing capacity?
- I assume that constraint (1d) should not be there, but if it should, then I do not understand what y_i is?

Authors' response: Thank you for carefully reading our paper. We have addressed each point as follows:

- The operational cost is indeed quadratic. In the previous version, $C_{i,z}^g$ denoted the cost per unit of dispatch, which was multiplied by $g_{i,z,t}$ in the objective, yielding the operational cost $\frac{A_{i,z}}{2}g_{i,z,t}^2 + B_{i,z}g_{i,z,t}$. We agree that this notation was not clear. Thus, we have revised the manuscript to define the operational cost directly, $C_{i,z}^g(g_{i,z,t}) = \frac{A_{i,z}}{2}g_{i,z,t}^2 + B_{i,z}g_{i,z,t}$. We corrected the unit of $A_{i,z}$ in Table I to €/MW²h.
- $Y_{i,z}$ denotes existing capacity and is a parameter.
- Eq.(1d) is part of the model, but contained a typo, making it look superfluous: y_i was a typographical error for $y_{i,z}$, which we have corrected. It bounds the total installed capacity from below by the existing capacity, $Y_{i,z} \leq y_{i,z}$. Together with the fixed-cost term $C_{i,z}^f(y_{i,z} - Y_{i,z})$ in Eq. (1a), this ensures that investment costs are incurred only on new build and that divestment of existing capacity cannot generate revenue. In our case study, dispatchable capacity investment is greenfield ($Y_{i,z} = 0$ for all $i \in \mathcal{I}$) while renewable capacity investment is exogenous.

Changes to the paper (p. 4, left column in the revised manuscript): Each generator $i \in \mathcal{I}$, located in zone $z \in \mathcal{Z}$ maximises profit (Eq. (1a)) by choosing $x_i = (g_{i,z,t}, y_{i,z}, y_{i,z}^{\text{cm}})$: (i) its dispatch $g_{i,z,t}$ for all time steps $t \in \mathcal{T}$, incurring variable operational cost $C_{i,z}^g(g_{i,z,t}) = \frac{A_{i,z}}{2}g_{i,z,t}^2 + B_{i,z}g_{i,z,t}$, where $A_{i,z}$ and $B_{i,z}$ are technology-specific parameters; (ii) total capacity $y_{i,z}$ in zone z at cost $C_{i,z}^f$, the annualised fixed cost of new investment per MW; and ~~(iii) its capacity offer in the capacity market, $y_{i,z}^{\text{cm}}$, up to its available capacity.~~ (iii) **for technologies eligible to participate in the capacity market, $i \in \mathcal{I}^{\text{cm}} \subseteq \mathcal{I}$, its capacity offer $y_{i,z}^{\text{cm}}$, up to its available capacity.** ... $\lambda_{z,t}^e$ and λ_z^{cm} denote the zonal electricity market clearing price (€/MWh) and the zonal capacity market clearing price (€/MW). **Existing installed capacity is denoted by $Y_{i,z}$.**

Changes to the paper (Eq. 1d, in the revised manuscript): $Y_{i,z} \leq y_{i,z}, \forall z \in \mathcal{Z}$,

2. Model (3):

It seems like this model is doing two things at the same time: zonal market clearing and implicit allocation of transmission capacity. In a decentralized model, I believe these two aspects should be disentangled. The market operator in this case would be the neoclassical Walrasian agent enforcing market clearing (equivalently, maximizing excess demand), and that would be with zonal granularity. The transmission system operator would maximize congestion rent. This is what corresponds to implicit allocation of transmission capacity as it is done currently in coupled European markets. Model 3 as presented in the manuscript would currently not work as the product of price and net position in the objective makes it nonlinear nonconvex (and so KKT conditions are not sufficient for optimality).

Authors' response: Thank you for your suggestion. We have adopted it. The previous presentation was written for conciseness. We have rewritten it to match our implementation and current practice in Europe. We emphasise that this is a change in presentation only. More precisely, we do the following:

- state the zonal energy market clearing as a complementary condition with the zonal price $\lambda_{z,t}^e$ as the associated market-clearing dual;
- introduce a separate price-taking transmission system operator, which maximises congestion rent and allocates transmission capacity.

For sake of brevity, we only list the objective of the transmission system operator's problem and the market clearing condition. For details, we invite the reviewer to inspect the revised manuscript.

Changes to the paper (Section III):

Previous formulation (removed; reproduced here only):

$$\begin{aligned} \max_{x_E \in \mathcal{X}_E} \quad & \sum_{z \in \mathcal{Z}} \lambda_{z,t}^e \left(\sum_{i \in \mathcal{I}} g_{i,z,t} + p_{z,t}^E - d_{z,t}^e \right) \\ \text{s.t.} \quad & 0 \leq \lambda_{z,t}^e \leq \bar{\lambda}_{z,t}^e, \quad \forall z \in \mathcal{Z}, t \in \mathcal{T} \end{aligned}$$

Revised formulation — transmission system operator (Section III-C):

$$\max_{x_E \in \mathcal{X}_E} \sum_{z \in \mathcal{Z}} \lambda_{z,t}^e p_{z,t}^E$$

with $x_E = (p_{z,t}^E, f_{t,\ell}, \hat{g}_{n,t}, \hat{y}_n, r_{n,t})$; the price $\lambda_{z,t}^e$ is no longer part of the decision vector.

Revised formulation — market clearing (Section III-E):

$$0 \leq \lambda_{z,t}^e \leq \bar{\lambda}_{z,t}^e \perp \sum_{i \in \mathcal{I}} g_{i,z,t} + p_{z,t}^E - d_{z,t}^e, \quad \forall z \in \mathcal{Z}, t \in \mathcal{T},$$

3. Model (4):

I believe model (4) suffers from the same drawback, as prices and allocations are multiplied in the objective, which makes the problem nonconvex. I would have a separate market clearing agent for the capacity market, and then the TSO problem in the capacity market (which could be different than the TSO problem in the zonal market).

Authors' response: We have adopted your recommendation in the same way we responded to comment no. 2 above. The previous presentation was written for conciseness. We have rewritten it to match its implementation. We emphasise that this is a change in presentation only. More precisely, we do the following:

- state the zonal capacity market clearing conditions explicitly as complementary conditions;
- introduce a separate price-taking transmission system operator, which maximises capacity congestion rent and allocates transmission capacity.

For sake of brevity, we only list the objective of the transmission system operator's problem and the market clearing condition. For details, we invite the reviewer to inspect the revised manuscript.

Changes to the paper (Section III):

Previous formulation (removed; reproduced here only):

$$\max_{x_{CO} \in \mathcal{X}_{CO}} \sum_{z \in \mathcal{Z}^{cm}} \lambda_{z,cm}^{cm} \left(y_{i,z}^{cm} + \sum_{z \in \mathcal{Z}} p_z^{CO} - D_z^{cm} \right)$$

Revised formulation — transmission system operator in the capacity market (Section III-D):

$$\max_{x_{CO} \in \mathcal{X}_{CO}} \sum_{z \in \mathcal{Z}^{cm}} \lambda_z^{cm} p_z^{CO}$$

with $x_{CO} = (p_z^{CO}, \hat{g}_{s,n}^{sc}, f_{s,\ell}^{cm}, \hat{y}_n^{cm}, r_{s,n})$; the price λ_z^{cm} is no longer part of the decision vector.

Revised formulation — capacity market clearing (Section III-E):

$$0 \leq \lambda_z^{cm} \perp \sum_{i \in \mathcal{I}^{cm}} y_{i,z}^{cm} + p_z^{CO} - D_z^{cm} \geq 0, \quad \forall z \in \mathcal{Z}^{cm}.$$

4. How does a zonal capacity price steer investment to deliverable locations?:

In addition, I am not sure I understand the statement that "CM-FBMC ensures that cleared capacity is deliverable w.r.t. network constraints under a set of defined scarcity scenarios". I do not understand how this can work if the capacity market price remains zonal, as it seems that there is no way to steer investment in the right location so that network constraints can be met during scarcity events when the capacity price is zonal. It would be nice if the paper could clarify that.

Authors' response: Thank you for your comment. The zonal capacity price provides an investment signal at the zonal,

and not at the nodal level. *CM-FBMC* does not specify where within a zone new capacity is located, because, as you pointed out, this would require a more granular price signal. However, it enforces network deliverability of procured generation capacity by limiting cross-zonal capacity exchange to the flows that the network can support during scarcity. Specifically, for a given nodal allocation of capacity and demand, *CM-FBMC* allows cross-zonal capacity obligations only up to the level for which the scarcity flows remain feasible with respect to the network constraints. Hence, the feasible amount of capacity exchange depends on where generation capacity is located within each zone. If a larger cross-zonal obligation would make the scarcity scenarios infeasible, the corresponding transmission capacity cannot be made available to the capacity market, and a larger share of the zonal capacity requirement must consequently be met by domestic capacity. *CM-FBMC* therefore does not steer investment toward particular nodes through capacity prices; rather, it limits the extent a zone can rely on foreign capacity based on the deliverability of the resulting flows during scarcity, given the realised nodal allocation of capacity. We have rephrased the description to clarify that "right location" refers to zones and not nodes within a zone.

Changes to the paper (p. 5, right column in the revised manuscript): Our proposed approach consists (i) a coupled capacity market clearing and (ii) a **simultaneous joint** scarcity assessment (Figure 2). The **flow-based coupling clearing** of the capacity markets **under flow-based coupling** is formalised in Problem 4. **In the proposed coupled capacity market, accepted capacity offers Y_z^{cm} in zone z must be deliverable to the relevant load centres under a set of anticipated scarcity scenarios $s \in \mathcal{S}$.** The capacity market coupling allows net cross-border obligations p_z^{CO} only if they are certified to be deliverable to the relevant load centres in each anticipated scarcity scenario $s \in \mathcal{S}$. It does so by limiting the transmission capacity made available for cross-zonal capacity exchange, ensuring the feasibility of network flows during all scarcity situations.

Thus, we require the existence of a feasible nodal dispatch $\hat{g}_{s,n}^{\text{sc}}$ of the committed firm capacity that meets the energy demand at every node n and satisfies all network constraints (including intra-zonal bottlenecks) in every scarcity scenario. As a result, any MW of capacity cleared in the auction can contribute to the nodal capacity requirement $D_{s,n}^{\text{cm}}$ when scarcity materialises. The auction yields for each zone z , a single annual capacity price λ_z^{cm} , the domestic capacity obligation Y_z^{cm} and the net cross-border capacity obligation p_z^{CO} that meets all nodal adequacy requirements $D_{s,n}^{\text{cm}}$ across the defined scarcity scenarios. In each scenario s , the zones in $\mathcal{Z}_s^{\text{sc}}$ are placed at their scarcity demand, above the capacity demand $D_{s,n}^{\text{cm}} = S_{n,z} D_z^{\text{cm}}$ allocated to them by the joint assessment, while the remaining zones are placed at their coincident demand (Figure 2, right). Due to the absence of a nodal price signal to optimally locate new capacity in zonal markets, we exploit the network operator's energy market problem (Problem 3) to obtain a feasible nodal allocation $\hat{g}_{i,n}$ of each generator's zonal capacity. This allocation provides the nodal representation of the zonal investment outcome used in the TSO's capacity market problem (Problem 4). Based on this nodal allocation, the capacity market problem determines how much cross-zonal capacity exchange can be supported while remaining deliverable across the prescribed scarcity scenarios.

For each scenario, there must exist a nodal dispatch $\hat{g}_{s,n}^{\text{sc}}$ that serves demand while respecting the network constraints, including intra-zonal lines at state-N thermal limits. The resulting zonal net positions must remain consistent with the cross-zonal capacity obligations. In particular, a zone that has sold capacity abroad must be able to sustain the corresponding export position when that capacity is called during the importing zone's scarcity event. The auction yields, for each zone z , a single annual capacity price λ_z^{cm} , the domestic capacity obligation Y_z^{cm} , and the net cross-border capacity obligation p_z^{CO} .

5. Compare implicit-vs-explicit conclusion with [11] (Menegatti and Meeus):

Since the paper, as a by-product of the simulation of their mechanism, get a comparison of implicit and explicit cross-border allocation, I believe it would be nice if the authors would compare their result with that of [11] and discuss whether they reach the same conclusion w.r.t. the implicit vs explicit capacity allocation question.

Authors' response: Thank you for your comment. Our work differs from Menegatti and Meeus [11] in the institutional setting that motivates the analysis, which limits a one-to-one comparison of the resulting implicit and explicit participation outcomes. Menegatti and Meeus study the externality created by a capacity market on neighbouring zone that has a energy-only market in a two-zone setting. In contrast, we consider a "multi-capacity-market" setting in which neighbouring Member States simultaneously operate capacity markets and ask how their procurement can be coordinated.

In [11], foreign generators are located in the energy-only zone. Under explicit participation, they may bid into the neighbouring capacity market, but the foreign capacity price is zero in their simulations. Explicit participation, therefore, provides no additional investment signal relative to implicit participation, which explains why the two mechanisms yield the same long-run investment outcome. This mechanism does not carry over to our setting. Because all zones operate capacity markets, foreign capacity has value in its domestic capacity market, and explicit cross-border participation can provide positive remuneration for capacity located in one zone but procured to satisfy capacity obligations in another. Menegatti and Meeus [11] explicitly identify this as a different question.

“Fourth, can explicit cross-border participation become relevant when the neighbouring zone also has a capacity market? In such a configuration, explicit cross-border participation can produce a non-zero price for capacity in the neighbouring zone. However, the regulatory objective would be different. The regulatory challenge is to coordinate between two (competing) capacity markets rather than to fix the externalities of a capacity market on the neighbouring zone.”

Our results, therefore, do not contradict the equivalence between implicit and explicit participation found in [11]. Rather, we address a coordination problem for interconnected energy markets with multiple national capacity markets, which they identify as a separate question to which their analysis and conclusions do not extend. In this setting, explicit participation connects cross-border capacity contributions to remuneration and scarcity events within a market-clearing problem. By contrast, implicit participation relies on ex-ante estimates of import contributions, creating a circular dependency because those contributions depend on the investment outcome that the import assumptions themselves induce. We show that our proposed explicit cross-border participation avoids this chicken-and-egg problem, enables investment to locate in lower-cost zones, and restores the economic value of interconnection by facilitating greater cross-border trade of energy and capacity.

6. Why does CM-Implicit exhibit ENS :

It's not fully clear to me why the CM-implicit mechanism exhibit energy-not-served. Perhaps the authors could explain a bit more the economic mechanism behind and discuss whether this is expected or an artefact of the model. Wouldn't it possible for MS to reduce this by fine tuning the method for calculating the external contribution?

Authors' response: Thank you for your question. Under implicit participation, each zone deducts from its capacity demand the imports it expects to receive from foreign capacity assumed to be available during scarcity events. In a long-run equilibrium setting, this creates a circular dependency. The import contribution used to determine capacity demand depends on the resulting investment outcome, while that investment outcome, in turn, depends on the capacity demand. In the long run, the unremunerated foreign capacity may fail to materialise, and if the assumed foreign contribution is not iteratively reassessed as investment reacts, this may lead to underprocurement and ENS. Our CM-Implicit results illustrate this issue.

However, in a model setting, one can indeed, as you suggest, iteratively tune the contributions from other zones until ENS disappears. We introduced another scenario, which we call *CM-ref*, that provides a useful benchmark for this issue. It uses the result of the joint scarcity assessment to compute capacity demand, thereby accounting for non-coincident scarcity across zones. *CM-Implicit* starts from these coordinated capacity demands but further deducts the expected imports observed under *CM-ref* during scarcity events, without remunerating the foreign capacity that supports those imports. We also clarify the sensitivity of *CM-Implicit* to these import assumptions. Higher import deductions reduce investment but increase ENS costs much more rapidly than the savings, while with no import deduction, the design collapses to *CM-ref*.

Changes to the paper (p. 7, right column in the revised manuscript):

CM-ref: A reference capacity market scenario whose capacity demands result from a joint scarcity assessment (Figure 2 (Right)), but without cross-border participation. The joint scarcity assessment considers three zonal scarcity scenarios, representing each zone's individual peak demand. The set of scarcity scenarios considers the residual demand valued at the price cap of 4,000 €/MWh across all zones at each time step, any zone experiences ENS under *EOM-cap*. The joint scarcity assessment sets the system-wide capacity demand to the maximum coincident peak residual demand. This capacity demand is distributed over the zones proportionally to the average share of the zone's residual demand to the overall system residual demand during scarcity situations. This scenario allows isolating the effect of coordinating capacity requirements from that of cross-border participation. The resulting capacity demand forms the basis for all scenarios considering cross-border participation.

CM-Implicit: A capacity market with implicit cross-border participation. ~~To determine capacity demand, expected scarcity imports from neighbouring zones are subtracted from the peak residual demand in each zone. This creates a circular dependency, as imports depend on available capacity. We use the investment outcome of the *CM-FBMC* to determine available imports, thereby enabling comparability with other scenarios.~~ Each zone's capacity demand is that determined in the joint scarcity assessment minus the average imports it receives in the scarcity scenarios under scenario *CM-ref*.

Changes to the paper (p. 10, left column in the revised manuscript): Coordinating the capacity requirement in *CM-ref* reduces the required capacity investments by 2.5 GW and lowers annualised investment costs by 181 M€/yr, limiting the total system-cost increase relative to *EOM-ref* to 3.1% (470 M€/yr). *CM-Implicit* reduces investment costs by a further 245 M€/yr, but this is largely offset by 232 M€/yr in unserved-energy costs, leaving total system costs only marginally below *CM-ref* and 2.9% above *EOM-ref*. Unserved energy under *CM-Implicit* is 12.4 GWh per year, equivalent to roughly 0.005% of annual demand (Table V). *CM-implicit* is sensitive to assumptions on the availability of imports during scarcity. Higher import assumptions would increase the volume of unserved energy, and because ENS costs rise faster than investment savings, the total system cost would increase. Without import deductions, the design collapses to *CM-ref*.

7. Numerical details:

Perhaps a bit more could be said about existence, uniqueness of the complementarity model and convergence of the ADMM algorithm.

Authors' response: Thank you for your comment. We have included an appendix, in which we've added numerical details as requested.

Changes to the paper (Appendix I in the revised manuscript): The equilibrium is computed using an exchange-form consensus ADMM implemented in Julia, in which each agent's convex quadratic subproblem is solved in parallel with Gurobi. Each iteration solves Problems (1)–(4) in parallel, given the prices, then updates $\lambda_{z,t}^e$ and λ_z^{cm} as duals of the zonal imbalances. The algorithm terminates when, for every zone and for both markets (electricity and capacity), the primal and dual residuals fall below an absolute tolerance $\epsilon = 0.1$ (in MW for primal residuals; in €/MWh and €/MW-year for EOM and CM dual residuals). The penalty parameter ρ is adapted per zone and market via residual balancing following Boyd et al [1]. Converged runs typically require 460-2,500 iterations (13-38 seconds of wall time), with each iteration taking residuals to decay steadily (Table). All runs are cold-started, with prices, quantities, and penalty terms initialised at zero.

Because each agent solves a convex quadratic problem, a converged ADMM solution satisfies the conditions for competitive equilibrium. As degeneracy exists in some variables (e.g., within-zone investment allocation, network operator capacity allocation, and unserved energy when price cap hits multiple price zones simultaneously), solutions may be non-unique. However, these multiple equilibria differ only in nodal quantities (except for unserved energy, which we treat ex-post, see Appendix II); aggregate installed capacity and zonal outcomes remain consistent across repeated runs.

Table VI: Final ADMM residuals and iteration counts across market designs. Reported residuals are maximum over zones. The convergence tolerance is $\epsilon = 0.1$

Design	Iterations	$\max r_E^p$	$\max r_E^d$	$\max r_{CM}^p$	$\max r_{CM}^d$
EOM-ref	1,504	0.012	0.100	0	0
EOM-cap	462	0.017	0.100	0	0
CM-NoCBP	498	0.047	0.099	0.031	0.077
CM-ref	845	0.084	0.038	0.092	0.024
CM-Implicit	2,223	0.027	0.100	0.017	0.029
CM-NTC	830	0.038	0.097	0.057	0.099
CM-FBMC	733	0.031	0.047	0.025	0.088